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Lang et al.

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(54) **ADMINISTERING A LYMPHOCYTIC CHORIOMENINGITIS VIRUS (LCMV) WITH IMPROVED TUMOR REGRESSION PROPERTIES**

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Related U.S. Application Data

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Feb. 4, 2019, now Pat. No. 11,801,294, which is a
division of application No. 15/567,343, filed as
application No. PCT/EP2016/058347 on Apr. 15,
2016, now abandoned.

(30) **Foreign Application Priority Data**

Apr. 17, 2015 (DE) 102015207036.0

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A61K 39/12 (2006.01)
C12N 7/00 (2006.01)
A61K 39/00 (2006.01)
A61P 35/00 (2006.01)

(52) **U.S. Cl.**

CPC **A61K 39/12** (2013.01); **A61K 35/76**
(2013.01); **A61K 35/768** (2013.01); **C12N 7/00**
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C12N 2760/10011 (2013.01); **C12N**
2760/10021 (2013.01); **C12N 2760/10032**
(2013.01); **C12N 2760/10033** (2013.01)

(58) **Field of Classification Search**

CPC **A61K 35/768**; **C12N 2760/10032**
See application file for complete search history.

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(57) **ABSTRACT**

The invention relates to arenaviruses for use in the treatment
and/or prevention of tumors and also a method for preparing
arenaviruses with (improved) tumor-regressive properties.

16 Claims, 9 Drawing Sheets

Specification includes a Sequence Listing.

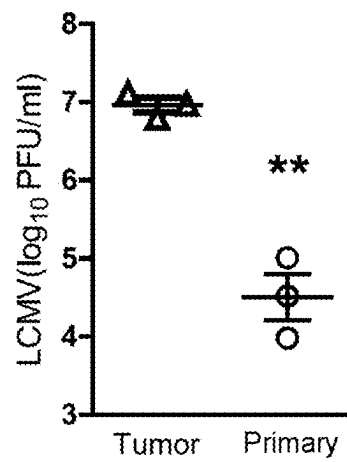


Figure 1

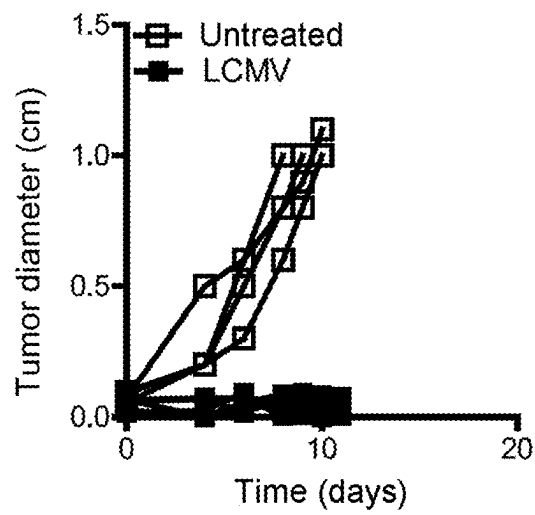


Figure 2

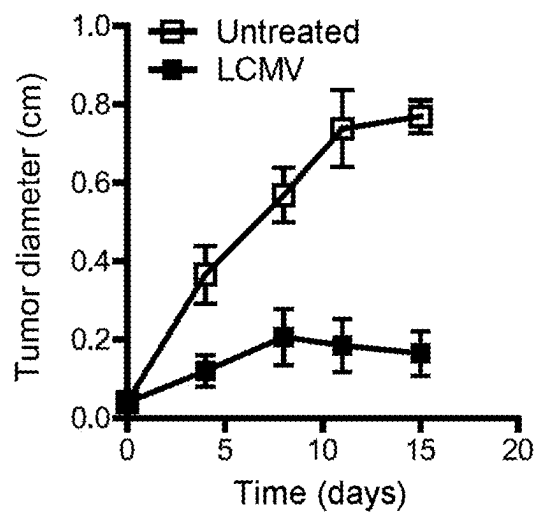


Figure 3

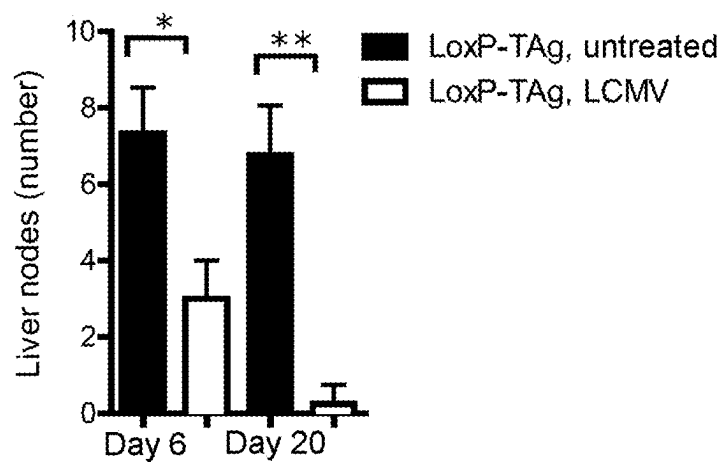


Figure 4

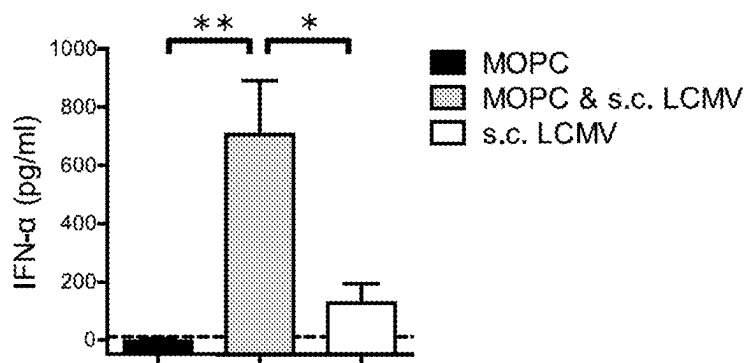


Figure 5

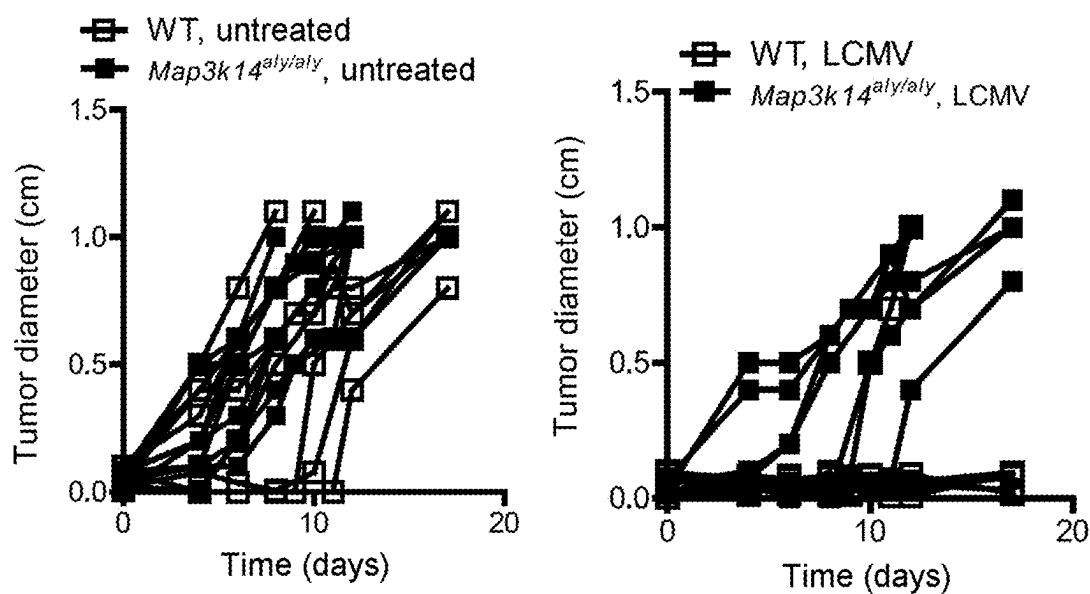


Figure 6

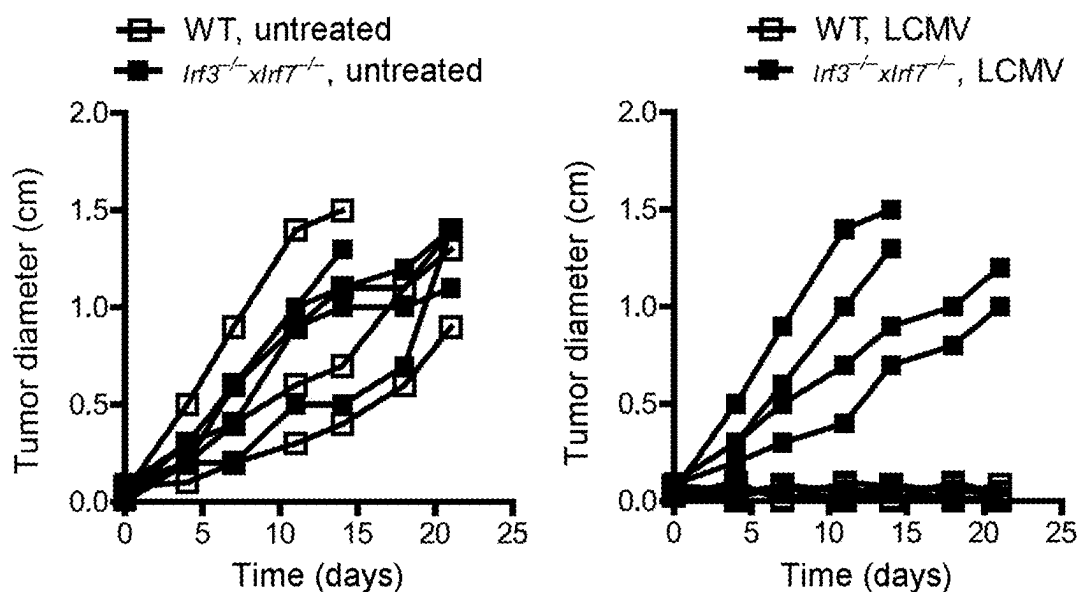


Figure 7

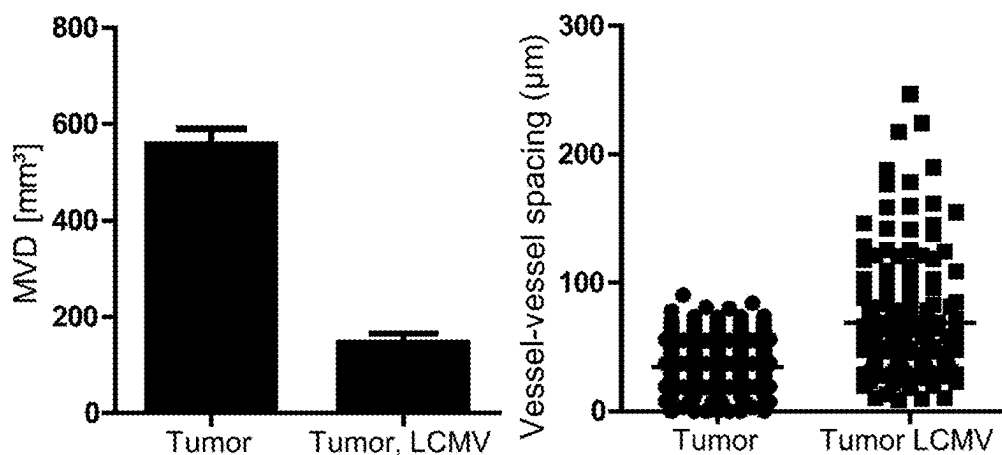


Figure 8A

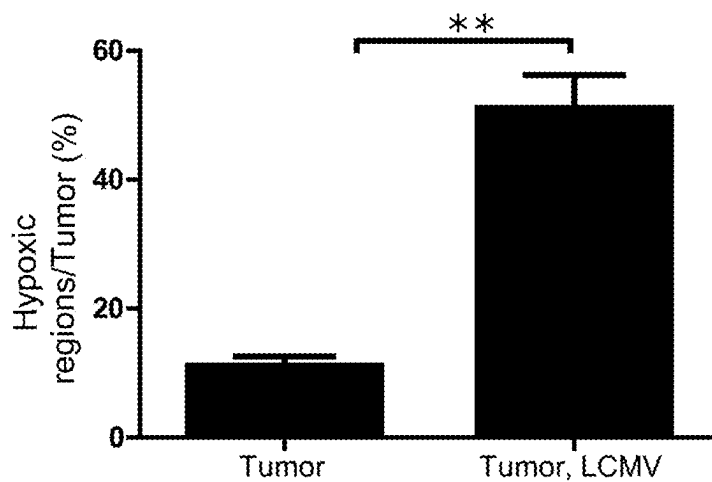


Figure 8B

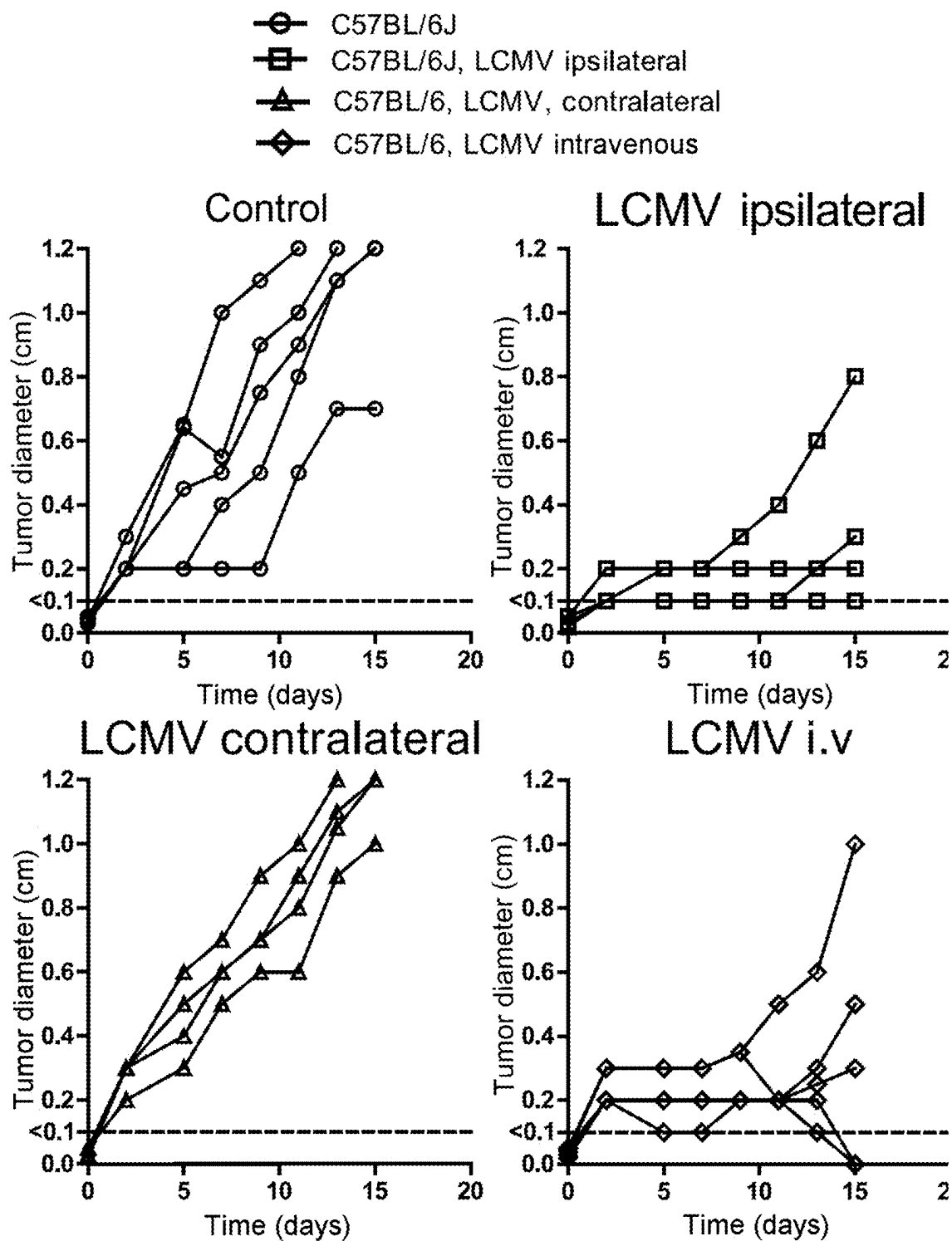


Figure 9

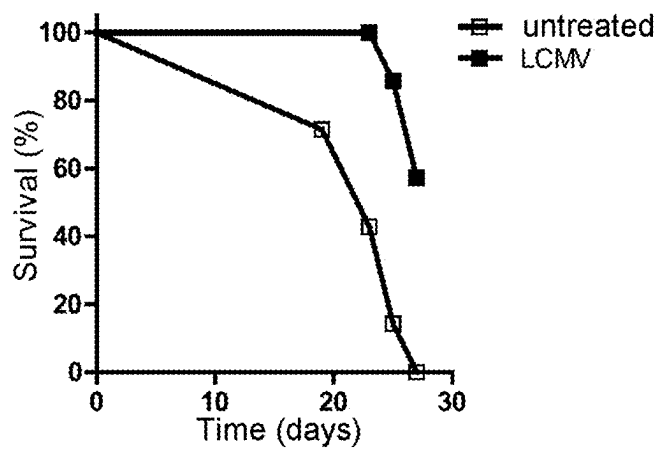


Figure 10

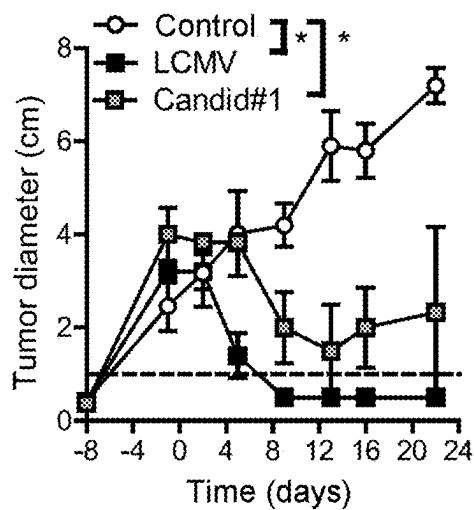


Figure 11

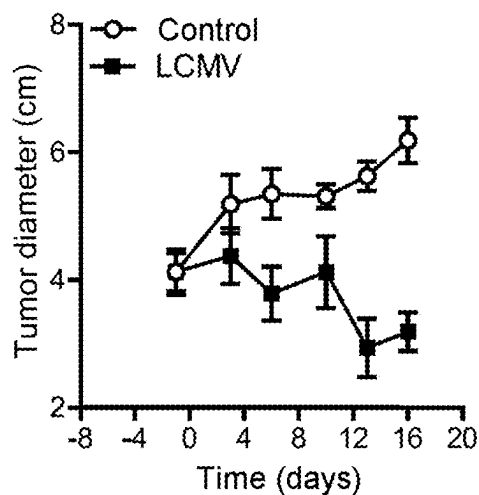


Figure 12

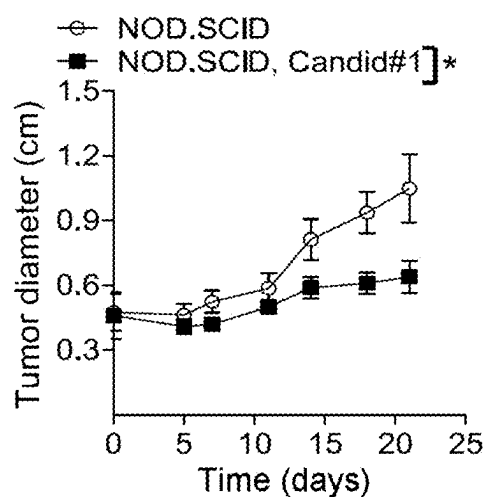


Figure 13

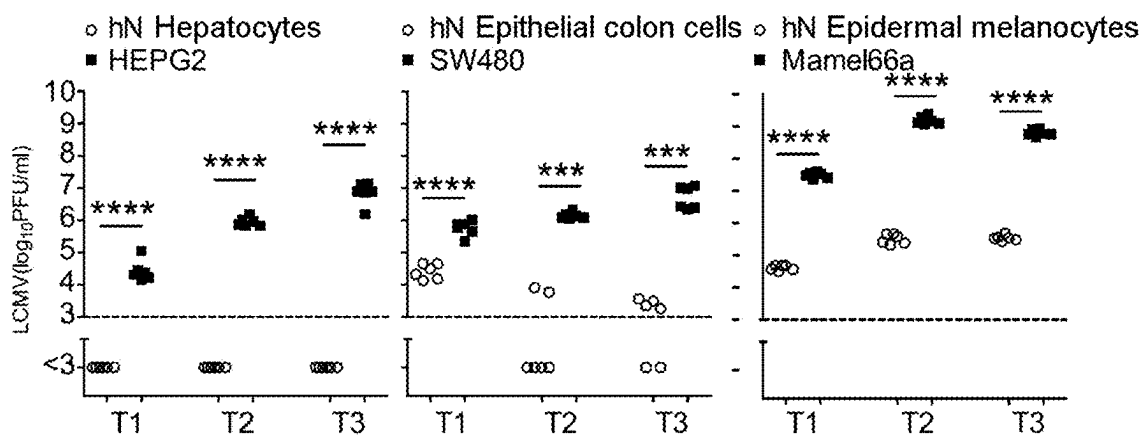


Figure 14

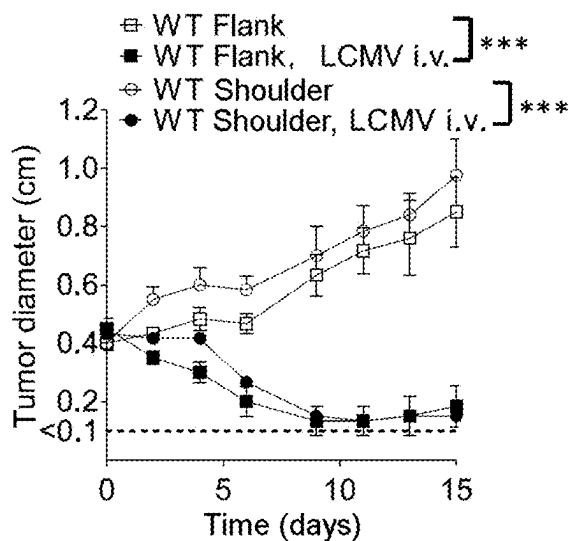


Figure 15A

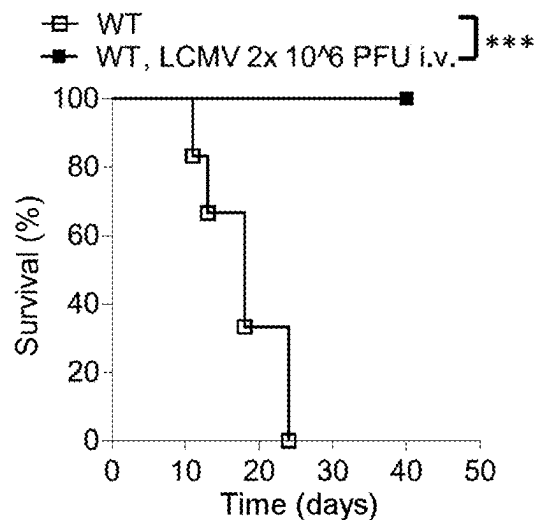


Figure 15B

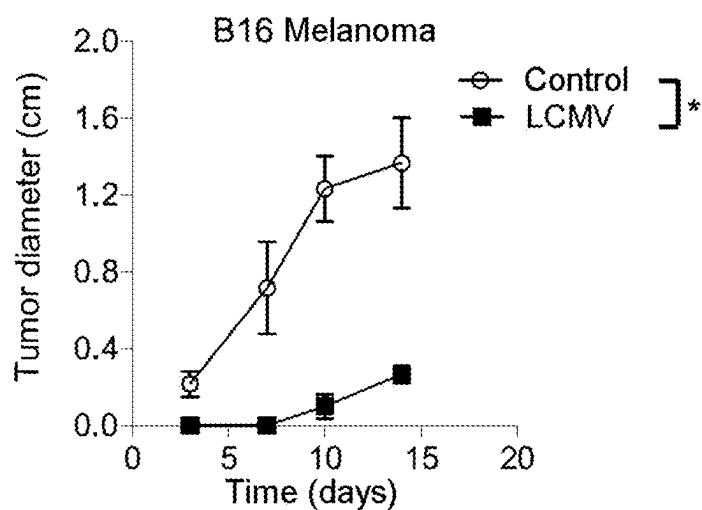


Figure 16

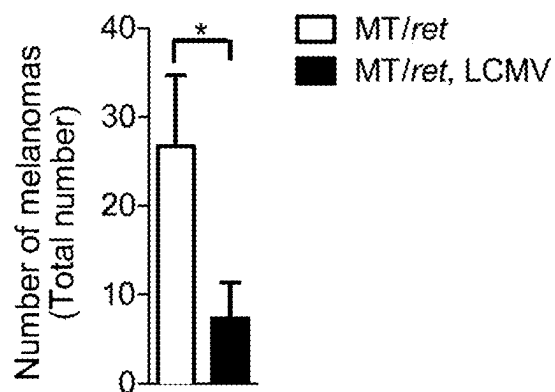


Figure 17

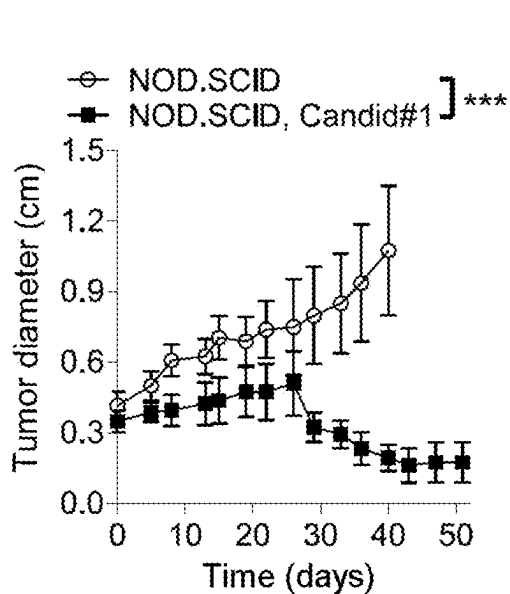


Figure 18A

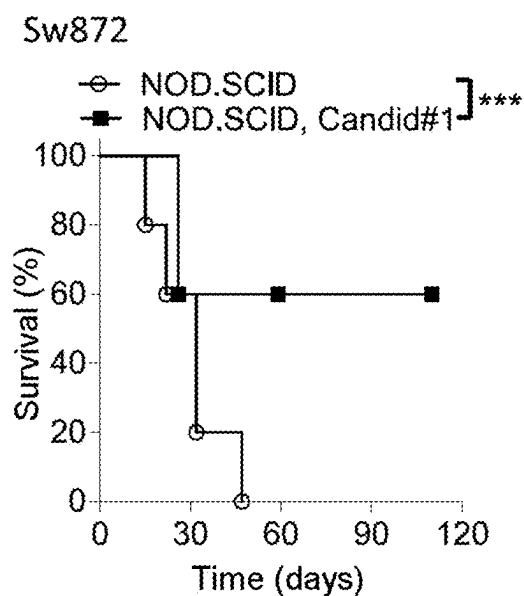


Figure 18B

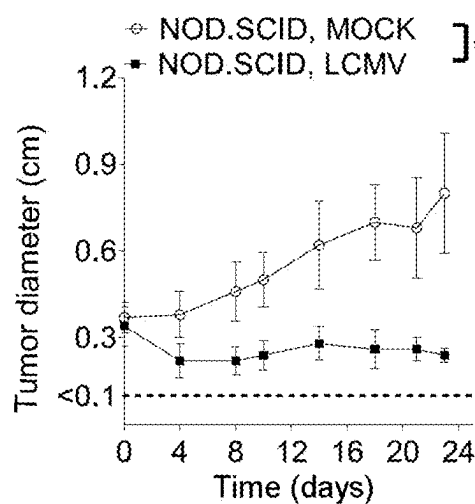


Figure 19A

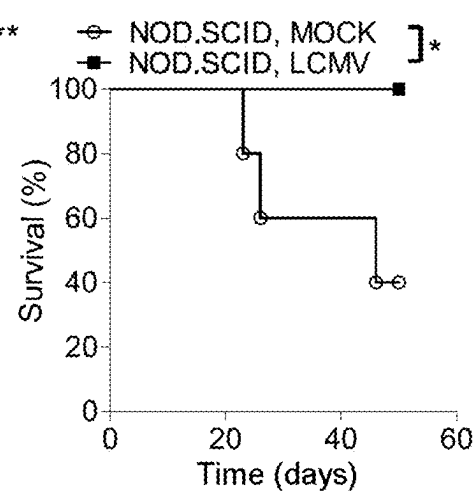


Figure 19B

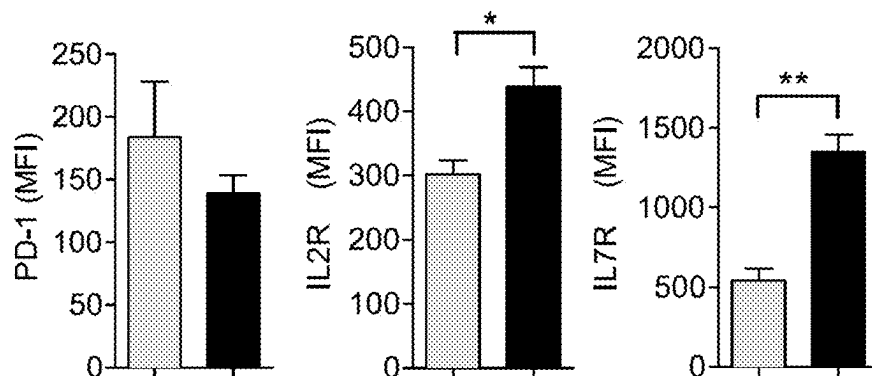


Figure 20

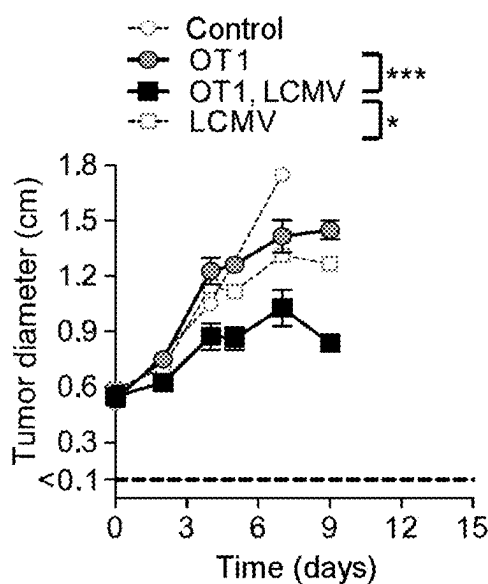


Figure 21A

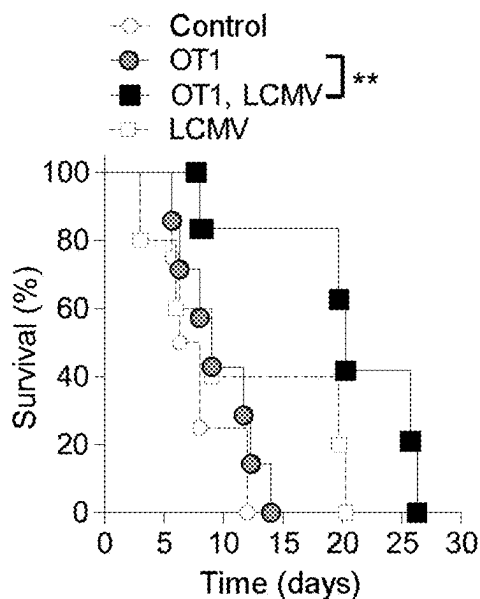
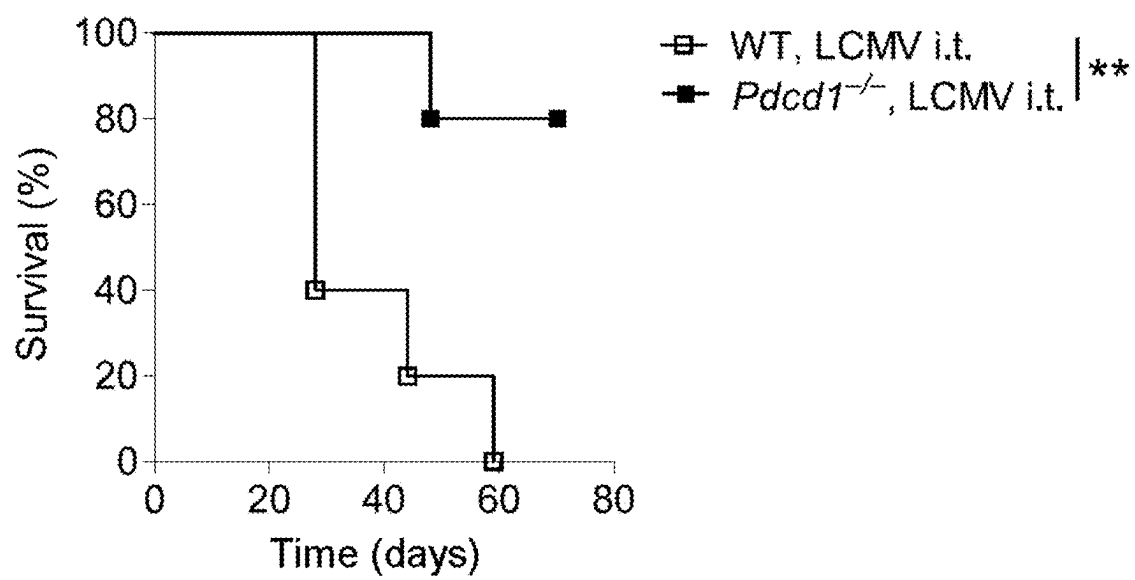


Figure 21B

**Figure 22**

ADMINISTERING A LYMPHOCYTIC CHORIOMENINGITIS VIRUS (LCMV) WITH IMPROVED TUMOR REGRESSION PROPERTIES

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a continuation of U.S. patent application Ser. No. 16/267,095, filed Feb. 4, 2019, now U.S. Pat. No. 11,801,294, which is a divisional application of U.S. patent application Ser. No. 15/567,343, filed Oct. 17, 2017, which is the United States National stage patent application of International Application No. PCT/EP2016/058347, filed Apr. 15, 2016, which claims the right of priority of German patent application No. 102015207036.0 filed Apr. 17, 2015, with the German Patent Office, the entire contents of each of which are incorporated herein for all purposes.

SEQUENCE LISTING

The application contains a Sequence Listing which has been submitted electronically in .XML format and is hereby incorporated by reference in its entirety. Said .XML copy, created on Mar. 27, 2024, is named "SCH-5000-CT1_SL.xml" and is 2,758,175 bytes in size. The sequence listing contained in this .XML file is part of the specification and is hereby incorporated by reference herein in its entirety.

FIELD OF THE INVENTION

The invention relates to arenaviruses for use in the treatment and/or prevention of tumors and also methods for preparing arenaviruses with (improved) tumor-regressive properties.

Arenaviruses belong to the family of human pathogenic, pleomorphic RNA viruses. Diseases with these viruses belong to the zoonoses due to their natural reservoir in animals, predominantly rodents. Zoonoses refer to diseases that can be transferred from the animal to humans and vice versa from humans to the animal.

At least eight arenaviruses are known to cause illness in humans. Typical are aseptic meningitis and haemorrhagic fever. Known viruses which can trigger a disease in humans are the lymphocytic choriomeningitis virus (LCMV), Guanarito virus (GTOV), Junin virus (JUNV), Lassa virus (LASV), Lujo virus (LUJV), Machupo virus (MACV), Sabia virus (SABV) and the Whitewater Arroyo virus (WWAV).

Arenaviruses are generally divided into two groups, namely the Old World arenaviruses and the New World arenaviruses. These groups differ geographically and genetically. Old World arenaviruses, such as the lymphocytic choriomeningitis virus, have been found in countries of the eastern hemisphere, such as European, Asian and African countries. In contrast, New World arenaviruses have been found in countries of the western hemisphere, such as Argentina, Bolivia, Venezuela, Brazil and the United States of America.

The name of the virus family is derived from the Latin arenosus (sandy) and arena (sand) to describe the sandy ribosomal structure within the virions. The virions of the arenaviruses have a round to irregular shape and have a diameter, depending on species and preparation of the test material, from 50 nm to 300 nm, usually between 110 nm and 130 nm. Club-shaped glycoprotein spikes, 8 nm to 10

nm long, are embedded in the virus envelope. The individual spikes consist of a tetramer of the viral envelope protein.

The virions also comprise two closed-ring capsids with helical symmetry. The length of the capsids varies between 450 nm and 1300 nm. One molecule of the viral RNA (ribonucleic acid) polymerase (L-protein) is attached to each of them.

Each capsid comprises one molecule of a single-stranded RNA with mixed (i.e. ambisense, +/−) polarity. The two single-stranded RNA molecules represent the viral genome. They are referred to as L (large) and S (small) and are about 7.5 kb (kilobases) or 3.5 kb (kilobases) large. Although the capsids are closed ring-shaped, the RNA strands are linear and thus not circular. A 19 to 30 base long sequence at the 3' end of the RNA is present on both strands and is also conserved within the virus family.

Very exceptional morphologically is the presence of an alternating number of cellular ribosomes within the virions, which give the viral particles their "sandy" appearance. Similarly, in purified virus preparations, a number of different cellular RNAs (also including ribosomal RNA) and also replicative forms of viral RNA are found, as are diverse viral mRNAs (messenger ribonucleic acids bound to the ribosomes) and complete complementary strands of the virus genome. These non-genomic RNAs are found in varying amounts all lying outside the abovementioned capsids.

The use of arenaviruses as vaccination vectors is known. A prominent example is the vaccination virus Candid #1 used against Argentinian hemorrhagic fever. This is a vaccination variant of the Junin virus.

Known from WO 2009/083210 A1 is the use of replication defects, i.e. genetically modified arenavirus particles (virions), inter alia, for the treatment of neo-plastic diseases such as, for example, melanoma, prostate carcinoma, breast carcinoma and lung carcinoma. In the publication "Development of replication-defective lymphocytic choriomeningitis virus vectors for the induction of potent CD8⁺ T cell immunity" (Nature Medicine, Vol. 16, No. 3, March 2010, pp. 339-345; doi: 10.1038/Nm.2104), cancer immunotherapy is mentioned as a potential area of application for such viral particles.

Furthermore, WO 2006/008074 A1 discloses the use of packaging cells, which produce retroviral virions pseudotyped with arenavirus glycoprotein, for gene therapy of solid tumors.

The methods for the treatment of tumors described in the prior art are based on the use of virus particles which are very complicated to produce by genetic engineering. In the case of gene therapy treatment methods, it is frequently not possible to achieve adequate, therapeutically effective transduction of the tumor tissue with genetically engineered virions or packaging cells which produce virions.

SUMMARY OF THE INVENTION

Object and Solution

The present invention is therefore based on the object of providing a simpler and, in particular, more efficient therapeutic solution for tumors, in particular carcinomas and sarcomas, compared to the prior art.

This object is achieved according to the invention by an arenavirus according to independent claim 1, by a medicament according to claim 14 and also by an in vitro method according to independent claim 15. Preferred embodiments are defined in the dependent claims. The wording of all claims is hereby incorporated by reference into the content

of the description. An additional subject of the invention, which achieves the object of the invention, is disclosed in the description.

According to a first aspect, the invention relates to an arenavirus for use in the treatment and/or prevention of a tumor, preferably a malignant tumor, in humans or animals.

The arenavirus is preferably characterized in that it is free of genomic foreign RNA, i.e. it does not comprise any genomic foreign RNA. In other words, the genome of the arenavirus is preferably free of foreign RNA or preferably comprises no foreign RNA.

In the context of the present invention, the expression "genomic foreign RNA" is intended to mean an RNA (ribonucleic acid) or RNA sequence which does not occur or is not present in the genome of a wild-type arenavirus or in the genome of a mutant of a wild-type arenavirus (mutated arenavirus), in particular in the genome of a natural mutant of a wild-type arenavirus (naturally mutated arenavirus). Examples of foreign RNA are artificial or synthetic RNA molecules, RNA of organisms and RNA from other viruses.

In the context of the present invention, (in accordance with the understanding of those skilled in the art), the expression "wild-type arenavirus" is understood to mean an arenavirus whose genome is the genetically normal form occurring in nature.

In the context of the present invention, (in accordance with the understanding of those skilled in the art), the expression "mutant of a wild-type arenavirus" or "mutated arenavirus" is understood to mean an arenavirus whose genome comprises a spontaneous, i.e. naturally-induced, modification or modification induced by mutagenesis, compared to the wild-type genome.

Accordingly, in the context of the present invention, (in accordance with the understanding of those skilled in the art), the expression "natural mutant of a wild-type arenavirus" or "naturally mutated arenavirus" is understood to mean an arenavirus whose genome comprises a spontaneous, i.e. naturally-induced, modification, compared to the wild-type genome. A naturally mutated arenavirus can be produced preferably by passage, in particular serial passage, which will be discussed in more detail below.

The invention is based on the surprising finding that arenaviruses without genomic foreign RNA are able to effect tumor regression. Tumor regression is due to an activation or stimulation of congenital and adaptive immune cells caused by the arenaviruses. The activated immune cells secrete increased antitumoral cytokines such as interferon- α and interferon- γ , thereby counteracting or repelling the tumor. A further surprising finding is the realization that the arenaviruses cause a significantly increased secretion of antitumoral cytokines in the case of a tumor manifestation. Arenaviruses without genomic foreign RNA are thus suitable for use in tumor treatment. This has been successfully verified by the applicant by means of animal experiments. For this purpose, mice were used, inter alia, in which growth of human tumors is possible.

In a further embodiment, the arenavirus is also free of non-genomic foreign RNA. In the context of the present invention, the expression "non-genomic foreign RNA" is intended to mean an RNA or RNA sequence, apart from the arenavirus genome, which does not occur or is not present in a wild-type arenavirus or a mutant of a wild-type arenavirus (mutated arenavirus), in particular of a natural mutant of a wild-type arenavirus (naturally mutated arenavirus). In other words, in this embodiment the arenavirus does not comprise overall any foreign RNA, i.e. neither genomic foreign RNA nor non-genomic foreign RNA.

In a preferred embodiment, the arenavirus is a wild-type arenavirus.

In a further embodiment, the arenavirus is a natural mutant of a wild-type arenavirus, i.e. a naturally mutated arenavirus.

The natural mutant or the naturally mutated arenavirus is preferably produced by passage, in particular multiple passage, in host animals and/or host cells.

The natural mutant or the naturally mutated arenavirus is particularly preferably produced by serial passage in host animals and/or host cells.

The arenavirus provided according to the invention is thus preferably an arenavirus which is produced starting from its wild-type form by passage, preferably serial passage, in host animals and/or host cells.

The host animals mentioned in the preceding paragraphs are preferably rodents, particularly mice. The host cells mentioned in the previous paragraphs, on the other hand, are preferably dendritic cells or tumor cells.

In the context of the present invention, in accordance with the understanding of those skilled in the art, the term "passage" is understood to mean a multiple, regular introduction of the arenavirus into host animals and/or host cells. In the context of the present invention, in accordance with the understanding of those skilled in the art, the expression "serial passage" is understood to mean a multiple, regular introduction of the arenavirus into different host animals, preferably of the same type, and/or different cells, preferably of the same type. Due to the multiple changes of environment (host animal and/or host cell), the arenavirus is subject to an increased adaptation pressure or mutational pressure, thereby increasing the likelihood of advantageous mutations occurring in the genome of the arenavirus from the perspective of tumor regression.

In a further embodiment, the tumor is selected from the group comprising or consisting of carcinoma, melanoma, blastoma, lymphoma and sarcoma.

In the context of the present invention, (in accordance with the understanding of those skilled in the art), the term "carcinoma" is intended to mean malignant neoplasia of epithelial origin.

In the context of the present invention, (in accordance with the understanding of those skilled in the art), the term "sarcoma" is intended to mean malignant neoplasia of mesodermal origin.

In the context of the present invention, (in accordance with the understanding of those skilled in the art), the term "melanoma" is intended to mean malignant neoplasia of melanocytic origin.

In the context of the present invention, (in accordance with the understanding of those skilled in the art), the term "lymphoma" is intended to mean malignant neoplasia of lymphocytic origin.

In the context of the present invention, (in accordance with the understanding of those skilled in the art), the term "blastoma" is intended to mean malignant neoplasia of embryonic origin.

In a preferred embodiment, the carcinoma is selected from the group comprising or consisting of anal carcinoma, bronchial carcinoma, lung carcinoma, endometrial carcinoma, gallbladder carcinoma, hepatocellular carcinoma, testicular carcinoma, colorectal carcinoma, laryngeal carcinoma, oesophageal cancer, gastric carcinoma, breast carcinoma, renal carcinoma, ovarian carcinoma, pancreas tumor, pharyngeal carcinoma, prostate carcinoma, thyroid carcinoma and cervical carcinoma.

In a preferred embodiment, the sarcoma is selected from the group comprising or consisting of angiosarcoma, chondrosarcoma, Ewing's sarcoma, fibrosarcoma, Kaposi's sarcoma, liposarcoma, leiomyosarcoma, malignant fibrous histiocytoma, neurogenic sarcoma, osteosarcoma and rhabdomyosarcoma.

In a preferred embodiment, the arenavirus is an Old World arenavirus which is preferably selected from the group comprising or consisting of Ippy virus (IP-PYV), Lassa virus (LASV), lymphocytic choriomeningitis virus (LCMV), Mobala virus (MOBV) and Mopeia virus (MOPV).

In a particularly preferred embodiment, the arenavirus is the lymphocytic choriomeningitis virus, preferably a strain which is selected from the group comprising or consisting of WE, Armstrong, Clone 13 and Docile.

In a preferred embodiment, the arenavirus is a New World arenavirus, which is preferably selected from the group comprising or consisting of Allpaahuayo virus (ALLV), Amapari virus (AMAV), Bear Canyon virus (BCNV), Chapare virus, Cupixi virus (CPXV), Flexal virus (FLEV), Guanarito virus (GTOV), Junin virus (JUNV), Latino virus (LATV), Machupo virus (MACV), Oliveros virus (OLVV), Parana virus (PARV), Pichinde virus (PICV), Pirital virus (PIRV), Sabia virus (SABV), Tacaribe virus (TCRV), Tamiami virus (TAMV) and Whitewara Arroyo virus (WWAV).

In a particularly preferred embodiment, the arenavirus is a Junin virus, in particular the strain Candid #1 (Candid No.1).

In a further embodiment, the Junin virus, in particular the strain Candid #1 (Candid No.1), has a nucleic acid sequence, in particular an S-ribonucleic acid sequence or ambisense sequence, according to SEQ ID No. 1 (according to sequence listing).

In a further embodiment, the Junin virus, in particular the strain Candid #1 (Candid No.1), has a nucleic acid sequence, in particular an L-ribonucleic acid sequence or ambisense sequence, according to SEQ ID No. 2 (according to sequence listing).

In a further embodiment, the Junin virus, in particular the strain Candid #1 (Candid No.1), has a nucleic acid sequence, in particular an S-ribonucleic acid sequence or ambisense sequence, according to SEQ ID No. 3 (according to sequence listing).

In a further embodiment, the Junin virus, in particular the strain Candid #1 (Candid No.1), has a nucleic acid sequence, in particular an L-ribonucleic acid sequence or ambisense sequence, according to SEQ ID No. 4 (according to sequence listing).

In a further embodiment, the lymphocytic choriomeningitis virus (LCMV virus) mentioned above has a nucleic acid sequence, in particular an S-ribonucleic acid sequence or ambisense sequence, according to SEQ ID No. 5 (according to sequence listing).

In a further embodiment, the lymphocytic choriomeningitis virus (LCMV virus) mentioned above has a nucleic acid sequence, in particular an L-ribonucleic acid sequence or ambisense sequence, according to SEQ ID No. 6 (according to sequence listing).

In a further embodiment, the arenavirus is isolated from tumor lysates, organ lysates, urine or blood.

In an alternative embodiment, the arenavirus is isolated from a cell culture medium, in particular from a human tumor cell line.

In a further embodiment, the arenavirus is for administration in the form of virions, i.e. in the form of arenavirus particles, which are outside a cell.

In a further embodiment, the arenavirus is provided, preferably prepared, for local, in particular intramuscular, intraperitoneal or subcutaneous administration. In other words, the arenavirus is used in a further embodiment for local, in particular intramuscular, intraperitoneal or subcutaneous administration.

Preferably the arenavirus is provided, preferably prepared, for local administration at a dose of 1 PFU (Plaque Forming Unit)/kg body weight to 10^{12} PFU/kg body weight, particularly 10^2 PFU/kg body weight to 10^6 PFU/kg body weight, preferably 10^3 PFU/kg body weight to 10^5 PFU/kg body weight. In other words, the arenavirus is used, preferably for local administration, at a dose of 1 PFU (Plaque Forming Unit)/kg body weight to 10^{12} PFU/kg body weight, particularly 10^2 PFU/kg body weight to 10^6 PFU/kg body weight, preferably 10^3 PFU/kg body weight to 10^5 PFU/kg body weight.

In an alternative embodiment, the arenavirus is provided, preferably prepared, for systemic, in particular intravenous, administration. In other words, the arenavirus is used in an alternative embodiment for systemic, in particular intravenous, administration.

Preferably, the arenavirus is provided, preferably prepared, for systemic administration at a dose of 1 PFU/kg body weight to 10^{12} PFU/kg body weight, particularly 10^2 PFU/kg body weight to 10^6 PFU/kg body weight, preferably 10^3 PFU/kg body weight to 10^5 PFU/kg body weight. In other words, the arenavirus is used, preferably for systemic administration, at a dose of 1 PFU/kg body weight to 10^{12} PFU/kg body weight, particularly 10^2 PFU/kg body weight to 10^6 PFU/kg body weight, preferably 10^3 PFU/kg body weight to 10^5 PFU/kg body weight.

According to a second aspect, the invention relates to a medicament for use in the treatment and/or prevention of a tumor, in particular a malignant tumor.

The medicament is characterized in particular by the fact that it has an arenavirus according to the first aspect of the invention.

The medicament preferably further comprises a pharmaceutically acceptable carrier. The carrier may in particular be selected from the group comprising or consisting of water, saline solution, buffer solution and cell culture medium.

In a further embodiment, the medicament also comprises an active ingredient. The active ingredient can in particular be a cytostatic agent, an antibody and/or a cytokine.

With regard to further features and advantages of the medicament, in particular of the arenavirus and also the tumor, reference is made fully to the present description in order to avoid repetitions.

According to a third aspect, the invention relates to a method for producing an arenavirus with tumor-regressive, i.e. tumor-repelling/counteracting properties or improved tumor-regressive properties.

The tumor is preferably a malignant tumor, preferably a carcinoma, melanoma, blastoma, lymphoma or sarcoma. Accordingly, the method is preferably a method for producing an arenavirus with (improved) carcinoma-, melanoma-, blastoma-, lymphoma- or sarcoma-regressive properties.

The method comprises the following steps:

- infected dendritic cells or tumor cells with an arenavirus,
- culturing the arenavirus in the infected dendritic cells or infected tumor cells and

c) isolating the cultured arenavirus or a subset of the cultured arenavirus from the infected dendritic cells or infected tumor cells.

The sequence of steps a) to c) may also be referred to as a (single) passage of the arenavirus in the dendritic cells or tumor cells.

In a preferred embodiment, the arenavirus is an arenavirus which has been subjected to a serial passage in host animals prior to carrying out step a). For further details and advantages, reference may be made to the corresponding statements made in the context of the fourth aspect of the invention.

In a further embodiment, the dendritic cells or tumor cells are in the form of a cell culture (dendritic cell culture or tumor cell culture).

The tumor cells are preferably malignant tumor cells, in particular carcinoma, melanoma, blastoma, lymphoma or sarcoma cells.

The carcinoma cells may be selected from the group comprising or consisting of anal carcinoma cells, bronchial carcinoma cells, lung carcinoma cells, endometrial carcinoma cells, gallbladder carcinoma cells, hepatocellular carcinoma cells, testicular carcinoma cells, colorectal carcinoma cells, laryngeal carcinoma cells, oesophageal carcinoma cells, gastric carcinoma cells, breast carcinoma cells, renal carcinoma cells, ovarian carcinoma cells, pancreas tumor cells, pharyngeal carcinoma cells, prostate carcinoma cells, thyroid carcinoma cells and cervical carcinoma cells.

The sarcoma cells may be selected from the group comprising or consisting of angiosarcoma cells, chondrosarcoma cells, Ewing's sarcoma cells, fibrosarcoma cells, Kaposi's sarcoma cells, liposarcoma cells, leiomyosarcoma cells, malignant fibrous histiocytoma cells, neurogenic sarcoma cells, osteosarcoma cells and rhabdo-myosarcoma cells.

In a further embodiment, the tumor cells are immortalized immune cells, in particular immortalized macrophages.

The dendritic cells or tumor cells are infected according to step a) preferably by adding the arenavirus to the cells.

In a preferred embodiment, the sequence of steps a) to c) is repeated with new, in particular non-infected, dendritic cells, preferably of the same type, or with new, in particular non-infected, tumor cells, preferably of the same type (the same tumor type).

In a particularly preferred embodiment, the sequence of steps a) to c) is repeated many times. By way of preference, new, in particular non-infected, dendritic cells, preferably of the same type, or new, in particular non-infected, tumor cells, preferably of the same type, are used for each repetition.

The sequence of steps a) to c) is preferably repeated once to 1000 times, particularly 10 times to 100 times, preferably 30 times to 60 times, wherein new, in particular non-infected, dendritic cells, preferably of the same type, or new, in particular non-infected, tumor cells, preferably of the same type, are used for each repetition.

By multiple repetition of steps a) to c), the arenavirus is constantly forced to adapt to a new environment, i.e. to new dendritic cells or tumor cells. This permanent adaptation pressure favors the occurrence of mutations, which can produce or improve the tumor-regressive properties of the arenavirus.

During the culturing of the arenavirus according to step b), a replication of the arenavirus genome and a propagation of the arenavirus occur within the dendritic cells or tumor cells.

The arenavirus according to step b) is preferably cultured under standard cell culture conditions.

The arenavirus is preferably cultured in the dendritic cells or tumor cells for a period of 1 minute to 1 year, in particular 10 hours to 1 month, preferably 24 hours to 72 hours.

The cultured arenavirus according to step c) is preferably isolated from a cell culture supernatant.

Preferably, the dendritic cells or tumor cells are sorted according to specific properties, preferably by means of a cell sorting device, and subsequently cultured, prior to isolating according to step c). The sorted cells are preferably cultured over a period of 24 hours.

In a preferred embodiment, the method further comprises the following steps:

d) cloning the isolated arenavirus and

e) sequencing the isolated arenavirus.

With regard to further features and advantages of the method, in particular of the arenavirus and also the tumor, reference is made to the present description.

According to a fourth aspect, the invention relates to a method for producing an arenavirus with tumor-regressive, i.e. tumor-repelling/counteracting, properties or improved tumor-regressive properties.

The tumor is preferably a malignant tumor, preferably a carcinoma, melanoma, blastoma, lymphoma or sarcoma. Accordingly, the method is preferably a method for producing an arenavirus with (improved) carcinoma-, melanoma-, blastoma-, lymphoma- or sarcoma-regressive properties.

The method comprises the following steps:

a) infecting a host animal, which has a tumor, with an arenavirus,

b) culturing the arenavirus in the infected host animal and

c) isolating the cultured arenavirus or a subset of the cultured arenavirus from the infected host animal.

The sequence of steps a) to c) may also be referred to as a (single) passage of the arenavirus in the host animal.

In a preferred embodiment, tumor tissue is transplanted to the host animal prior to carrying out step a).

In an alternative embodiment, a genetically modified host animal is used which spontaneously develops the tumor.

Rodents in particular, preferably mice, can be used as host animals. For example, NOD SCID mice or LoxP-Tag mice can be used as host animals. In the Nod SCID mice, the SCID (Severe Combined Immunodeficiency) mutation is combined with a NOD (non-obese diabetic) type. In mice that are homozygous for the SCID mutation, no functional T-cells or B-cells are formed. By means of this immunodeficiency, these animals are exceptionally suitable for tolerating foreign body cells, for example transplanted tumors.

The host animal can be infected according to step a), for example by systemic, particularly intravenous, or local, for example subcutaneous, administration of the arenavirus.

In particular, the host animal can be infected by administration, preferably injection, of the arenavirus into the tumor of the host animal.

In a preferred embodiment, the sequence of steps a) to c) is repeated with a new, in particular non-infected, host animal, preferably of the same type.

In a particularly preferred embodiment, the sequence of steps a) to c) is repeated many times. A new, in particular non-infected host animal, preferably of the same type, is preferably used for each repetition. In other words, it is preferred according to the invention if the arenavirus is subjected to serial passage in host animals, preferably of the same type.

The sequence of steps a) to c) is preferably repeated once to 1000 times, particularly 10 times to 100 times, preferably

30 times to 60 times, wherein a new, in particular non-infected, host animal, preferably of the same type, is used for each repetition.

By multiple repetition of steps a) to c), the arenavirus is constantly forced to adapt to a new environment, i.e. to a new host animal. This permanent adaptation pressure favors the occurrence of mutations, which can produce or improve the tumor-regressive properties of the arenavirus.

During the culturing of the arenavirus according to step b), a replication of the arenavirus genome and a propagation of the arenavirus occur within the host animal.

The arenavirus is preferably cultured in the host animal for a period of 1 minute to 500 days, in particular 10 minutes to 100 days, preferably 1 hour to 30 days.

In a preferred embodiment, the arenavirus is isolated from urine, blood, the tumor, or organ lysates of the host animal.

In a further embodiment, the method further comprises the following steps:

- d) infecting dendritic cells or tumor cells with the arenavirus isolated according to step c),
- e) culturing the arenavirus in the infected dendritic cells or infected tumor cells and
- f) isolating the cultured arenavirus or a subset of the cultured arenavirus from the infected dendritic cells or infected tumor cells.

The sequence of steps d) to f) may also be referred to as a (single) passage of the arenavirus in the dendritic cells or tumor cells.

The dendritic cells or tumor cells are infected according to step d) preferably by adding the arenavirus to the cells.

In a preferred embodiment, the sequence of steps d) to f) is repeated with new, in particular non-infected, dendritic cells, preferably of the same type, or with new, in particular non-infected, tumor cells, preferably of the same type (the same tumor type).

In a particularly preferred embodiment, the sequence of steps d) to f) is repeated many times. By way of preference, new, in particular non-infected, dendritic cells, preferably of the same type, or new, in particular non-infected, tumor cells, preferably of the same type, are used for each repetition. In other words, it is particularly preferred according to the invention if the arenavirus is additionally subjected to a serial passage in dendritic cells, preferably of the same type, or tumor cells, preferably of the same type.

The sequence of steps d) to f) is preferably repeated once to 1000 times, particularly 10 times to 100 times, preferably 30 times to 60 times, wherein new, in particular non-infected, dendritic cells, preferably of the same type, or new, in particular non-infected, tumor cells, preferably of the same type, are preferably used for each repetition.

The combination of a serial passage of the arenavirus in host animals, preferably of the same type, with a serial passage of the arenavirus in dendritic cells, preferably of the same type, or tumor cells, preferably of the same type, is suitable, because of the additionally increased adaptation pressure or mutation pressure in a particular manner, for producing arenaviruses with (improved) tumor-regressive properties.

The arenavirus is preferably cultured in the dendritic cells or tumor cells for a period of 1 minute to 500 days, in particular 10 minutes to 100 days, preferably 1 hour to 30 days.

Preferably, the dendritic cells or tumor cells are sorted according to specific properties, preferably by means of a cell sorting device, and subsequently cultured, prior to isolating according to step f). The sorted cells are preferably cultured over a period of 24 hours.

In a preferred embodiment, the method further comprises the following steps:

- g) cloning the arenavirus isolated according to step f) and
- h) sequencing the isolated arenavirus.

Alternatively, it can be provided according to the invention that the arenavirus isolated according to step c) is cloned and subsequently sequenced.

With regard to other features and advantages of the method, in particular the arenavirus, the tumor and also the dendritic cells and tumor cells, the present description is also fully incorporated by way of reference in order to avoid unnecessary repetitions.

Further features and advantages of the invention will emerge from the following description of preferred embodiments in the form of working examples, the associated figures and the claims. The embodiments described below are merely for the purpose of illustration and for the better understanding of the invention and are in no way to be understood as limiting.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 shows LCMV (WE strain) replication results in immortalized macrophages ("tumour") and macrophages ("primary") generated from primary bone marrow.

FIG. 2 shows the effect of LCMV (WE strain) administration on MOPC (mouse plasmacytoma cell) tumor growth in C57BL/6 mice.

FIG. 3 shows the effect of LCMV (WE strain) administration on MC38 (mouse colon adenocarcinoma cell) tumor growth in C57BL/6 mice.

FIG. 4 shows the effect of LCMV (WE strain) administration on spontaneous liver carcinoma tumor growth in LoxP-Tag mice.

FIG. 5 shows the effect of LCMV (WE strain) on interferon- γ secretion in MOPC tumor-bearing C57BL/6 mice.

FIG. 6 shows the effect of LCMV (WE strain) administration on MOPC tumor growth in Map3k14^{aly/aly} mice compared to corresponding wild type mice.

FIG. 7 shows the effect of LCMV (WE strain) administration on MOPC tumor growth in Irf3 $\times$ Ir7^{-/-} mice compared to corresponding wild type mice.

FIG. 8A shows effect of LCMV (WE strain) administration on tumor microvessel density (MDV) and vessel-vessel spacing in MOPC tumor-bearing wild type mice.

FIG. 8B shows effect of LCMV (WE strain) administration on tumor hypoxic regions in MOPC tumor-bearing wild type mice.

FIG. 9 shows the effect of LCMV (WE strain) administration (ipsilateral, contralateral, and intravenous) on MOPC tumor growth in C57BL/6 mice.

FIG. 10 shows effect of LCMV (WE strain) administration on survival in A431 (squamous carcinoma) tumor-bearing NOD.SCID mice.

FIG. 11 shows effect of LCMV (WE strain) administration on tumor growth in A431 (colon adenocarcinoma) tumor-bearing NOD.SCID mice.

FIG. 12 shows effect of Candid #1 administration on tumor growth in HeLa cell-bearing NOD.SCID mice.

FIG. 13 shows effect of Candid #1 administration on tumor growth in HepG2 (hepatocellular carcinoma) tumor-bearing NOD.SCID mice.

FIG. 14 shows LCMV (MOI 1) replication results in primary hepatocytes, colon epithelial cells, and melanocytes and in tumour cells from the same tissue source.

FIG. 15A shows the effect of LCMV administration on metastatic MOPC tumor growth in C57BL/6 mice.

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FIG. 15B shows the effect of LCMV administration on survival in MOPC metastatic tumor-bearing C57BL/6 mice.

FIG. 16 shows the effect of LCMV administration on B16F10 (melanoma) tumor growth in C57BL/6 mice.

FIG. 17 shows the effect of LCMV administration on spontaneous melanoma tumor growth in MT/ret mice.

FIG. 18A shows the effect of LCMV administration on Sw872 (fibrosarcoma) tumor growth in NOD.SCID mice.

FIG. 18B shows the effect of LCMV administration on survival in Sw872 (fibrosarcoma) tumor-bearing NOD.SCID mice.

FIG. 19A shows the effect of LCMV administration on FaDu (pharyngeal carcinoma) tumor growth in NOD.SCID mice.

FIG. 19B shows the effect of LCMV administration on survival in FaDu (pharyngeal carcinoma) tumor-bearing NOD.SCID mice.

FIG. 20 shows the effect of LCMV on T-cell expression of PD-1, IL2 receptor, and IL7 receptor in B16F10 tumor-bearing mice.

FIG. 21A shows the effect of LCMV administration on EL4 (lymphoma) tumor growth in C57BL/6 mice.

FIG. 21B shows the effect of LCMV administration on survival in EL4 (lymphoma) tumor-bearing C57BL/6 mice.

FIG. 22 shows the effect of LCMV administration on survival in MOPC tumor-bearing C57BL/6 and Pdc1^{-/-} mice.

EXPERIMENTAL SECTION

1. Methods and Materials

1.1 Mice

Unless mentioned otherwise, the mice used were from a C57BL/6 background. Map3k14^{aly/aly} mice lack NF- κ B signals and are therefore highly immunosuppressed. Irf3 $\times$ Irf7^{-/-} mice cannot produce any interferon. NOD.SCID mice have no adaptive immune system. Therefore, it is possible to grow human tumors in these mice. LoxP-Tag mice spontaneously develop liver tumors.

1.2 Cell Lines and Reagents

MOPC cells are murine oropharynx carcinoma cells. Mc38 are murine colon carcinoma cells. Raw cells are immortalized macrophages. A431 are human lung carcinoma cells; Sw40 are human colon carcinoma cells, Hela are human cervical carcinoma cells. Primary macrophages were cultured from bone marrow precursor cells by means of M-CSF. Cells were maintained in Dulbecco's modified Eagle's medium with 10% fetal bovine serum (Sigma-Aldrich), 2 mmol/l L-glutamine and 100 U/ml penicillin. All cells were cultured in 5% CO₂.

1.3 Viruses

The LCMV strain WE was obtained from the laboratory of Prof. Zinkernagel (Experimental Immunology, Zurich, Switzerland) and was propagated in L929 cells. Candid #1 was obtained from Professor Paula Cannon (University of Southern California).

1.4 Tumor Growth and Treatments

Approximately 5 $\times$ 10⁵ tumor cells (in 100 μ L) were injected subcutaneously into the right flank of 6 to 12 week old mice. The longest tumor diameter was measured by. Mice were treated by peritumoral injections of 2 $\times$ 10⁴ PFU LCMV-WE or Candid #1 (in 100-200 μ L).

1.5 Morphometric Analysis of Tumor Vessels

Morphometric analyses were performed with successive frozen sections, in which the endothelial cell marker CD31 was stained. Quantification of the microvessel density

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(MVD) was calculated using the mean of three tumor sections. MVD was calculated as the number of vessels per tumor area.

1.6 Detection of Hypoxia

Hypoxic tumor regions were detected by the formation of pimonidazole adducts after injection of pimonidazole into tumor-transplanted animals for 30 min. The tumor sections were stained using the Hypoxyprobe-1 Plus kit according to the manufacturer's instructions (Pharmacia Natur International, Inc.).

1.7 IFN- α ELISA

Serum IFN- α levels were determined by ELISA according to the manufacturer's data (Research Diagnostics RDI, Flanders, NJ).

1.8 Statistical Analysis

The mean values were compared using an unpaired two-sided student t-test. The data are shown as mean $\pm$ SEM. The level of statistical significance was set at p<0.05.

2. Investigations

2.1 Immortalized Macrophages (Tumour Cells) and Macrophages (Primary) generated from primary bone marrow were infected with LCMV (WE strain). Replication was measured after 24 hours (n=3).

It could be shown that LCMV (WE strain) replicates both in immortalized and healthy cells. The results obtained are shown graphically in FIG. 1.

FIG. 1 has the following legend:

Ordinate: LCMV (log₁₀ PFU/ml)

Abscissa: Tumor cells/healthy macrophages (primary)

2.2 WT C57BL/6 mice were treated with 5 $\times$ 10⁵ MOPC cells (day 3). One group of mice was additionally treated peritumorally with 2 $\times$ 10⁴ PFU LCMV (WE strain) (day 0). Tumor growth was observed.

It could be shown that the treatment with LCMV caused almost complete tumor regression. The results obtained are shown graphically in FIG. 2.

FIG. 2 has the following legend:

Ordinate: Tumor diameter (cm)

Abscissa: Time (days)

2.3 WT C57BL/6 mice were treated with 5 $\times$ 10⁵ MC38 cells (day 3). One group of mice was additionally treated peritumorally with 2 $\times$ 10⁴ PFU LCMV (WE strain) (day 0). Tumor growth was observed.

It could be shown that the treatment with LCMV caused a significant tumor regression. The results obtained are shown graphically in FIG. 3.

FIG. 3 has the following legend:

Ordinate: Tumor diameter (cm)

Abscissa: Time (days)

2.4 About nine month old LoxP-Tag mice with spontaneously developed liver carcinomas were infected intravenously with 2 $\times$ 10⁶ PFU LCMV or left untreated. The tumor nodes (diameters $\geq$ 3 mm) were quantified macroscopically on day 6 (n=3) and day 20 (n=4-5).

It could be shown that the treatment with LCMV significantly reduced the number of carcinomatous liver nodes. The results obtained are shown graphically in FIG. 4.

FIG. 4 has the following legend:

Ordinate: Liver nodes (number)

Abscissa: Time

2.5 WT C57BL/6 mice (n=4/group) were injected subcutaneously with 5 $\times$ 10⁵ MOPC cells (day -3) or LCMV (WE strain) 2 $\times$ 10⁴ PFU (day 0) or both 5 $\times$ 10⁵ MOPC cells (day -3) and 2 $\times$ 10⁴ PFU LCMV (day 0). Serum samples were collected on day 3 and an IFN- α -ELISA was performed.

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It could be shown that the LCMV caused a drastically increased secretion of interferon- γ in experimental animals which were simultaneously administered carcinoma cells. The results obtained are shown graphically in FIG. 5.

FIG. 5 has the following legend:

Ordinate: IFN- α (pg/ml)

2.6 Map3k14^{aly/aly} mice and WT mice were treated with 5×10^5 MOPC cells (day 3). One group of mice was additionally treated peritumorally with 2×10^4 PFU LCMV (WE strain) (day 0). Tumor growth was observed.

It could be shown that the treatment with LCMV caused tumor regression. The results obtained are shown graphically in FIG. 6.

FIG. 6 has the following legend:

Ordinate: Tumor diameter (cm)

Abscissa: Time (days)

2.7 Irf3 $\times$ Irf7^{-/-} mice and WT mice were treated with 5×10^5 MOPC cells (day 3). One group of mice was additionally treated peritumorally with 2×10^4 PFU LCMV (WE strain) (day 0). Tumor growth was observed.

It could be shown that the treatment with LCMV caused tumor regression. The results obtained are shown graphically in FIG. 7.

FIG. 7 has the following legend:

Ordinate: Tumor diameter (cm)

Abscissa: Time (days)

2.8 WT mice were treated with 5×10^5 MOPC cells (day 3). One group of mice was additionally treated peritumorally with 2×10^4 PFU LCMV (WE strain) (day 0). On day 9 after the tumor graft, the tumors were analyzed histologically with CD31 staining. The microvessel density (MDV) and the vessel-vessel spacing were quantified.

It could be shown that the treatment with LCMV caused a decrease in tumor vessel density. The results obtained are shown graphically in FIG. 8A.

FIG. 8A has the following legends:

Left side: Right side:

Ordinate: MVD [mm^3]

Ordinate: Vessel-vessel spacing (μm)

Abscissa: Tumor/Tumor LCMV Abscissa: Tumor/Tumor LCMV

2.9 WT mice were treated with 5×10^5 MOPC cells (day 3). One group of mice was additionally treated peritumorally with 2×10^4 PFU LCMV (WE strain) (day 0). On day 9, the animals were injected with pimonidazole, and the tumors were then analyzed histologically for hypoxic regions.

It could be shown that the treatment with LCMV caused oxygen deficiency in the carcinoma tissue. The results obtained are shown graphically in FIG. 8B.

FIG. 8B has the following legend:

Ordinate: Hypoxic regions/tumor (%)

Abscissa: Tumor/Tumor LCMV

2.10 WT C57BL/6 mice were injected subcutaneously with 5×10^5 MOPC cells in the right flank (day 3). On day 0, a group of animals were treated with 2×10^4 PFU LCMV (WE strain) in the right flank (ipsilateral), left flank (contralateral) or intravenously. Tumor growth was observed.

It could be shown that the treatment with LCMV caused tumor regression even with systemic administration. The results obtained are shown graphically in FIG. 9.

FIG. 9 has the following legend:

Ordinate: Tumor diameter (cm)

Abscissa: Time (days)

2.11 NOD.SCID mice were injected subcutaneously with 5×10^5 A431 cells (day -3) and then either left untreated or treated with 2×10^4 PFU LCMV (WE strain). The tumor size

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(longest diameter) was measured on the specified day. The mice were sacrificed when the tumor size reached 12 mm.

It could be shown that the treatment with LCMV increased the survival rate in the experimental animals. The results obtained are shown graphically in FIG. 10.

FIG. 10 has the following legend:

Ordinate: Survival (%)

Abscissa: Time (days)

2.12 NOD.SCID mice were treated with 5×10^5 Sw40 cells (day 0). A group of mice was additionally treated peritumorally with 2×10^4 PFU LCMV (WE strain) or 2×10^4 PFU Candid #1 (day 0). Tumor growth was observed.

It could be shown that the treatment with LCMV and Candid #1 caused tumor regression. The results obtained are shown graphically in FIG. 11.

FIG. 11 has the following legend:

Ordinate: Tumor diameter (cm)

Abscissa: Time (days)

2.13 NOD.SCID mice were treated with 5×10^5 Hela cells (day 0). A group of mice was additionally treated peritumorally with 2×10^4 PFU LCMV (WE strain) (day 3). Tumor growth was observed.

It could be shown that the treatment with LCMV caused tumor regression. The results obtained are shown graphically in FIG. 12.

FIG. 12 has the following legend:

Ordinate: Tumor diameter (cm)

Abscissa: Time (days)

2.14 NOD.SCID mice were treated with 5×10^5 HepG2 cells (day 10) and then additionally treated peritumorally with or without 2×10^4 PFU Candid #1 (day 0). Tumor growth was observed.

It could be shown that the treatment with Candid #1 caused tumor regression with this tumor type. The results obtained are shown graphically in FIG. 13.

FIG. 13 has the following legend:

Ordinate: Tumor diameter (cm)

Abscissa: Time (days)

2.15 Primary human cells (hepatocytes, colon epithelial cells, melanocytes) and tumour cells from the same tissue source were infected with LCMV (MOI 1). The amount of virus was measured in the supernatant after 1, 2 and 3 days.

In this experiment it was shown that arenaviruses are replicated in tumor cells in comparison to healthy tissue.

FIG. 14 has the following legend:

Ordinate: Infectious virus in cell culture supernatant (logarithmic plaque forming units)

Abscissa: Time (days)

2.16 Tumor diameter (A) and survival (B) of C57BL/6 mice bearing a metastasis in the shoulder and a metastasis in the flank (MOPC cells), which were left untreated or had been treated intravenously with 2×10^6 PFU LCMV.

It could be shown in this experiment that intravenous therapy of LCMV acts very efficiently on two local metastases and thus prolongs survival.

FIG. 15A has the following legend:

Ordinate: Tumor diameter of both metastases (cm)

Abscissa: Time (days)

FIG. 15B has the following legend:

Ordinate: Survival in percent

Abscissa: Time (days)

2.17 Tumor diameter of C57BL/6 mice bearing a melanoma (B16F10 cells) which were left untreated or were treated intratumorally with 2×10^4 PFU LCMV.

It could be shown in this experiment that local therapy with LCMV is very efficient in melanoma.

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FIG. 16 has the following legend:

Ordinate: Tumor diameter of the melanoma (cm)

Abscissa: Time (days)

2.18 Number of melanomas of MT/ret mice (develop endogenous melanomas), which were left untreated or were treated intravenously with 2×10^6 PFU LCMV.

It could be shown in this experiment that systemic therapy with LCMV is very efficient in melanoma.

FIG. 17 has the following legend:

Ordinate: Number of melanomas

2.19 Tumor diameter (A) and survival (B) of NOD.SCID mice bearing a human fibrosarcoma (Sw872 cells), which were left untreated or were treated intratumorally with 2×10^6 PFU Candid #1.

It could be shown in this experiment that Candid #1 acts very efficiently also in the case of fibrosarcoma and thus prolongs survival.

FIG. 18A has the following legend:

Ordinate: Tumor diameter (cm)

Abscissa: Time (days)

FIG. 18B has the following legend:

Ordinate: Survival in percent

Abscissa: Time (days)

2.20 Tumor diameter (A) and survival (B) of NOD.SCID mice bearing a human pharyngeal carcinoma (FaDu cells), which were left untreated or were treated intratumorally with 2×10^6 PFU LCMV.

It could be shown in this experiment that LCMV acts very efficiently also in the case of pharyngeal carcinoma and thus prolongs survival.

FIG. 19A has the following legend:

Ordinate: Tumor diameter (cm)

Abscissa: Time (days)

FIG. 19B has the following legend:

Ordinate: Survival in percent

Abscissa: Time (days)

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2.21 Expression of receptors on tumor-specific T cells (PD-1, IL2 receptor, IL7 receptor), which influence the function of T cells. Tumor-specific T cells are derived from the blood of mice with B16F10 tumors, which were additionally treated intratumorally with or without LCMV.

It could be shown in this experiment that LCMV positively influences the tumor-specific T cells.

FIG. 20 has the following legend:

Ordinate: Potency of the expression of the different receptors (mean fluorescence intensity)

2.22 Tumor diameter (A) and survival (B) of C57BL/6 mice bearing a murine subcutaneous lymphoma (EL4 cells) which were treated with or without tumor-specific T cells (OT1 cells) and additionally intratumorally with or without LCMV (2×10^6 PFU).

It could be shown in this experiment that LCMV acts synergistically with T cell therapy.

FIG. 21A has the following legend:

Ordinate: Tumor diameter (cm)

Abscissa: Time (days)

FIG. 21B has the following legend:

Ordinate: Survival in percent

Abscissa: Time (days)

2.23 Survival of C57BL/6 mice and PD-1 deficient mice (Pdc1^{-/-} mice) bearing a murine pharyngeal carcinoma (MOPC cells) and which were treated intratumorally with LCMV (2×10^4 PFU).

It could be shown in this experiment that LCMV acts synergistically with a PD-1 blockade.

FIG. 22 has the following legend:

Ordinate: Survival in percent

Abscissa: Time (days)

The nucleic acid sequences SEQ ID No. 1 to SEQ ID No. 6 mentioned in the general description correspond to the nucleic acid sequences disclosed in the following sequence listing.

SEQUENCE LISTING

```
Sequence total quantity: 6
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modified_base    15
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modified_base    18
                  mod_base = OTHER
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modified_base    25..28
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modified_base    37
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modified_base    39
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modified_base    42..43
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modified_base    45
                  mod_base = OTHER
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-continued

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modified_base	note = thymine 102..103 mod_base = OTHER
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modified_base	note = thymine 180 mod_base = OTHER
modified_base	note = thymine 183..184 mod_base = OTHER
	note = thymine

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modified_base	192
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	mod_base = OTHER
	note = thymine
modified_base	308
	mod_base = OTHER
	note = thymine
modified_base	311..312
	mod_base = OTHER
	note = thymine
modified_base	314
	mod_base = OTHER
	note = thymine
modified_base	318

-continued

	mod_base = OTHER
	note = thymine
modified_base	321
	mod_base = OTHER
	note = thymine
modified_base	325
	mod_base = OTHER
	note = thymine
modified_base	327
	mod_base = OTHER
	note = thymine
modified_base	329..332
	mod_base = OTHER
	note = thymine
modified_base	340
	mod_base = OTHER
	note = thymine
modified_base	346
	mod_base = OTHER
	note = thymine
modified_base	351
	mod_base = OTHER
	note = thymine
modified_base	355..357
	mod_base = OTHER
	note = thymine
modified_base	359..360
	mod_base = OTHER
	note = thymine
modified_base	362
	mod_base = OTHER
	note = thymine
modified_base	364
	mod_base = OTHER
	note = thymine
modified_base	368..369
	mod_base = OTHER
	note = thymine
modified_base	382
	mod_base = OTHER
	note = thymine
modified_base	384..386
	mod_base = OTHER
	note = thymine
modified_base	390..391
	mod_base = OTHER
	note = thymine
modified_base	403
	mod_base = OTHER
	note = thymine
modified_base	406..407
	mod_base = OTHER
	note = thymine
modified_base	410..412
	mod_base = OTHER
	note = thymine
modified_base	417
	mod_base = OTHER
	note = thymine
modified_base	422..424
	mod_base = OTHER
	note = thymine
modified_base	427
	mod_base = OTHER
	note = thymine
modified_base	430
	mod_base = OTHER
	note = thymine
modified_base	432..433
	mod_base = OTHER
	note = thymine
modified_base	438
	mod_base = OTHER
	note = thymine
modified_base	440..441
	mod_base = OTHER
	note = thymine
modified_base	443..444
	mod_base = OTHER

-continued

modified_base	note = thymine 452 mod_base = OTHER note = thymine
modified_base	454 mod_base = OTHER note = thymine
modified_base	457 mod_base = OTHER note = thymine
modified_base	459..460 mod_base = OTHER note = thymine
modified_base	462 mod_base = OTHER note = thymine
modified_base	465 mod_base = OTHER note = thymine
modified_base	472 mod_base = OTHER note = thymine
modified_base	481 mod_base = OTHER note = thymine
modified_base	483 mod_base = OTHER note = thymine
modified_base	488 mod_base = OTHER note = thymine
modified_base	491 mod_base = OTHER note = thymine
modified_base	493..494 mod_base = OTHER note = thymine
modified_base	502 mod_base = OTHER note = thymine
modified_base	505 mod_base = OTHER note = thymine
modified_base	508 mod_base = OTHER note = thymine
modified_base	513..515 mod_base = OTHER note = thymine
modified_base	518..519 mod_base = OTHER note = thymine
modified_base	521 mod_base = OTHER note = thymine
modified_base	523 mod_base = OTHER note = thymine
modified_base	527 mod_base = OTHER note = thymine
modified_base	530..531 mod_base = OTHER note = thymine
modified_base	534 mod_base = OTHER note = thymine
modified_base	538 mod_base = OTHER note = thymine
modified_base	541 mod_base = OTHER note = thymine
modified_base	543 mod_base = OTHER note = thymine
modified_base	550 mod_base = OTHER note = thymine

-continued

modified_base	553..554
	mod_base = OTHER
	note = thymine
modified_base	559
	mod_base = OTHER
	note = thymine
modified_base	561
	mod_base = OTHER
	note = thymine
modified_base	572..574
	mod_base = OTHER
	note = thymine
modified_base	576
	mod_base = OTHER
	note = thymine
modified_base	578
	mod_base = OTHER
	note = thymine
modified_base	580
	mod_base = OTHER
	note = thymine
modified_base	589
	mod_base = OTHER
	note = thymine
modified_base	605..606
	mod_base = OTHER
	note = thymine
modified_base	609
	mod_base = OTHER
	note = thymine
modified_base	611..613
	mod_base = OTHER
	note = thymine
modified_base	618
	mod_base = OTHER
	note = thymine
modified_base	626
	mod_base = OTHER
	note = thymine
modified_base	634
	mod_base = OTHER
	note = thymine
modified_base	637
	mod_base = OTHER
	note = thymine
modified_base	639
	mod_base = OTHER
	note = thymine
modified_base	643
	mod_base = OTHER
	note = thymine
modified_base	649..650
	mod_base = OTHER
	note = thymine
modified_base	652
	mod_base = OTHER
	note = thymine
modified_base	655
	mod_base = OTHER
	note = thymine
modified_base	662..664
	mod_base = OTHER
	note = thymine
modified_base	670
	mod_base = OTHER
	note = thymine
modified_base	675
	mod_base = OTHER
	note = thymine
modified_base	679
	mod_base = OTHER
	note = thymine
modified_base	682..684
	mod_base = OTHER
	note = thymine
modified_base	686
	mod_base = OTHER
	note = thymine
modified_base	688

-continued

	mod_base = OTHER
	note = thymine
modified_base	695
	mod_base = OTHER
	note = thymine
modified_base	697
	mod_base = OTHER
	note = thymine
modified_base	700
	mod_base = OTHER
	note = thymine
modified_base	706..707
	mod_base = OTHER
	note = thymine
modified_base	710..711
	mod_base = OTHER
	note = thymine
modified_base	715
	mod_base = OTHER
	note = thymine
modified_base	723
	mod_base = OTHER
	note = thymine
modified_base	725
	mod_base = OTHER
	note = thymine
modified_base	728..729
	mod_base = OTHER
	note = thymine
modified_base	732
	mod_base = OTHER
	note = thymine
modified_base	745
	mod_base = OTHER
	note = thymine
modified_base	751..752
	mod_base = OTHER
	note = thymine
modified_base	757
	mod_base = OTHER
	note = thymine
modified_base	759
	mod_base = OTHER
	note = thymine
modified_base	764
	mod_base = OTHER
	note = thymine
modified_base	766
	mod_base = OTHER
	note = thymine
modified_base	771
	mod_base = OTHER
	note = thymine
modified_base	780..781
	mod_base = OTHER
	note = thymine
modified_base	788..789
	mod_base = OTHER
	note = thymine
modified_base	794..795
	mod_base = OTHER
	note = thymine
modified_base	798..799
	mod_base = OTHER
	note = thymine
modified_base	808
	mod_base = OTHER
	note = thymine
modified_base	816..817
	mod_base = OTHER
	note = thymine
modified_base	822..823
	mod_base = OTHER
	note = thymine
modified_base	833
	mod_base = OTHER
	note = thymine
modified_base	836..837
	mod_base = OTHER

-continued

modified_base	note = thymine 845..846 mod_base = OTHER
modified_base	note = thymine 848..849 mod_base = OTHER
modified_base	note = thymine 851 mod_base = OTHER
modified_base	note = thymine 854 mod_base = OTHER
modified_base	note = thymine 857 mod_base = OTHER
modified_base	note = thymine 859..861 mod_base = OTHER
modified_base	note = thymine 869 mod_base = OTHER
modified_base	note = thymine 872 mod_base = OTHER
modified_base	note = thymine 883 mod_base = OTHER
modified_base	note = thymine 889 mod_base = OTHER
modified_base	note = thymine 896 mod_base = OTHER
modified_base	note = thymine 898..899 mod_base = OTHER
modified_base	note = thymine 901 mod_base = OTHER
modified_base	note = thymine 903 mod_base = OTHER
modified_base	note = thymine 911 mod_base = OTHER
modified_base	note = thymine 915 mod_base = OTHER
modified_base	note = thymine 918 mod_base = OTHER
modified_base	note = thymine 921 mod_base = OTHER
modified_base	note = thymine 933 mod_base = OTHER
modified_base	note = thymine 938 mod_base = OTHER
modified_base	note = thymine 940..943 mod_base = OTHER
modified_base	note = thymine 949 mod_base = OTHER
modified_base	note = thymine 952 mod_base = OTHER
modified_base	note = thymine 955 mod_base = OTHER
modified_base	note = thymine 957 mod_base = OTHER
modified_base	note = thymine 965 mod_base = OTHER
	note = thymine

-continued

modified_base	970..972
	mod_base = OTHER
	note = thymine
modified_base	976
	mod_base = OTHER
	note = thymine
modified_base	979
	mod_base = OTHER
	note = thymine
modified_base	983
	mod_base = OTHER
	note = thymine
modified_base	985
	mod_base = OTHER
	note = thymine
modified_base	989..990
	mod_base = OTHER
	note = thymine
modified_base	992
	mod_base = OTHER
	note = thymine
modified_base	994
	mod_base = OTHER
	note = thymine
modified_base	999
	mod_base = OTHER
	note = thymine
modified_base	1001..1002
	mod_base = OTHER
	note = thymine
modified_base	1008
	mod_base = OTHER
	note = thymine
modified_base	1010..1012
	mod_base = OTHER
	note = thymine
modified_base	1015..1016
	mod_base = OTHER
	note = thymine
modified_base	1027
	mod_base = OTHER
	note = thymine
modified_base	1030
	mod_base = OTHER
	note = thymine
modified_base	1032
	mod_base = OTHER
	note = thymine
modified_base	1041
	mod_base = OTHER
	note = thymine
modified_base	1045
	mod_base = OTHER
	note = thymine
modified_base	1048
	mod_base = OTHER
	note = thymine
modified_base	1054
	mod_base = OTHER
	note = thymine
modified_base	1065
	mod_base = OTHER
	note = thymine
modified_base	1069
	mod_base = OTHER
	note = thymine
modified_base	1071
	mod_base = OTHER
	note = thymine
modified_base	1074
	mod_base = OTHER
	note = thymine
modified_base	1086
	mod_base = OTHER
	note = thymine
modified_base	1090
	mod_base = OTHER
	note = thymine
modified_base	1095

-continued

	mod_base = OTHER
	note = thymine
modified_base	1098
	mod_base = OTHER
	note = thymine
modified_base	1100
	mod_base = OTHER
	note = thymine
modified_base	1102
	mod_base = OTHER
	note = thymine
modified_base	1108..1110
	mod_base = OTHER
	note = thymine
modified_base	1112..1113
	mod_base = OTHER
	note = thymine
modified_base	1116
	mod_base = OTHER
	note = thymine
modified_base	1128..1129
	mod_base = OTHER
	note = thymine
modified_base	1137
	mod_base = OTHER
	note = thymine
modified_base	1140
	mod_base = OTHER
	note = thymine
modified_base	1144
	mod_base = OTHER
	note = thymine
modified_base	1146
	mod_base = OTHER
	note = thymine
modified_base	1150..1151
	mod_base = OTHER
	note = thymine
modified_base	1154
	mod_base = OTHER
	note = thymine
modified_base	1159..1160
	mod_base = OTHER
	note = thymine
modified_base	1169..1172
	mod_base = OTHER
	note = thymine
modified_base	1175
	mod_base = OTHER
	note = thymine
modified_base	1177
	mod_base = OTHER
	note = thymine
modified_base	1179
	mod_base = OTHER
	note = thymine
modified_base	1191..1193
	mod_base = OTHER
	note = thymine
modified_base	1205
	mod_base = OTHER
	note = thymine
modified_base	1208..1209
	mod_base = OTHER
	note = thymine
modified_base	1217
	mod_base = OTHER
	note = thymine
modified_base	1220
	mod_base = OTHER
	note = thymine
modified_base	1223..1224
	mod_base = OTHER
	note = thymine
modified_base	1227
	mod_base = OTHER
	note = thymine
modified_base	1241
	mod_base = OTHER

-continued

modified_base	note = thymine 1243..1245 mod_base = OTHER note = thymine
modified_base	1251 mod_base = OTHER note = thymine
modified_base	1253 mod_base = OTHER note = thymine
modified_base	1255 mod_base = OTHER note = thymine
modified_base	1259..1260 mod_base = OTHER note = thymine
modified_base	1264 mod_base = OTHER note = thymine
modified_base	1267 mod_base = OTHER note = thymine
modified_base	1271 mod_base = OTHER note = thymine
modified_base	1275 mod_base = OTHER note = thymine
modified_base	1277..1278 mod_base = OTHER note = thymine
modified_base	1285 mod_base = OTHER note = thymine
modified_base	1289..1290 mod_base = OTHER note = thymine
modified_base	1292..1293 mod_base = OTHER note = thymine
modified_base	1296..1298 mod_base = OTHER note = thymine
modified_base	1300 mod_base = OTHER note = thymine
modified_base	1305 mod_base = OTHER note = thymine
modified_base	1308 mod_base = OTHER note = thymine
modified_base	1319 mod_base = OTHER note = thymine
modified_base	1321..1322 mod_base = OTHER note = thymine
modified_base	1336 mod_base = OTHER note = thymine
modified_base	1342 mod_base = OTHER note = thymine
modified_base	1345..1347 mod_base = OTHER note = thymine
modified_base	1351..1353 mod_base = OTHER note = thymine
modified_base	1356..1357 mod_base = OTHER note = thymine
modified_base	1362 mod_base = OTHER note = thymine
modified_base	1364 mod_base = OTHER note = thymine

-continued

modified_base	1366 mod_base = OTHER note = thymine
modified_base	1368..1370 mod_base = OTHER note = thymine
modified_base	1380 mod_base = OTHER note = thymine
modified_base	1382..1383 mod_base = OTHER note = thymine
modified_base	1385..1386 mod_base = OTHER note = thymine
modified_base	1394 mod_base = OTHER note = thymine
modified_base	1398 mod_base = OTHER note = thymine
modified_base	1400..1401 mod_base = OTHER note = thymine
modified_base	1404..1405 mod_base = OTHER note = thymine
modified_base	1409..1410 mod_base = OTHER note = thymine
modified_base	1413 mod_base = OTHER note = thymine
modified_base	1417 mod_base = OTHER note = thymine
modified_base	1419 mod_base = OTHER note = thymine
modified_base	1424 mod_base = OTHER note = thymine
modified_base	1437 mod_base = OTHER note = thymine
modified_base	1451 mod_base = OTHER note = thymine
modified_base	1456..1458 mod_base = OTHER note = thymine
modified_base	1469..1470 mod_base = OTHER note = thymine
modified_base	1478..1479 mod_base = OTHER note = thymine
modified_base	1483 mod_base = OTHER note = thymine
modified_base	1486..1487 mod_base = OTHER note = thymine
modified_base	1493 mod_base = OTHER note = thymine
modified_base	1495 mod_base = OTHER note = thymine
modified_base	1498 mod_base = OTHER note = thymine
modified_base	1502 mod_base = OTHER note = thymine
modified_base	1510 mod_base = OTHER note = thymine
modified_base	1512 mod_base = OTHER note = thymine

-continued

	mod_base = OTHER
	note = thymine
modified_base	1527..1529
	mod_base = OTHER
	note = thymine
modified_base	1534
	mod_base = OTHER
	note = thymine
modified_base	1544
	mod_base = OTHER
	note = thymine
modified_base	1551
	mod_base = OTHER
	note = thymine
modified_base	1554
	mod_base = OTHER
	note = thymine
modified_base	1560
	mod_base = OTHER
	note = thymine
modified_base	1579
	mod_base = OTHER
	note = thymine
modified_base	1588
	mod_base = OTHER
	note = thymine
modified_base	1591
	mod_base = OTHER
	note = thymine
modified_base	1593
	mod_base = OTHER
	note = thymine
modified_base	1604
	mod_base = OTHER
	note = thymine
modified_base	1613
	mod_base = OTHER
	note = thymine
modified_base	1624
	mod_base = OTHER
	note = thymine
modified_base	1635
	mod_base = OTHER
	note = thymine
modified_base	1638..1639
	mod_base = OTHER
	note = thymine
modified_base	1644
	mod_base = OTHER
	note = thymine
modified_base	1648
	mod_base = OTHER
	note = thymine
modified_base	1653
	mod_base = OTHER
	note = thymine
modified_base	1657..1658
	mod_base = OTHER
	note = thymine
modified_base	1676
	mod_base = OTHER
	note = thymine
modified_base	1680
	mod_base = OTHER
	note = thymine
modified_base	1683
	mod_base = OTHER
	note = thymine
modified_base	1686
	mod_base = OTHER
	note = thymine
modified_base	1692
	mod_base = OTHER
	note = thymine
modified_base	1694
	mod_base = OTHER
	note = thymine
modified_base	1698..1699
	mod_base = OTHER

-continued

modified_base	note = thymine 1702 mod_base = OTHER note = thymine
modified_base	1705 mod_base = OTHER note = thymine
modified_base	1708 mod_base = OTHER note = thymine
modified_base	1714..1715 mod_base = OTHER note = thymine
modified_base	1720 mod_base = OTHER note = thymine
modified_base	1722 mod_base = OTHER note = thymine
modified_base	1728 mod_base = OTHER note = thymine
modified_base	1732 mod_base = OTHER note = thymine
modified_base	1739..1740 mod_base = OTHER note = thymine
modified_base	1742 mod_base = OTHER note = thymine
modified_base	1747 mod_base = OTHER note = thymine
modified_base	1749 mod_base = OTHER note = thymine
modified_base	1762 mod_base = OTHER note = thymine
modified_base	1764 mod_base = OTHER note = thymine
modified_base	1767..1768 mod_base = OTHER note = thymine
modified_base	1772 mod_base = OTHER note = thymine
modified_base	1783..1784 mod_base = OTHER note = thymine
modified_base	1793 mod_base = OTHER note = thymine
modified_base	1795 mod_base = OTHER note = thymine
modified_base	1798..1801 mod_base = OTHER note = thymine
modified_base	1805 mod_base = OTHER note = thymine
modified_base	1807..1811 mod_base = OTHER note = thymine
modified_base	1813..1814 mod_base = OTHER note = thymine
modified_base	1816..1817 mod_base = OTHER note = thymine
modified_base	1827 mod_base = OTHER note = thymine
modified_base	1831..1832 mod_base = OTHER note = thymine

-continued

modified_base	1834 mod_base = OTHER note = thymine
modified_base	1837 mod_base = OTHER note = thymine
modified_base	1839..1841 mod_base = OTHER note = thymine
modified_base	1845 mod_base = OTHER note = thymine
modified_base	1849 mod_base = OTHER note = thymine
modified_base	1855 mod_base = OTHER note = thymine
modified_base	1857..1859 mod_base = OTHER note = thymine
modified_base	1861 mod_base = OTHER note = thymine
modified_base	1870 mod_base = OTHER note = thymine
modified_base	1872..1873 mod_base = OTHER note = thymine
modified_base	1876 mod_base = OTHER note = thymine
modified_base	1882 mod_base = OTHER note = thymine
modified_base	1886 mod_base = OTHER note = thymine
modified_base	1890..1891 mod_base = OTHER note = thymine
modified_base	1893..1894 mod_base = OTHER note = thymine
modified_base	1899 mod_base = OTHER note = thymine
modified_base	1901 mod_base = OTHER note = thymine
modified_base	1906..1907 mod_base = OTHER note = thymine
modified_base	1912 mod_base = OTHER note = thymine
modified_base	1920..1922 mod_base = OTHER note = thymine
modified_base	1927 mod_base = OTHER note = thymine
modified_base	1929 mod_base = OTHER note = thymine
modified_base	1931..1932 mod_base = OTHER note = thymine
modified_base	1934..1935 mod_base = OTHER note = thymine
modified_base	1939 mod_base = OTHER note = thymine
modified_base	1945 mod_base = OTHER note = thymine
modified_base	1954..1955

-continued

	mod_base = OTHER
	note = thymine
modified_base	1958
	mod_base = OTHER
	note = thymine
modified_base	1961
	mod_base = OTHER
	note = thymine
modified_base	1963
	mod_base = OTHER
	note = thymine
modified_base	1966
	mod_base = OTHER
	note = thymine
modified_base	1971
	mod_base = OTHER
	note = thymine
modified_base	1974..1975
	mod_base = OTHER
	note = thymine
modified_base	1979
	mod_base = OTHER
	note = thymine
modified_base	1982
	mod_base = OTHER
	note = thymine
modified_base	1991
	mod_base = OTHER
	note = thymine
modified_base	1993
	mod_base = OTHER
	note = thymine
modified_base	1995
	mod_base = OTHER
	note = thymine
modified_base	2012
	mod_base = OTHER
	note = thymine
modified_base	2017
	mod_base = OTHER
	note = thymine
modified_base	2021
	mod_base = OTHER
	note = thymine
modified_base	2028
	mod_base = OTHER
	note = thymine
modified_base	2031
	mod_base = OTHER
	note = thymine
modified_base	2036..2037
	mod_base = OTHER
	note = thymine
modified_base	2041
	mod_base = OTHER
	note = thymine
modified_base	2043..2044
	mod_base = OTHER
	note = thymine
modified_base	2046
	mod_base = OTHER
	note = thymine
modified_base	2048
	mod_base = OTHER
	note = thymine
modified_base	2050
	mod_base = OTHER
	note = thymine
modified_base	2053..2054
	mod_base = OTHER
	note = thymine
modified_base	2057..2058
	mod_base = OTHER
	note = thymine
modified_base	2063
	mod_base = OTHER
	note = thymine
modified_base	2068
	mod_base = OTHER

-continued

modified_base	note = thymine
	2074
	mod_base = OTHER
modified_base	note = thymine
	2076
	mod_base = OTHER
modified_base	note = thymine
	2078
	mod_base = OTHER
modified_base	note = thymine
	2081
	mod_base = OTHER
modified_base	note = thymine
	2086..2090
	mod_base = OTHER
modified_base	note = thymine
	2097..2099
	mod_base = OTHER
modified_base	note = thymine
	2101
	mod_base = OTHER
modified_base	note = thymine
	2104
	mod_base = OTHER
modified_base	note = thymine
	2107
	mod_base = OTHER
modified_base	note = thymine
	2109
	mod_base = OTHER
modified_base	note = thymine
	2112..2115
	mod_base = OTHER
modified_base	note = thymine
	2117
	mod_base = OTHER
modified_base	note = thymine
	2125
	mod_base = OTHER
modified_base	note = thymine
	2129
	mod_base = OTHER
modified_base	note = thymine
	2131
	mod_base = OTHER
modified_base	note = thymine
	2134
	mod_base = OTHER
modified_base	note = thymine
	2137..2138
	mod_base = OTHER
modified_base	note = thymine
	2142
	mod_base = OTHER
modified_base	note = thymine
	2148..2149
	mod_base = OTHER
modified_base	note = thymine
	2154
	mod_base = OTHER
modified_base	note = thymine
	2157
	mod_base = OTHER
modified_base	note = thymine
	2162
	mod_base = OTHER
modified_base	note = thymine
	2164
	mod_base = OTHER
modified_base	note = thymine
	2173
	mod_base = OTHER
modified_base	note = thymine
	2177
	mod_base = OTHER
modified_base	note = thymine
	2181
	mod_base = OTHER
	note = thymine

-continued

modified_base	2184
	mod_base = OTHER
	note = thymine
modified_base	2187..2188
	mod_base = OTHER
	note = thymine
modified_base	2192
	mod_base = OTHER
	note = thymine
modified_base	2194
	mod_base = OTHER
	note = thymine
modified_base	2207
	mod_base = OTHER
	note = thymine
modified_base	2209..2210
	mod_base = OTHER
	note = thymine
modified_base	2213
	mod_base = OTHER
	note = thymine
modified_base	2218
	mod_base = OTHER
	note = thymine
modified_base	2223..2225
	mod_base = OTHER
	note = thymine
modified_base	2232
	mod_base = OTHER
	note = thymine
modified_base	2236
	mod_base = OTHER
	note = thymine
modified_base	2238
	mod_base = OTHER
	note = thymine
modified_base	2240
	mod_base = OTHER
	note = thymine
modified_base	2243
	mod_base = OTHER
	note = thymine
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	mod_base = OTHER
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modified_base	2253..2254
	mod_base = OTHER
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modified_base	2256
	mod_base = OTHER
	note = thymine
modified_base	2258
	mod_base = OTHER
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modified_base	2260
	mod_base = OTHER
	note = thymine
modified_base	2262
	mod_base = OTHER
	note = thymine
modified_base	2264
	mod_base = OTHER
	note = thymine
modified_base	2269..2270
	mod_base = OTHER
	note = thymine
modified_base	2277
	mod_base = OTHER
	note = thymine
modified_base	2279
	mod_base = OTHER
	note = thymine
modified_base	2284..2285
	mod_base = OTHER
	note = thymine
modified_base	2287
	mod_base = OTHER
	note = thymine
modified_base	2293..2299

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	note = thymine
modified_base	2307
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	note = thymine
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	mod_base = OTHER
	note = thymine
modified_base	2322
	mod_base = OTHER
	note = thymine
modified_base	2325..2327
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	note = thymine
modified_base	2330..2331
	mod_base = OTHER
	note = thymine
modified_base	2333
	mod_base = OTHER
	note = thymine
modified_base	2338
	mod_base = OTHER
	note = thymine
modified_base	2342
	mod_base = OTHER
	note = thymine
modified_base	2345..2346
	mod_base = OTHER
	note = thymine
modified_base	2348
	mod_base = OTHER
	note = thymine
modified_base	2350..2351
	mod_base = OTHER
	note = thymine
modified_base	2353
	mod_base = OTHER
	note = thymine
modified_base	2358
	mod_base = OTHER
	note = thymine
modified_base	2363
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	note = thymine
modified_base	2367
	mod_base = OTHER
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modified_base	2369
	mod_base = OTHER
	note = thymine
modified_base	2372..2374
	mod_base = OTHER
	note = thymine
modified_base	2376
	mod_base = OTHER
	note = thymine
modified_base	2379
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	note = thymine
modified_base	2381
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	mod_base = OTHER
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modified_base	2392
	mod_base = OTHER
	note = thymine
modified_base	2404
	mod_base = OTHER
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modified_base	2421..2423
	mod_base = OTHER
	note = thymine
modified_base	2425
	mod_base = OTHER

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modified_base	2460..2462 mod_base = OTHER note = thymine
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modified_base	2467 mod_base = OTHER note = thymine
modified_base	2471 mod_base = OTHER note = thymine
modified_base	2477 mod_base = OTHER note = thymine
modified_base	2481..2482 mod_base = OTHER note = thymine
modified_base	2484 mod_base = OTHER note = thymine
modified_base	2486..2487 mod_base = OTHER note = thymine
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modified_base	2501 mod_base = OTHER note = thymine
modified_base	2504..2505 mod_base = OTHER note = thymine
modified_base	2508..2511 mod_base = OTHER note = thymine
modified_base	2513 mod_base = OTHER note = thymine
modified_base	2527..2530 mod_base = OTHER note = thymine
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modified_base	2546..2547 mod_base = OTHER note = thymine

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modified_base	2576 mod_base = OTHER note = thymine
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modified_base	2590..2592 mod_base = OTHER note = thymine
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modified_base	2609 mod_base = OTHER note = thymine
modified_base	2614..2615 mod_base = OTHER note = thymine
modified_base	2617 mod_base = OTHER note = thymine
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modified_base	2629..2630 mod_base = OTHER note = thymine
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modified_base	2681

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modified_base	2793
	mod_base = OTHER

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modified_base	2895..2896 mod_base = OTHER note = thymine
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modified_base	2920 mod_base = OTHER note = thymine
modified_base	2922 mod_base = OTHER note = thymine

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modified_base	2937..2938
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	note = thymine
modified_base	2993..2994
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modified_base	3035
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	note = thymine
modified_base	3047

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modified_base	3117..3119
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modified_base	3121..3122
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	mod_base = OTHER
	note = thymine
modified_base	3131
	mod_base = OTHER
	note = thymine
modified_base	3136..3137
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	note = thymine
modified_base	3139..3140
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modified_base	3147..3148
	mod_base = OTHER
	note = thymine
modified_base	3150..3151
	mod_base = OTHER
	note = thymine
modified_base	3157..3161
	mod_base = OTHER
	note = thymine
modified_base	3164
	mod_base = OTHER

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modified_base	3187 mod_base = OTHER note = thymine
modified_base	3189 mod_base = OTHER note = thymine
modified_base	3195..3196 mod_base = OTHER note = thymine
modified_base	3199..3200 mod_base = OTHER note = thymine
modified_base	3205 mod_base = OTHER note = thymine
modified_base	3209 mod_base = OTHER note = thymine
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modified_base	3214 mod_base = OTHER note = thymine
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modified_base	3232 mod_base = OTHER note = thymine
modified_base	3235..3236 mod_base = OTHER note = thymine
modified_base	3244 mod_base = OTHER note = thymine
modified_base	3246 mod_base = OTHER note = thymine
modified_base	3250..3251 mod_base = OTHER note = thymine
modified_base	3257 mod_base = OTHER note = thymine
modified_base	3259 mod_base = OTHER note = thymine
modified_base	3263 mod_base = OTHER note = thymine
modified_base	3267..3268 mod_base = OTHER note = thymine
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modified_base	3279 mod_base = OTHER note = thymine
modified_base	3281 mod_base = OTHER note = thymine

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modified_base	3296
	mod_base = OTHER
	note = thymine
modified_base	3300
	mod_base = OTHER
	note = thymine
modified_base	3302
	mod_base = OTHER
	note = thymine
modified_base	3308..3309
	mod_base = OTHER
	note = thymine
modified_base	3316
	mod_base = OTHER
	note = thymine
modified_base	3319..3320
	mod_base = OTHER
	note = thymine
modified_base	3325
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	note = thymine
modified_base	3327
	mod_base = OTHER
	note = thymine
modified_base	3332
	mod_base = OTHER
	note = thymine
modified_base	3341..3342
	mod_base = OTHER
	note = thymine
modified_base	3344
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modified_base	3347
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modified_base	3350..3353
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modified_base	3358
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modified_base	3363
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modified_base	3368..3369
	mod_base = OTHER
	note = thymine
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	mod_base = OTHER
	note = thymine
modified_base	3376
	mod_base = OTHER
	note = thymine
modified_base	3382
	mod_base = OTHER
	note = thymine
modified_base	3384..3385
	mod_base = OTHER
	note = thymine
modified_base	3392
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	note = thymine
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modified_base	3398
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	note = thymine
modified_base	3403

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note = thymine
modified_base 3410
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note = Junin virus strain Candid-1 segment S - AY746353 -
      virusesM ssrna negative-strand virusesM arenaviridaeM
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source 1..3413
mol_type = other RNA
organism = unidentified

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SEQ ID NO: 2      moltype = RNA length = 7114
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                  note = thymine

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	mod_base = OTHER
	note = thymine
modified_base	179

-continued

	mod_base = OTHER
	note = thymine
modified_base	181
	mod_base = OTHER
	note = thymine
modified_base	183
	mod_base = OTHER
	note = thymine
modified_base	185
	mod_base = OTHER
	note = thymine
modified_base	188
	mod_base = OTHER
	note = thymine
modified_base	192
	mod_base = OTHER
	note = thymine
modified_base	194
	mod_base = OTHER
	note = thymine
modified_base	198
	mod_base = OTHER
	note = thymine
modified_base	200
	mod_base = OTHER
	note = thymine
modified_base	204
	mod_base = OTHER
	note = thymine
modified_base	207
	mod_base = OTHER
	note = thymine
modified_base	210
	mod_base = OTHER
	note = thymine
modified_base	213..215
	mod_base = OTHER
	note = thymine
modified_base	218
	mod_base = OTHER
	note = thymine
modified_base	221
	mod_base = OTHER
	note = thymine
modified_base	227..229
	mod_base = OTHER
	note = thymine
modified_base	232
	mod_base = OTHER
	note = thymine
modified_base	237
	mod_base = OTHER
	note = thymine
modified_base	239
	mod_base = OTHER
	note = thymine
modified_base	242
	mod_base = OTHER
	note = thymine
modified_base	245
	mod_base = OTHER
	note = thymine
modified_base	249
	mod_base = OTHER
	note = thymine
modified_base	253..255
	mod_base = OTHER
	note = thymine
modified_base	257..259
	mod_base = OTHER
	note = thymine
modified_base	264
	mod_base = OTHER
	note = thymine
modified_base	269
	mod_base = OTHER
	note = thymine
modified_base	275
	mod_base = OTHER

-continued

modified_base	note = thymine
	277
	mod_base = OTHER
	note = thymine
modified_base	279..280
	mod_base = OTHER
	note = thymine
modified_base	287..288
	mod_base = OTHER
	note = thymine
modified_base	293
	mod_base = OTHER
	note = thymine
modified_base	295
	mod_base = OTHER
	note = thymine
modified_base	297
	mod_base = OTHER
	note = thymine
modified_base	302
	mod_base = OTHER
	note = thymine
modified_base	304
	mod_base = OTHER
	note = thymine
modified_base	306
	mod_base = OTHER
	note = thymine
modified_base	309
	mod_base = OTHER
	note = thymine
modified_base	319
	mod_base = OTHER
	note = thymine
modified_base	331
	mod_base = OTHER
	note = thymine
modified_base	337
	mod_base = OTHER
	note = thymine
modified_base	343
	mod_base = OTHER
	note = thymine
modified_base	366
	mod_base = OTHER
	note = thymine
modified_base	375
	mod_base = OTHER
	note = thymine
modified_base	384
	mod_base = OTHER
	note = thymine
modified_base	412
	mod_base = OTHER
	note = thymine
modified_base	425
	mod_base = OTHER
	note = thymine
modified_base	442
	mod_base = OTHER
	note = thymine
modified_base	445
	mod_base = OTHER
	note = thymine
modified_base	452..453
	mod_base = OTHER
	note = thymine
modified_base	456
	mod_base = OTHER
	note = thymine
modified_base	460
	mod_base = OTHER
	note = thymine
modified_base	462
	mod_base = OTHER
	note = thymine
modified_base	464..466
	mod_base = OTHER
	note = thymine

-continued

modified_base	469
	mod_base = OTHER
	note = thymine
modified_base	471
	mod_base = OTHER
	note = thymine
modified_base	476..478
	mod_base = OTHER
	note = thymine
modified_base	480
	mod_base = OTHER
	note = thymine
modified_base	483..484
	mod_base = OTHER
	note = thymine
modified_base	488
	mod_base = OTHER
	note = thymine
modified_base	493
	mod_base = OTHER
	note = thymine
modified_base	497
	mod_base = OTHER
	note = thymine
modified_base	499
	mod_base = OTHER
	note = thymine
modified_base	506..508
	mod_base = OTHER
	note = thymine
modified_base	512
	mod_base = OTHER
	note = thymine
modified_base	514
	mod_base = OTHER
	note = thymine
modified_base	519
	mod_base = OTHER
	note = thymine
modified_base	524..525
	mod_base = OTHER
	note = thymine
modified_base	531
	mod_base = OTHER
	note = thymine
modified_base	533
	mod_base = OTHER
	note = thymine
modified_base	536
	mod_base = OTHER
	note = thymine
modified_base	538
	mod_base = OTHER
	note = thymine
modified_base	546
	mod_base = OTHER
	note = thymine
modified_base	549..551
	mod_base = OTHER
	note = thymine
modified_base	557
	mod_base = OTHER
	note = thymine
modified_base	559
	mod_base = OTHER
	note = thymine
modified_base	562
	mod_base = OTHER
	note = thymine
modified_base	566
	mod_base = OTHER
	note = thymine
modified_base	573..574
	mod_base = OTHER
	note = thymine
modified_base	579..581
	mod_base = OTHER
	note = thymine
modified_base	587..588

-continued

	mod_base = OTHER
	note = thymine
modified_base	597
	mod_base = OTHER
	note = thymine
modified_base	605..607
	mod_base = OTHER
	note = thymine
modified_base	609
	mod_base = OTHER
	note = thymine
modified_base	614
	mod_base = OTHER
	note = thymine
modified_base	622
	mod_base = OTHER
	note = thymine
modified_base	627
	mod_base = OTHER
	note = thymine
modified_base	630
	mod_base = OTHER
	note = thymine
modified_base	632..634
	mod_base = OTHER
	note = thymine
modified_base	636
	mod_base = OTHER
	note = thymine
modified_base	638
	mod_base = OTHER
	note = thymine
modified_base	642
	mod_base = OTHER
	note = thymine
modified_base	645
	mod_base = OTHER
	note = thymine
modified_base	660
	mod_base = OTHER
	note = thymine
modified_base	663
	mod_base = OTHER
	note = thymine
modified_base	665..666
	mod_base = OTHER
	note = thymine
modified_base	669
	mod_base = OTHER
	note = thymine
modified_base	674
	mod_base = OTHER
	note = thymine
modified_base	677
	mod_base = OTHER
	note = thymine
modified_base	679..680
	mod_base = OTHER
	note = thymine
modified_base	682..684
	mod_base = OTHER
	note = thymine
modified_base	686
	mod_base = OTHER
	note = thymine
modified_base	689
	mod_base = OTHER
	note = thymine
modified_base	691..692
	mod_base = OTHER
	note = thymine
modified_base	694..698
	mod_base = OTHER
	note = thymine
modified_base	702
	mod_base = OTHER
	note = thymine
modified_base	710
	mod_base = OTHER

-continued

modified_base	note = thymine 713..714 mod_base = OTHER
modified_base	note = thymine 718..719 mod_base = OTHER
modified_base	note = thymine 722 mod_base = OTHER
modified_base	note = thymine 727..729 mod_base = OTHER
modified_base	note = thymine 736 mod_base = OTHER
modified_base	note = thymine 738 mod_base = OTHER
modified_base	note = thymine 742 mod_base = OTHER
modified_base	note = thymine 744 mod_base = OTHER
modified_base	note = thymine 747 mod_base = OTHER
modified_base	note = thymine 749 mod_base = OTHER
modified_base	note = thymine 752 mod_base = OTHER
modified_base	note = thymine 755 mod_base = OTHER
modified_base	note = thymine 760..761 mod_base = OTHER
modified_base	note = thymine 765..766 mod_base = OTHER
modified_base	note = thymine 770 mod_base = OTHER
modified_base	note = thymine 772 mod_base = OTHER
modified_base	note = thymine 777 mod_base = OTHER
modified_base	note = thymine 789 mod_base = OTHER
modified_base	note = thymine 794 mod_base = OTHER
modified_base	note = thymine 796 mod_base = OTHER
modified_base	note = thymine 798 mod_base = OTHER
modified_base	note = thymine 800 mod_base = OTHER
modified_base	note = thymine 802..803 mod_base = OTHER
modified_base	note = thymine 811..813 mod_base = OTHER
modified_base	note = thymine 816..817 mod_base = OTHER
modified_base	note = thymine 821 mod_base = OTHER
	note = thymine

-continued

modified_base	823
	mod_base = OTHER
	note = thymine
modified_base	825
	mod_base = OTHER
	note = thymine
modified_base	832
	mod_base = OTHER
	note = thymine
modified_base	837
	mod_base = OTHER
	note = thymine
modified_base	839..841
	mod_base = OTHER
	note = thymine
modified_base	854..855
	mod_base = OTHER
	note = thymine
modified_base	857
	mod_base = OTHER
	note = thymine
modified_base	860
	mod_base = OTHER
	note = thymine
modified_base	864..865
	mod_base = OTHER
	note = thymine
modified_base	867
	mod_base = OTHER
	note = thymine
modified_base	869
	mod_base = OTHER
	note = thymine
modified_base	877
	mod_base = OTHER
	note = thymine
modified_base	882
	mod_base = OTHER
	note = thymine
modified_base	885
	mod_base = OTHER
	note = thymine
modified_base	887
	mod_base = OTHER
	note = thymine
modified_base	891
	mod_base = OTHER
	note = thymine
modified_base	894..895
	mod_base = OTHER
	note = thymine
modified_base	897..898
	mod_base = OTHER
	note = thymine
modified_base	901..902
	mod_base = OTHER
	note = thymine
modified_base	905..906
	mod_base = OTHER
	note = thymine
modified_base	912
	mod_base = OTHER
	note = thymine
modified_base	918
	mod_base = OTHER
	note = thymine
modified_base	927..928
	mod_base = OTHER
	note = thymine
modified_base	933..934
	mod_base = OTHER
	note = thymine
modified_base	939
	mod_base = OTHER
	note = thymine
modified_base	943
	mod_base = OTHER
	note = thymine
modified_base	949

-continued

	mod_base = OTHER
	note = thymine
modified_base	951..952
	mod_base = OTHER
	note = thymine
modified_base	954
	mod_base = OTHER
	note = thymine
modified_base	957
	mod_base = OTHER
	note = thymine
modified_base	960..962
	mod_base = OTHER
	note = thymine
modified_base	964
	mod_base = OTHER
	note = thymine
modified_base	972
	mod_base = OTHER
	note = thymine
modified_base	977
	mod_base = OTHER
	note = thymine
modified_base	981
	mod_base = OTHER
	note = thymine
modified_base	985
	mod_base = OTHER
	note = thymine
modified_base	998
	mod_base = OTHER
	note = thymine
modified_base	1002
	mod_base = OTHER
	note = thymine
modified_base	1009
	mod_base = OTHER
	note = thymine
modified_base	1011
	mod_base = OTHER
	note = thymine
modified_base	1014
	mod_base = OTHER
	note = thymine
modified_base	1019
	mod_base = OTHER
	note = thymine
modified_base	1021
	mod_base = OTHER
	note = thymine
modified_base	1028
	mod_base = OTHER
	note = thymine
modified_base	1031
	mod_base = OTHER
	note = thymine
modified_base	1036
	mod_base = OTHER
	note = thymine
modified_base	1042
	mod_base = OTHER
	note = thymine
modified_base	1049
	mod_base = OTHER
	note = thymine
modified_base	1056
	mod_base = OTHER
	note = thymine
modified_base	1064
	mod_base = OTHER
	note = thymine
modified_base	1068
	mod_base = OTHER
	note = thymine
modified_base	1072..1073
	mod_base = OTHER
	note = thymine
modified_base	1077
	mod_base = OTHER

-continued

modified_base	note = thymine 1094..1095 mod_base = OTHER note = thymine
modified_base	1099 mod_base = OTHER note = thymine
modified_base	1102..1103 mod_base = OTHER note = thymine
modified_base	1107 mod_base = OTHER note = thymine
modified_base	1113..1114 mod_base = OTHER note = thymine
modified_base	1119 mod_base = OTHER note = thymine
modified_base	1121 mod_base = OTHER note = thymine
modified_base	1124..1125 mod_base = OTHER note = thymine
modified_base	1132 mod_base = OTHER note = thymine
modified_base	1140 mod_base = OTHER note = thymine
modified_base	1143 mod_base = OTHER note = thymine
modified_base	1148 mod_base = OTHER note = thymine
modified_base	1151 mod_base = OTHER note = thymine
modified_base	1154 mod_base = OTHER note = thymine
modified_base	1159 mod_base = OTHER note = thymine
modified_base	1165 mod_base = OTHER note = thymine
modified_base	1168 mod_base = OTHER note = thymine
modified_base	1170 mod_base = OTHER note = thymine
modified_base	1174 mod_base = OTHER note = thymine
modified_base	1176 mod_base = OTHER note = thymine
modified_base	1183 mod_base = OTHER note = thymine
modified_base	1185..1187 mod_base = OTHER note = thymine
modified_base	1189..1190 mod_base = OTHER note = thymine
modified_base	1192 mod_base = OTHER note = thymine
modified_base	1196..1198 mod_base = OTHER note = thymine
modified_base	1202..1205 mod_base = OTHER note = thymine

-continued

modified_base	1207
	mod_base = OTHER
	note = thymine
modified_base	1214
	mod_base = OTHER
	note = thymine
modified_base	1217
	mod_base = OTHER
	note = thymine
modified_base	1219
	mod_base = OTHER
	note = thymine
modified_base	1221
	mod_base = OTHER
	note = thymine
modified_base	1223..1224
	mod_base = OTHER
	note = thymine
modified_base	1226
	mod_base = OTHER
	note = thymine
modified_base	1229..1230
	mod_base = OTHER
	note = thymine
modified_base	1234
	mod_base = OTHER
	note = thymine
modified_base	1237
	mod_base = OTHER
	note = thymine
modified_base	1241..1244
	mod_base = OTHER
	note = thymine
modified_base	1247..1249
	mod_base = OTHER
	note = thymine
modified_base	1256..1257
	mod_base = OTHER
	note = thymine
modified_base	1259..1260
	mod_base = OTHER
	note = thymine
modified_base	1272..1273
	mod_base = OTHER
	note = thymine
modified_base	1285
	mod_base = OTHER
	note = thymine
modified_base	1287
	mod_base = OTHER
	note = thymine
modified_base	1293
	mod_base = OTHER
	note = thymine
modified_base	1297
	mod_base = OTHER
	note = thymine
modified_base	1300
	mod_base = OTHER
	note = thymine
modified_base	1302
	mod_base = OTHER
	note = thymine
modified_base	1304..1305
	mod_base = OTHER
	note = thymine
modified_base	1308
	mod_base = OTHER
	note = thymine
modified_base	1324
	mod_base = OTHER
	note = thymine
modified_base	1326..1327
	mod_base = OTHER
	note = thymine
modified_base	1329
	mod_base = OTHER
	note = thymine
modified_base	1344

-continued

	mod_base = OTHER
	note = thymine
modified_base	1356..1357
	mod_base = OTHER
	note = thymine
modified_base	1360
	mod_base = OTHER
	note = thymine
modified_base	1362..1363
	mod_base = OTHER
	note = thymine
modified_base	1365
	mod_base = OTHER
	note = thymine
modified_base	1372
	mod_base = OTHER
	note = thymine
modified_base	1379
	mod_base = OTHER
	note = thymine
modified_base	1383
	mod_base = OTHER
	note = thymine
modified_base	1390..1391
	mod_base = OTHER
	note = thymine
modified_base	1396
	mod_base = OTHER
	note = thymine
modified_base	1398
	mod_base = OTHER
	note = thymine
modified_base	1400..1401
	mod_base = OTHER
	note = thymine
modified_base	1411..1412
	mod_base = OTHER
	note = thymine
modified_base	1415
	mod_base = OTHER
	note = thymine
modified_base	1417..1418
	mod_base = OTHER
	note = thymine
modified_base	1421
	mod_base = OTHER
	note = thymine
modified_base	1424..1426
	mod_base = OTHER
	note = thymine
modified_base	1428..1429
	mod_base = OTHER
	note = thymine
modified_base	1431..1432
	mod_base = OTHER
	note = thymine
modified_base	1435
	mod_base = OTHER
	note = thymine
modified_base	1439
	mod_base = OTHER
	note = thymine
modified_base	1447
	mod_base = OTHER
	note = thymine
modified_base	1452
	mod_base = OTHER
	note = thymine
modified_base	1454
	mod_base = OTHER
	note = thymine
modified_base	1456
	mod_base = OTHER
	note = thymine
modified_base	1460..1461
	mod_base = OTHER
	note = thymine
modified_base	1466
	mod_base = OTHER

-continued

modified_base	note = thymine 1470 mod_base = OTHER note = thymine
modified_base	1472 mod_base = OTHER note = thymine
modified_base	1475..1476 mod_base = OTHER note = thymine
modified_base	1480 mod_base = OTHER note = thymine
modified_base	1482 mod_base = OTHER note = thymine
modified_base	1485 mod_base = OTHER note = thymine
modified_base	1487..1488 mod_base = OTHER note = thymine
modified_base	1490 mod_base = OTHER note = thymine
modified_base	1494 mod_base = OTHER note = thymine
modified_base	1496 mod_base = OTHER note = thymine
modified_base	1502..1503 mod_base = OTHER note = thymine
modified_base	1506..1507 mod_base = OTHER note = thymine
modified_base	1523..1525 mod_base = OTHER note = thymine
modified_base	1535 mod_base = OTHER note = thymine
modified_base	1537..1538 mod_base = OTHER note = thymine
modified_base	1543 mod_base = OTHER note = thymine
modified_base	1547 mod_base = OTHER note = thymine
modified_base	1550 mod_base = OTHER note = thymine
modified_base	1554..1556 mod_base = OTHER note = thymine
modified_base	1558 mod_base = OTHER note = thymine
modified_base	1561 mod_base = OTHER note = thymine
modified_base	1568 mod_base = OTHER note = thymine
modified_base	1577..1578 mod_base = OTHER note = thymine
modified_base	1585 mod_base = OTHER note = thymine
modified_base	1590..1591 mod_base = OTHER note = thymine
modified_base	1594 mod_base = OTHER note = thymine

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modified_base	1601..1602
	mod_base = OTHER
	note = thymine
modified_base	1605
	mod_base = OTHER
	note = thymine
modified_base	1607
	mod_base = OTHER
	note = thymine
modified_base	1612
	mod_base = OTHER
	note = thymine
modified_base	1615
	mod_base = OTHER
	note = thymine
modified_base	1622..1623
	mod_base = OTHER
	note = thymine
modified_base	1629..1631
	mod_base = OTHER
	note = thymine
modified_base	1633
	mod_base = OTHER
	note = thymine
modified_base	1635
	mod_base = OTHER
	note = thymine
modified_base	1638
	mod_base = OTHER
	note = thymine
modified_base	1640
	mod_base = OTHER
	note = thymine
modified_base	1644
	mod_base = OTHER
	note = thymine
modified_base	1646
	mod_base = OTHER
	note = thymine
modified_base	1648
	mod_base = OTHER
	note = thymine
modified_base	1653
	mod_base = OTHER
	note = thymine
modified_base	1658
	mod_base = OTHER
	note = thymine
modified_base	1668
	mod_base = OTHER
	note = thymine
modified_base	1672
	mod_base = OTHER
	note = thymine
modified_base	1674
	mod_base = OTHER
	note = thymine
modified_base	1676..1678
	mod_base = OTHER
	note = thymine
modified_base	1684
	mod_base = OTHER
	note = thymine
modified_base	1686
	mod_base = OTHER
	note = thymine
modified_base	1692..1693
	mod_base = OTHER
	note = thymine
modified_base	1695..1696
	mod_base = OTHER
	note = thymine
modified_base	1698
	mod_base = OTHER
	note = thymine
modified_base	1700
	mod_base = OTHER
	note = thymine
modified_base	1702..1703

-continued

	mod_base = OTHER
	note = thymine
modified_base	1706..1708
	mod_base = OTHER
	note = thymine
modified_base	1712
	mod_base = OTHER
	note = thymine
modified_base	1716
	mod_base = OTHER
	note = thymine
modified_base	1720..1721
	mod_base = OTHER
	note = thymine
modified_base	1723
	mod_base = OTHER
	note = thymine
modified_base	1726
	mod_base = OTHER
	note = thymine
modified_base	1730
	mod_base = OTHER
	note = thymine
modified_base	1734..1737
	mod_base = OTHER
	note = thymine
modified_base	1740..1741
	mod_base = OTHER
	note = thymine
modified_base	1744
	mod_base = OTHER
	note = thymine
modified_base	1747
	mod_base = OTHER
	note = thymine
modified_base	1749
	mod_base = OTHER
	note = thymine
modified_base	1751
	mod_base = OTHER
	note = thymine
modified_base	1767
	mod_base = OTHER
	note = thymine
modified_base	1769
	mod_base = OTHER
	note = thymine
modified_base	1776..1777
	mod_base = OTHER
	note = thymine
modified_base	1781
	mod_base = OTHER
	note = thymine
modified_base	1783..1784
	mod_base = OTHER
	note = thymine
modified_base	1789
	mod_base = OTHER
	note = thymine
modified_base	1791
	mod_base = OTHER
	note = thymine
modified_base	1799
	mod_base = OTHER
	note = thymine
modified_base	1801
	mod_base = OTHER
	note = thymine
modified_base	1803..1804
	mod_base = OTHER
	note = thymine
modified_base	1808..1809
	mod_base = OTHER
	note = thymine
modified_base	1815
	mod_base = OTHER
	note = thymine
modified_base	1818..1819
	mod_base = OTHER

-continued

modified_base	note = thymine 1822 mod_base = OTHER note = thymine
modified_base	1829 mod_base = OTHER note = thymine
modified_base	1836 mod_base = OTHER note = thymine
modified_base	1838..1839 mod_base = OTHER note = thymine
modified_base	1841..1842 mod_base = OTHER note = thymine
modified_base	1847 mod_base = OTHER note = thymine
modified_base	1850..1852 mod_base = OTHER note = thymine
modified_base	1857 mod_base = OTHER note = thymine
modified_base	1860..1862 mod_base = OTHER note = thymine
modified_base	1864..1865 mod_base = OTHER note = thymine
modified_base	1868 mod_base = OTHER note = thymine
modified_base	1870 mod_base = OTHER note = thymine
modified_base	1875 mod_base = OTHER note = thymine
modified_base	1880 mod_base = OTHER note = thymine
modified_base	1884 mod_base = OTHER note = thymine
modified_base	1887..1888 mod_base = OTHER note = thymine
modified_base	1891 mod_base = OTHER note = thymine
modified_base	1895 mod_base = OTHER note = thymine
modified_base	1898..1899 mod_base = OTHER note = thymine
modified_base	1902 mod_base = OTHER note = thymine
modified_base	1904..1905 mod_base = OTHER note = thymine
modified_base	1907 mod_base = OTHER note = thymine
modified_base	1909 mod_base = OTHER note = thymine
modified_base	1911 mod_base = OTHER note = thymine
modified_base	1913 mod_base = OTHER note = thymine
modified_base	1916 mod_base = OTHER note = thymine

-continued

modified_base	1919
	mod_base = OTHER
	note = thymine
modified_base	1922..1924
	mod_base = OTHER
	note = thymine
modified_base	1928..1930
	mod_base = OTHER
	note = thymine
modified_base	1947..1948
	mod_base = OTHER
	note = thymine
modified_base	1950..1951
	mod_base = OTHER
	note = thymine
modified_base	1953..1954
	mod_base = OTHER
	note = thymine
modified_base	1958
	mod_base = OTHER
	note = thymine
modified_base	1974
	mod_base = OTHER
	note = thymine
modified_base	1976
	mod_base = OTHER
	note = thymine
modified_base	1981
	mod_base = OTHER
	note = thymine
modified_base	1983..1984
	mod_base = OTHER
	note = thymine
modified_base	1987
	mod_base = OTHER
	note = thymine
modified_base	2004
	mod_base = OTHER
	note = thymine
modified_base	2009
	mod_base = OTHER
	note = thymine
modified_base	2013
	mod_base = OTHER
	note = thymine
modified_base	2022..2024
	mod_base = OTHER
	note = thymine
modified_base	2026
	mod_base = OTHER
	note = thymine
modified_base	2043
	mod_base = OTHER
	note = thymine
modified_base	2046
	mod_base = OTHER
	note = thymine
modified_base	2056..2057
	mod_base = OTHER
	note = thymine
modified_base	2060..2062
	mod_base = OTHER
	note = thymine
modified_base	2069..2070
	mod_base = OTHER
	note = thymine
modified_base	2073..2074
	mod_base = OTHER
	note = thymine
modified_base	2078..2079
	mod_base = OTHER
	note = thymine
modified_base	2081..2083
	mod_base = OTHER
	note = thymine
modified_base	2087
	mod_base = OTHER
	note = thymine
modified_base	2090..2094

-continued

	mod_base = OTHER
	note = thymine
modified_base	2100
	mod_base = OTHER
	note = thymine
modified_base	2104
	mod_base = OTHER
	note = thymine
modified_base	2108..2110
	mod_base = OTHER
	note = thymine
modified_base	2115..2117
	mod_base = OTHER
	note = thymine
modified_base	2119
	mod_base = OTHER
	note = thymine
modified_base	2121
	mod_base = OTHER
	note = thymine
modified_base	2128..2129
	mod_base = OTHER
	note = thymine
modified_base	2141
	mod_base = OTHER
	note = thymine
modified_base	2146..2147
	mod_base = OTHER
	note = thymine
modified_base	2149..2150
	mod_base = OTHER
	note = thymine
modified_base	2155
	mod_base = OTHER
	note = thymine
modified_base	2159
	mod_base = OTHER
	note = thymine
modified_base	2161
	mod_base = OTHER
	note = thymine
modified_base	2163
	mod_base = OTHER
	note = thymine
modified_base	2166..2168
	mod_base = OTHER
	note = thymine
modified_base	2170
	mod_base = OTHER
	note = thymine
modified_base	2172
	mod_base = OTHER
	note = thymine
modified_base	2180
	mod_base = OTHER
	note = thymine
modified_base	2186..2188
	mod_base = OTHER
	note = thymine
modified_base	2193
	mod_base = OTHER
	note = thymine
modified_base	2197
	mod_base = OTHER
	note = thymine
modified_base	2205..2206
	mod_base = OTHER
	note = thymine
modified_base	2209
	mod_base = OTHER
	note = thymine
modified_base	2215
	mod_base = OTHER
	note = thymine
modified_base	2218
	mod_base = OTHER
	note = thymine
modified_base	2220..2221
	mod_base = OTHER

-continued

modified_base	note = thymine 2233 mod_base = OTHER note = thymine
modified_base	2240 mod_base = OTHER note = thymine
modified_base	2256 mod_base = OTHER note = thymine
modified_base	2261 mod_base = OTHER note = thymine
modified_base	2267..2269 mod_base = OTHER note = thymine
modified_base	2271..2272 mod_base = OTHER note = thymine
modified_base	2275..2276 mod_base = OTHER note = thymine
modified_base	2280 mod_base = OTHER note = thymine
modified_base	2282 mod_base = OTHER note = thymine
modified_base	2290 mod_base = OTHER note = thymine
modified_base	2292..2293 mod_base = OTHER note = thymine
modified_base	2295..2296 mod_base = OTHER note = thymine
modified_base	2303..2306 mod_base = OTHER note = thymine
modified_base	2308 mod_base = OTHER note = thymine
modified_base	2313..2315 mod_base = OTHER note = thymine
modified_base	2317..2318 mod_base = OTHER note = thymine
modified_base	2326 mod_base = OTHER note = thymine
modified_base	2331 mod_base = OTHER note = thymine
modified_base	2333 mod_base = OTHER note = thymine
modified_base	2335..2336 mod_base = OTHER note = thymine
modified_base	2338 mod_base = OTHER note = thymine
modified_base	2341..2342 mod_base = OTHER note = thymine
modified_base	2344 mod_base = OTHER note = thymine
modified_base	2348 mod_base = OTHER note = thymine
modified_base	2350..2351 mod_base = OTHER note = thymine
modified_base	2356 mod_base = OTHER note = thymine

-continued

modified_base	2358
	mod_base = OTHER
	note = thymine
modified_base	2360..2361
	mod_base = OTHER
	note = thymine
modified_base	2363..2365
	mod_base = OTHER
	note = thymine
modified_base	2367
	mod_base = OTHER
	note = thymine
modified_base	2369
	mod_base = OTHER
	note = thymine
modified_base	2375
	mod_base = OTHER
	note = thymine
modified_base	2377..2378
	mod_base = OTHER
	note = thymine
modified_base	2383..2384
	mod_base = OTHER
	note = thymine
modified_base	2386
	mod_base = OTHER
	note = thymine
modified_base	2389
	mod_base = OTHER
	note = thymine
modified_base	2393..2395
	mod_base = OTHER
	note = thymine
modified_base	2398
	mod_base = OTHER
	note = thymine
modified_base	2401
	mod_base = OTHER
	note = thymine
modified_base	2413
	mod_base = OTHER
	note = thymine
modified_base	2418
	mod_base = OTHER
	note = thymine
modified_base	2428
	mod_base = OTHER
	note = thymine
modified_base	2432
	mod_base = OTHER
	note = thymine
modified_base	2436..2437
	mod_base = OTHER
	note = thymine
modified_base	2444
	mod_base = OTHER
	note = thymine
modified_base	2446
	mod_base = OTHER
	note = thymine
modified_base	2448
	mod_base = OTHER
	note = thymine
modified_base	2451
	mod_base = OTHER
	note = thymine
modified_base	2453
	mod_base = OTHER
	note = thymine
modified_base	2458..2460
	mod_base = OTHER
	note = thymine
modified_base	2462
	mod_base = OTHER
	note = thymine
modified_base	2464
	mod_base = OTHER
	note = thymine
modified_base	2467..2469

-continued

	mod_base = OTHER
	note = thymine
modified_base	2472
	mod_base = OTHER
	note = thymine
modified_base	2474..2475
	mod_base = OTHER
	note = thymine
modified_base	2479
	mod_base = OTHER
	note = thymine
modified_base	2490
	mod_base = OTHER
	note = thymine
modified_base	2492..2493
	mod_base = OTHER
	note = thymine
modified_base	2501
	mod_base = OTHER
	note = thymine
modified_base	2507
	mod_base = OTHER
	note = thymine
modified_base	2514
	mod_base = OTHER
	note = thymine
modified_base	2517
	mod_base = OTHER
	note = thymine
modified_base	2522
	mod_base = OTHER
	note = thymine
modified_base	2527
	mod_base = OTHER
	note = thymine
modified_base	2530..2531
	mod_base = OTHER
	note = thymine
modified_base	2533
	mod_base = OTHER
	note = thymine
modified_base	2538..2539
	mod_base = OTHER
	note = thymine
modified_base	2542
	mod_base = OTHER
	note = thymine
modified_base	2547..2550
	mod_base = OTHER
	note = thymine
modified_base	2553
	mod_base = OTHER
	note = thymine
modified_base	2560..2563
	mod_base = OTHER
	note = thymine
modified_base	2567
	mod_base = OTHER
	note = thymine
modified_base	2569
	mod_base = OTHER
	note = thymine
modified_base	2571..2572
	mod_base = OTHER
	note = thymine
modified_base	2575
	mod_base = OTHER
	note = thymine
modified_base	2577
	mod_base = OTHER
	note = thymine
modified_base	2580..2581
	mod_base = OTHER
	note = thymine
modified_base	2588..2590
	mod_base = OTHER
	note = thymine
modified_base	2593
	mod_base = OTHER

-continued

modified_base	note = thymine 2601..2602 mod_base = OTHER
modified_base	note = thymine 2605..2606 mod_base = OTHER
modified_base	note = thymine 2608 mod_base = OTHER
modified_base	note = thymine 2614..2615 mod_base = OTHER
modified_base	note = thymine 2617 mod_base = OTHER
modified_base	note = thymine 2619 mod_base = OTHER
modified_base	note = thymine 2622..2623 mod_base = OTHER
modified_base	note = thymine 2625 mod_base = OTHER
modified_base	note = thymine 2628 mod_base = OTHER
modified_base	note = thymine 2638 mod_base = OTHER
modified_base	note = thymine 2642..2643 mod_base = OTHER
modified_base	note = thymine 2647..2650 mod_base = OTHER
modified_base	note = thymine 2653 mod_base = OTHER
modified_base	note = thymine 2655 mod_base = OTHER
modified_base	note = thymine 2657 mod_base = OTHER
modified_base	note = thymine 2664 mod_base = OTHER
modified_base	note = thymine 2669 mod_base = OTHER
modified_base	note = thymine 2671 mod_base = OTHER
modified_base	note = thymine 2679 mod_base = OTHER
modified_base	note = thymine 2688 mod_base = OTHER
modified_base	note = thymine 2690..2692 mod_base = OTHER
modified_base	note = thymine 2697 mod_base = OTHER
modified_base	note = thymine 2699..2700 mod_base = OTHER
modified_base	note = thymine 2702..2703 mod_base = OTHER
modified_base	note = thymine 2705..2706 mod_base = OTHER
modified_base	note = thymine 2710 mod_base = OTHER
	note = thymine

-continued

modified_base	2714
	mod_base = OTHER
	note = thymine
modified_base	2725
	mod_base = OTHER
	note = thymine
modified_base	2727
	mod_base = OTHER
	note = thymine
modified_base	2729
	mod_base = OTHER
	note = thymine
modified_base	2731
	mod_base = OTHER
	note = thymine
modified_base	2733
	mod_base = OTHER
	note = thymine
modified_base	2735
	mod_base = OTHER
	note = thymine
modified_base	2742
	mod_base = OTHER
	note = thymine
modified_base	2744
	mod_base = OTHER
	note = thymine
modified_base	2746
	mod_base = OTHER
	note = thymine
modified_base	2749
	mod_base = OTHER
	note = thymine
modified_base	2753..2754
	mod_base = OTHER
	note = thymine
modified_base	2757..2758
	mod_base = OTHER
	note = thymine
modified_base	2762
	mod_base = OTHER
	note = thymine
modified_base	2764
	mod_base = OTHER
	note = thymine
modified_base	2767
	mod_base = OTHER
	note = thymine
modified_base	2773
	mod_base = OTHER
	note = thymine
modified_base	2775..2777
	mod_base = OTHER
	note = thymine
modified_base	2783
	mod_base = OTHER
	note = thymine
modified_base	2785
	mod_base = OTHER
	note = thymine
modified_base	2787
	mod_base = OTHER
	note = thymine
modified_base	2791
	mod_base = OTHER
	note = thymine
modified_base	2794
	mod_base = OTHER
	note = thymine
modified_base	2802..2803
	mod_base = OTHER
	note = thymine
modified_base	2809
	mod_base = OTHER
	note = thymine
modified_base	2813..2814
	mod_base = OTHER
	note = thymine
modified_base	2817

-continued

	mod_base = OTHER
	note = thymine
modified_base	2824..2825
	mod_base = OTHER
	note = thymine
modified_base	2827
	mod_base = OTHER
	note = thymine
modified_base	2829
	mod_base = OTHER
	note = thymine
modified_base	2831
	mod_base = OTHER
	note = thymine
modified_base	2833
	mod_base = OTHER
	note = thymine
modified_base	2836
	mod_base = OTHER
	note = thymine
modified_base	2838
	mod_base = OTHER
	note = thymine
modified_base	2840
	mod_base = OTHER
	note = thymine
modified_base	2846..2847
	mod_base = OTHER
	note = thymine
modified_base	2849
	mod_base = OTHER
	note = thymine
modified_base	2851..2852
	mod_base = OTHER
	note = thymine
modified_base	2857..2860
	mod_base = OTHER
	note = thymine
modified_base	2865..2866
	mod_base = OTHER
	note = thymine
modified_base	2871..2872
	mod_base = OTHER
	note = thymine
modified_base	2874
	mod_base = OTHER
	note = thymine
modified_base	2876
	mod_base = OTHER
	note = thymine
modified_base	2895..2896
	mod_base = OTHER
	note = thymine
modified_base	2899
	mod_base = OTHER
	note = thymine
modified_base	2903
	mod_base = OTHER
	note = thymine
modified_base	2911
	mod_base = OTHER
	note = thymine
modified_base	2913
	mod_base = OTHER
	note = thymine
modified_base	2915..2916
	mod_base = OTHER
	note = thymine
modified_base	2923
	mod_base = OTHER
	note = thymine
modified_base	2933
	mod_base = OTHER
	note = thymine
modified_base	2935..2936
	mod_base = OTHER
	note = thymine
modified_base	2940..2941
	mod_base = OTHER

-continued

modified_base	note = thymine 2946 mod_base = OTHER note = thymine
modified_base	2959..2960 mod_base = OTHER note = thymine
modified_base	2963 mod_base = OTHER note = thymine
modified_base	2965 mod_base = OTHER note = thymine
modified_base	2971 mod_base = OTHER note = thymine
modified_base	2973..2975 mod_base = OTHER note = thymine
modified_base	2980 mod_base = OTHER note = thymine
modified_base	2983 mod_base = OTHER note = thymine
modified_base	2988..2989 mod_base = OTHER note = thymine
modified_base	2991..2993 mod_base = OTHER note = thymine
modified_base	2996..2998 mod_base = OTHER note = thymine
modified_base	3005 mod_base = OTHER note = thymine
modified_base	3011..3012 mod_base = OTHER note = thymine
modified_base	3015 mod_base = OTHER note = thymine
modified_base	3028..3030 mod_base = OTHER note = thymine
modified_base	3038..3039 mod_base = OTHER note = thymine
modified_base	3041 mod_base = OTHER note = thymine
modified_base	3043 mod_base = OTHER note = thymine
modified_base	3046 mod_base = OTHER note = thymine
modified_base	3048 mod_base = OTHER note = thymine
modified_base	3050..3051 mod_base = OTHER note = thymine
modified_base	3057..3060 mod_base = OTHER note = thymine
modified_base	3063 mod_base = OTHER note = thymine
modified_base	3067 mod_base = OTHER note = thymine
modified_base	3079..3083 mod_base = OTHER note = thymine
modified_base	3085 mod_base = OTHER note = thymine

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modified_base	3091
	mod_base = OTHER
	note = thymine
modified_base	3096
	mod_base = OTHER
	note = thymine
modified_base	3099
	mod_base = OTHER
	note = thymine
modified_base	3102
	mod_base = OTHER
	note = thymine
modified_base	3106
	mod_base = OTHER
	note = thymine
modified_base	3112
	mod_base = OTHER
	note = thymine
modified_base	3114
	mod_base = OTHER
	note = thymine
modified_base	3118
	mod_base = OTHER
	note = thymine
modified_base	3125
	mod_base = OTHER
	note = thymine
modified_base	3127
	mod_base = OTHER
	note = thymine
modified_base	3132
	mod_base = OTHER
	note = thymine
modified_base	3135
	mod_base = OTHER
	note = thymine
modified_base	3140
	mod_base = OTHER
	note = thymine
modified_base	3144
	mod_base = OTHER
	note = thymine
modified_base	3149
	mod_base = OTHER
	note = thymine
modified_base	3153
	mod_base = OTHER
	note = thymine
modified_base	3164..3165
	mod_base = OTHER
	note = thymine
modified_base	3167..3168
	mod_base = OTHER
	note = thymine
modified_base	3172
	mod_base = OTHER
	note = thymine
modified_base	3175..3176
	mod_base = OTHER
	note = thymine
modified_base	3183
	mod_base = OTHER
	note = thymine
modified_base	3185
	mod_base = OTHER
	note = thymine
modified_base	3189
	mod_base = OTHER
	note = thymine
modified_base	3191
	mod_base = OTHER
	note = thymine
modified_base	3194
	mod_base = OTHER
	note = thymine
modified_base	3196
	mod_base = OTHER
	note = thymine
modified_base	3198..3199

-continued

	mod_base = OTHER
	note = thymine
modified_base	3201
	mod_base = OTHER
	note = thymine
modified_base	3208..3209
	mod_base = OTHER
	note = thymine
modified_base	3213
	mod_base = OTHER
	note = thymine
modified_base	3220
	mod_base = OTHER
	note = thymine
modified_base	3222
	mod_base = OTHER
	note = thymine
modified_base	3235..3236
	mod_base = OTHER
	note = thymine
modified_base	3238
	mod_base = OTHER
	note = thymine
modified_base	3240
	mod_base = OTHER
	note = thymine
modified_base	3244
	mod_base = OTHER
	note = thymine
modified_base	3252
	mod_base = OTHER
	note = thymine
modified_base	3255
	mod_base = OTHER
	note = thymine
modified_base	3259..3260
	mod_base = OTHER
	note = thymine
modified_base	3264
	mod_base = OTHER
	note = thymine
modified_base	3270
	mod_base = OTHER
	note = thymine
modified_base	3273
	mod_base = OTHER
	note = thymine
modified_base	3276
	mod_base = OTHER
	note = thymine
modified_base	3285
	mod_base = OTHER
	note = thymine
modified_base	3287..3288
	mod_base = OTHER
	note = thymine
modified_base	3298
	mod_base = OTHER
	note = thymine
modified_base	3301
	mod_base = OTHER
	note = thymine
modified_base	3306
	mod_base = OTHER
	note = thymine
modified_base	3316..3317
	mod_base = OTHER
	note = thymine
modified_base	3323
	mod_base = OTHER
	note = thymine
modified_base	3329
	mod_base = OTHER
	note = thymine
modified_base	3331
	mod_base = OTHER
	note = thymine
modified_base	3333..3334
	mod_base = OTHER

-continued

modified_base	note = thymine 3338..3340 mod_base = OTHER note = thymine
modified_base	3344 mod_base = OTHER note = thymine
modified_base	3351 mod_base = OTHER note = thymine
modified_base	3356..3357 mod_base = OTHER note = thymine
modified_base	3359 mod_base = OTHER note = thymine
modified_base	3366..3367 mod_base = OTHER note = thymine
modified_base	3370..3371 mod_base = OTHER note = thymine
modified_base	3377..3378 mod_base = OTHER note = thymine
modified_base	3387..3388 mod_base = OTHER note = thymine
modified_base	3390 mod_base = OTHER note = thymine
modified_base	3396 mod_base = OTHER note = thymine
modified_base	3401..3403 mod_base = OTHER note = thymine
modified_base	3409 mod_base = OTHER note = thymine
modified_base	3411..3412 mod_base = OTHER note = thymine
modified_base	3415..3416 mod_base = OTHER note = thymine
modified_base	3419 mod_base = OTHER note = thymine
modified_base	3422..3423 mod_base = OTHER note = thymine
modified_base	3429..3430 mod_base = OTHER note = thymine
modified_base	3435..3437 mod_base = OTHER note = thymine
modified_base	3439..3440 mod_base = OTHER note = thymine
modified_base	3442 mod_base = OTHER note = thymine
modified_base	3444 mod_base = OTHER note = thymine
modified_base	3446 mod_base = OTHER note = thymine
modified_base	3448 mod_base = OTHER note = thymine
modified_base	3453..3454 mod_base = OTHER note = thymine
modified_base	3456..3457 mod_base = OTHER note = thymine

-continued

modified_base	3461..3462
	mod_base = OTHER
	note = thymine
modified_base	3464
	mod_base = OTHER
	note = thymine
modified_base	3467
	mod_base = OTHER
	note = thymine
modified_base	3469
	mod_base = OTHER
	note = thymine
modified_base	3471
	mod_base = OTHER
	note = thymine
modified_base	3491
	mod_base = OTHER
	note = thymine
modified_base	3494
	mod_base = OTHER
	note = thymine
modified_base	3499
	mod_base = OTHER
	note = thymine
modified_base	3502
	mod_base = OTHER
	note = thymine
modified_base	3505..3506
	mod_base = OTHER
	note = thymine
modified_base	3509
	mod_base = OTHER
	note = thymine
modified_base	3515..3517
	mod_base = OTHER
	note = thymine
modified_base	3522
	mod_base = OTHER
	note = thymine
modified_base	3525
	mod_base = OTHER
	note = thymine
modified_base	3528
	mod_base = OTHER
	note = thymine
modified_base	3530
	mod_base = OTHER
	note = thymine
modified_base	3543
	mod_base = OTHER
	note = thymine
modified_base	3549..3550
	mod_base = OTHER
	note = thymine
modified_base	3552..3553
	mod_base = OTHER
	note = thymine
modified_base	3562..3565
	mod_base = OTHER
	note = thymine
modified_base	3568
	mod_base = OTHER
	note = thymine
modified_base	3570
	mod_base = OTHER
	note = thymine
modified_base	3577..3578
	mod_base = OTHER
	note = thymine
modified_base	3584..3585
	mod_base = OTHER
	note = thymine
modified_base	3590..3591
	mod_base = OTHER
	note = thymine
modified_base	3594..3597
	mod_base = OTHER
	note = thymine
modified_base	3599

-continued

	mod_base = OTHER
	note = thymine
modified_base	3603..3604
	mod_base = OTHER
	note = thymine
modified_base	3611
	mod_base = OTHER
	note = thymine
modified_base	3613
	mod_base = OTHER
	note = thymine
modified_base	3615
	mod_base = OTHER
	note = thymine
modified_base	3617..3619
	mod_base = OTHER
	note = thymine
modified_base	3622..3623
	mod_base = OTHER
	note = thymine
modified_base	3628
	mod_base = OTHER
	note = thymine
modified_base	3638..3639
	mod_base = OTHER
	note = thymine
modified_base	3641
	mod_base = OTHER
	note = thymine
modified_base	3648
	mod_base = OTHER
	note = thymine
modified_base	3650..3651
	mod_base = OTHER
	note = thymine
modified_base	3656
	mod_base = OTHER
	note = thymine
modified_base	3661
	mod_base = OTHER
	note = thymine
modified_base	3664
	mod_base = OTHER
	note = thymine
modified_base	3667
	mod_base = OTHER
	note = thymine
modified_base	3672..3673
	mod_base = OTHER
	note = thymine
modified_base	3676
	mod_base = OTHER
	note = thymine
modified_base	3678..3679
	mod_base = OTHER
	note = thymine
modified_base	3684
	mod_base = OTHER
	note = thymine
modified_base	3686
	mod_base = OTHER
	note = thymine
modified_base	3693
	mod_base = OTHER
	note = thymine
modified_base	3698..3699
	mod_base = OTHER
	note = thymine
modified_base	3701
	mod_base = OTHER
	note = thymine
modified_base	3703
	mod_base = OTHER
	note = thymine
modified_base	3705..3707
	mod_base = OTHER
	note = thymine
modified_base	3717
	mod_base = OTHER

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ggattaccac	acttttatgt	cttgacagat	ccagagctgt	gctagtgtatg	ttaggctctg	4680
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agaaggacaa	aactctctta	cacactcctg	gaatttgagt	atctgaggaa	gtcttagcct	4800
ctttggaaaa	gaatctgttc	aactctctta	tcatgggtgc	ctcttgttcc	agtggttagac	4860
tccacttag	aggggggttt	acaacaacac	aatcaaacct	gaatttgggc	tcaataaact	4920
tctcaaaaaca	cttgatttga	tctgtcaggc	gatcaggtgt	ctctttgggt	accaagtgc	4980

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ccccctcacc	caaaatttgc	gttcaaaagt	ccacagccat	cctctgtaac	agacactctg	5100
atccagacat	gcagctgcac	cttcaacttg	atcaccaatc	acacatgattg	atttgatttg	5160
catatgccat	gaagaaatct	cttagacctt	gtaaaaatgt	ctgggtctct	ttggaagggg	5220
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cattcccaaa	atatctagaa	tactctcttt	cattgatttaa	caatcgggag	gaaaaatgatg	5580
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gctcaccttc	ctaactagct	ctgaatggct	gatgtctaca	gattttctca	aattctctgt	5700
ttaatagtata	tctcatcact	ggctgtgtcag	aaacaagtgc	ctgagctaaa	atcatcaagc	5760
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ttagtttaat	cgctccaga	atatctagaa	gtctgtcatt	gactaacatt	aacatttgtc	5940
caacaactat	tccgcgatt	cttaacctta	caattgcattc	atcatcggtt	ttgaaaagat	6000
ccaagaatgaa	attgagttaa	actaaagtcca	gaacaagttaa	agttgttctc	tctgtgttga	6060
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gcacatcact	gtctttgttg	cttccctaata	agatcaagtc	atttataagc	ttagactttt	6300
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ctaatcccaa	gctctggttt	tttgtgtgtg	acaaactatg	gatcaatcgc	tggtattttt	6540
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ccaataggat	gagtgttttc	ctgtgagtt	tgaagccatc	cggaaatgact	tttgaagggg	6840
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cttccaacaa	cagagagctg	gggtgatgtt	gagataaaaa	gatgtccctt	tggtatgcta	7020
gctctctgtc	ttctggaaaa	tgccttctaa	taaggctttt	tatttcattt	actgattctt	7080
ccatgctcaa	tctccqctca	gqatccactg	tacq			7140

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modified_base                               22..25
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modified_base                               28..29
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modified_base                               34
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modified_base                               39..40
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modified_base                               46
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modified_base                               48..51
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modified_base                               53
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modified_base                               55..57
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modified base                               66

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modified_base	153..154
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modified_base	171..172
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modified_base	note = thymine 308..309 mod_base = OTHER
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modified_base	459

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modified_base	502
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modified_base	505
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modified_base	515..516
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modified_base	524
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modified_base	note = thymine 676 mod_base = OTHER
modified_base	note = thymine 679..681 mod_base = OTHER
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modified_base	note = thymine 692 mod_base = OTHER
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modified_base	768
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modified_base	785..786
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modified_base	795..796
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modified_base	819..820
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modified_base	854

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modified_base	note = thymine 1029 mod_base = OTHER
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modified_base	1137 mod_base = OTHER note = thymine
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modified_base	1143 mod_base = OTHER note = thymine
modified_base	1147..1148 mod_base = OTHER note = thymine
modified_base	1151 mod_base = OTHER note = thymine
modified_base	1156..1157 mod_base = OTHER note = thymine
modified_base	1166..1169 mod_base = OTHER note = thymine
modified_base	1172 mod_base = OTHER note = thymine
modified_base	1174 mod_base = OTHER note = thymine
modified_base	1176 mod_base = OTHER note = thymine
modified_base	1188..1190 mod_base = OTHER note = thymine
modified_base	1202 mod_base = OTHER note = thymine
modified_base	1205..1206 mod_base = OTHER note = thymine
modified_base	1214 mod_base = OTHER note = thymine
modified_base	1217 mod_base = OTHER note = thymine
modified_base	1220..1221 mod_base = OTHER note = thymine
modified_base	1224 mod_base = OTHER note = thymine
modified_base	1238 mod_base = OTHER note = thymine
modified_base	1240..1242 mod_base = OTHER note = thymine
modified_base	1248 mod_base = OTHER note = thymine
modified_base	1250 mod_base = OTHER note = thymine
modified_base	1252 mod_base = OTHER note = thymine
modified_base	1256..1257

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modified_base	1268
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modified_base	1272
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modified_base	1274..1275
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modified_base	1282
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modified_base	1286..1287
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	note = thymine
modified_base	1289..1290
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modified_base	1293..1295
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modified_base	1297
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modified_base	1305
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modified_base	1316
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	note = thymine
modified_base	1318..1319
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modified_base	1339
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modified_base	1342..1344
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	note = thymine
modified_base	1348..1350
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modified_base	1365..1367
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modified_base	1377
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	note = thymine
modified_base	1379..1380
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modified_base	1382..1383
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modified_base	1397..1398 mod_base = OTHER note = thymine
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modified_base	1414 mod_base = OTHER note = thymine
modified_base	1416 mod_base = OTHER note = thymine
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modified_base	1448 mod_base = OTHER note = thymine
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modified_base	1480 mod_base = OTHER note = thymine
modified_base	1483..1484 mod_base = OTHER note = thymine
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modified_base	1492 mod_base = OTHER note = thymine
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modified_base	1509 mod_base = OTHER note = thymine
modified_base	1524..1526 mod_base = OTHER note = thymine
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modified_base	1541 mod_base = OTHER note = thymine
modified_base	1548 mod_base = OTHER note = thymine

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modified_base	1557
	mod_base = OTHER
	note = thymine
modified_base	1576
	mod_base = OTHER
	note = thymine
modified_base	1585
	mod_base = OTHER
	note = thymine
modified_base	1588
	mod_base = OTHER
	note = thymine
modified_base	1590
	mod_base = OTHER
	note = thymine
modified_base	1601
	mod_base = OTHER
	note = thymine
modified_base	1610
	mod_base = OTHER
	note = thymine
modified_base	1621
	mod_base = OTHER
	note = thymine
modified_base	1632
	mod_base = OTHER
	note = thymine
modified_base	1635..1636
	mod_base = OTHER
	note = thymine
modified_base	1641
	mod_base = OTHER
	note = thymine
modified_base	1645
	mod_base = OTHER
	note = thymine
modified_base	1650
	mod_base = OTHER
	note = thymine
modified_base	1654..1655
	mod_base = OTHER
	note = thymine
modified_base	1673
	mod_base = OTHER
	note = thymine
modified_base	1677
	mod_base = OTHER
	note = thymine
modified_base	1680
	mod_base = OTHER
	note = thymine
modified_base	1683
	mod_base = OTHER
	note = thymine
modified_base	1689
	mod_base = OTHER
	note = thymine
modified_base	1691
	mod_base = OTHER
	note = thymine
modified_base	1695..1696
	mod_base = OTHER
	note = thymine
modified_base	1699
	mod_base = OTHER
	note = thymine
modified_base	1702
	mod_base = OTHER
	note = thymine
modified_base	1705
	mod_base = OTHER
	note = thymine
modified_base	1711..1712
	mod_base = OTHER
	note = thymine
modified_base	1717

-continued

	mod_base = OTHER
	note = thymine
modified_base	1719
	mod_base = OTHER
	note = thymine
modified_base	1725
	mod_base = OTHER
	note = thymine
modified_base	1729
	mod_base = OTHER
	note = thymine
modified_base	1736..1737
	mod_base = OTHER
	note = thymine
modified_base	1739
	mod_base = OTHER
	note = thymine
modified_base	1744
	mod_base = OTHER
	note = thymine
modified_base	1746
	mod_base = OTHER
	note = thymine
modified_base	1759
	mod_base = OTHER
	note = thymine
modified_base	1761
	mod_base = OTHER
	note = thymine
modified_base	1764..1765
	mod_base = OTHER
	note = thymine
modified_base	1769
	mod_base = OTHER
	note = thymine
modified_base	1780..1781
	mod_base = OTHER
	note = thymine
modified_base	1790
	mod_base = OTHER
	note = thymine
modified_base	1792
	mod_base = OTHER
	note = thymine
modified_base	1795..1798
	mod_base = OTHER
	note = thymine
modified_base	1802
	mod_base = OTHER
	note = thymine
modified_base	1804..1808
	mod_base = OTHER
	note = thymine
modified_base	1810..1811
	mod_base = OTHER
	note = thymine
modified_base	1813..1814
	mod_base = OTHER
	note = thymine
modified_base	1824
	mod_base = OTHER
	note = thymine
modified_base	1828..1829
	mod_base = OTHER
	note = thymine
modified_base	1831
	mod_base = OTHER
	note = thymine
modified_base	1834
	mod_base = OTHER
	note = thymine
modified_base	1836..1838
	mod_base = OTHER
	note = thymine
modified_base	1842
	mod_base = OTHER
	note = thymine
modified_base	1846
	mod_base = OTHER

-continued

modified_base	note = thymine 1852 mod_base = OTHER note = thymine
modified_base	1854..1856 mod_base = OTHER note = thymine
modified_base	1858 mod_base = OTHER note = thymine
modified_base	1867 mod_base = OTHER note = thymine
modified_base	1869..1870 mod_base = OTHER note = thymine
modified_base	1873 mod_base = OTHER note = thymine
modified_base	1879 mod_base = OTHER note = thymine
modified_base	1883 mod_base = OTHER note = thymine
modified_base	1887..1888 mod_base = OTHER note = thymine
modified_base	1890..1891 mod_base = OTHER note = thymine
modified_base	1896 mod_base = OTHER note = thymine
modified_base	1898 mod_base = OTHER note = thymine
modified_base	1903..1904 mod_base = OTHER note = thymine
modified_base	1909 mod_base = OTHER note = thymine
modified_base	1917..1919 mod_base = OTHER note = thymine
modified_base	1924 mod_base = OTHER note = thymine
modified_base	1926 mod_base = OTHER note = thymine
modified_base	1928..1929 mod_base = OTHER note = thymine
modified_base	1931..1932 mod_base = OTHER note = thymine
modified_base	1936 mod_base = OTHER note = thymine
modified_base	1942 mod_base = OTHER note = thymine
modified_base	1951..1952 mod_base = OTHER note = thymine
modified_base	1955 mod_base = OTHER note = thymine
modified_base	1958 mod_base = OTHER note = thymine
modified_base	1960 mod_base = OTHER note = thymine
modified_base	1963 mod_base = OTHER note = thymine

-continued

modified_base	1968
	mod_base = OTHER
	note = thymine
modified_base	1971..1972
	mod_base = OTHER
	note = thymine
modified_base	1976
	mod_base = OTHER
	note = thymine
modified_base	1979
	mod_base = OTHER
	note = thymine
modified_base	1988
	mod_base = OTHER
	note = thymine
modified_base	1990
	mod_base = OTHER
	note = thymine
modified_base	1992
	mod_base = OTHER
	note = thymine
modified_base	2009
	mod_base = OTHER
	note = thymine
modified_base	2014
	mod_base = OTHER
	note = thymine
modified_base	2018
	mod_base = OTHER
	note = thymine
modified_base	2025
	mod_base = OTHER
	note = thymine
modified_base	2028
	mod_base = OTHER
	note = thymine
modified_base	2033..2034
	mod_base = OTHER
	note = thymine
modified_base	2038
	mod_base = OTHER
	note = thymine
modified_base	2040..2041
	mod_base = OTHER
	note = thymine
modified_base	2043
	mod_base = OTHER
	note = thymine
modified_base	2045
	mod_base = OTHER
	note = thymine
modified_base	2047
	mod_base = OTHER
	note = thymine
modified_base	2049..2051
	mod_base = OTHER
	note = thymine
modified_base	2054..2055
	mod_base = OTHER
	note = thymine
modified_base	2060
	mod_base = OTHER
	note = thymine
modified_base	2065
	mod_base = OTHER
	note = thymine
modified_base	2071
	mod_base = OTHER
	note = thymine
modified_base	2073
	mod_base = OTHER
	note = thymine
modified_base	2075
	mod_base = OTHER
	note = thymine
modified_base	2078
	mod_base = OTHER
	note = thymine
modified_base	2083..2087

-continued

	mod_base = OTHER
	note = thymine
modified_base	2094..2096
	mod_base = OTHER
	note = thymine
modified_base	2098
	mod_base = OTHER
	note = thymine
modified_base	2101
	mod_base = OTHER
	note = thymine
modified_base	2104
	mod_base = OTHER
	note = thymine
modified_base	2106
	mod_base = OTHER
	note = thymine
modified_base	2109..2112
	mod_base = OTHER
	note = thymine
modified_base	2114
	mod_base = OTHER
	note = thymine
modified_base	2122
	mod_base = OTHER
	note = thymine
modified_base	2126
	mod_base = OTHER
	note = thymine
modified_base	2128
	mod_base = OTHER
	note = thymine
modified_base	2131
	mod_base = OTHER
	note = thymine
modified_base	2134..2135
	mod_base = OTHER
	note = thymine
modified_base	2139
	mod_base = OTHER
	note = thymine
modified_base	2146
	mod_base = OTHER
	note = thymine
modified_base	2159
	mod_base = OTHER
	note = thymine
modified_base	2161
	mod_base = OTHER
	note = thymine
modified_base	2170
	mod_base = OTHER
	note = thymine
modified_base	2174
	mod_base = OTHER
	note = thymine
modified_base	2178
	mod_base = OTHER
	note = thymine
modified_base	2181
	mod_base = OTHER
	note = thymine
modified_base	2184..2185
	mod_base = OTHER
	note = thymine
modified_base	2189
	mod_base = OTHER
	note = thymine
modified_base	2191
	mod_base = OTHER
	note = thymine
modified_base	2204
	mod_base = OTHER
	note = thymine
modified_base	2206..2207
	mod_base = OTHER
	note = thymine
modified_base	2210
	mod_base = OTHER

-continued

modified_base	note = thymine 2215 mod_base = OTHER
modified_base	note = thymine 2220..2222 mod_base = OTHER
modified_base	note = thymine 2229 mod_base = OTHER
modified_base	note = thymine 2233 mod_base = OTHER
modified_base	note = thymine 2235 mod_base = OTHER
modified_base	note = thymine 2237 mod_base = OTHER
modified_base	note = thymine 2240 mod_base = OTHER
modified_base	note = thymine 2247..2248 mod_base = OTHER
modified_base	note = thymine 2250..2251 mod_base = OTHER
modified_base	note = thymine 2253 mod_base = OTHER
modified_base	note = thymine 2255 mod_base = OTHER
modified_base	note = thymine 2257 mod_base = OTHER
modified_base	note = thymine 2259 mod_base = OTHER
modified_base	note = thymine 2261 mod_base = OTHER
modified_base	note = thymine 2266..2267 mod_base = OTHER
modified_base	note = thymine 2274 mod_base = OTHER
modified_base	note = thymine 2276 mod_base = OTHER
modified_base	note = thymine 2281..2282 mod_base = OTHER
modified_base	note = thymine 2284 mod_base = OTHER
modified_base	note = thymine 2290..2291 mod_base = OTHER
modified_base	note = thymine 2293..2296 mod_base = OTHER
modified_base	note = thymine 2301..2302 mod_base = OTHER
modified_base	note = thymine 2304 mod_base = OTHER
modified_base	note = thymine 2306 mod_base = OTHER
modified_base	note = thymine 2310 mod_base = OTHER
modified_base	note = thymine 2319 mod_base = OTHER
	note = thymine

-continued

modified_base	2322..2324
	mod_base = OTHER
	note = thymine
modified_base	2327..2328
	mod_base = OTHER
	note = thymine
modified_base	2330
	mod_base = OTHER
	note = thymine
modified_base	2335
	mod_base = OTHER
	note = thymine
modified_base	2339
	mod_base = OTHER
	note = thymine
modified_base	2342..2343
	mod_base = OTHER
	note = thymine
modified_base	2345
	mod_base = OTHER
	note = thymine
modified_base	2347..2348
	mod_base = OTHER
	note = thymine
modified_base	2350
	mod_base = OTHER
	note = thymine
modified_base	2355
	mod_base = OTHER
	note = thymine
modified_base	2360
	mod_base = OTHER
	note = thymine
modified_base	2364
	mod_base = OTHER
	note = thymine
modified_base	2366
	mod_base = OTHER
	note = thymine
modified_base	2369..2371
	mod_base = OTHER
	note = thymine
modified_base	2373
	mod_base = OTHER
	note = thymine
modified_base	2376
	mod_base = OTHER
	note = thymine
modified_base	2378
	mod_base = OTHER
	note = thymine
modified_base	2387
	mod_base = OTHER
	note = thymine
modified_base	2389
	mod_base = OTHER
	note = thymine
modified_base	2401
	mod_base = OTHER
	note = thymine
modified_base	2418..2420
	mod_base = OTHER
	note = thymine
modified_base	2422
	mod_base = OTHER
	note = thymine
modified_base	2431..2434
	mod_base = OTHER
	note = thymine
modified_base	2437
	mod_base = OTHER
	note = thymine
modified_base	2443..2445
	mod_base = OTHER
	note = thymine
modified_base	2447
	mod_base = OTHER
	note = thymine
modified_base	2449..2450

-continued

	mod_base = OTHER
	note = thymine
modified_base	2454
	mod_base = OTHER
	note = thymine
modified_base	2457..2459
	mod_base = OTHER
	note = thymine
modified_base	2461
	mod_base = OTHER
	note = thymine
modified_base	2464
	mod_base = OTHER
	note = thymine
modified_base	2468
	mod_base = OTHER
	note = thymine
modified_base	2474
	mod_base = OTHER
	note = thymine
modified_base	2478..2479
	mod_base = OTHER
	note = thymine
modified_base	2481
	mod_base = OTHER
	note = thymine
modified_base	2483..2484
	mod_base = OTHER
	note = thymine
modified_base	2487..2489
	mod_base = OTHER
	note = thymine
modified_base	2494
	mod_base = OTHER
	note = thymine
modified_base	2498
	mod_base = OTHER
	note = thymine
modified_base	2501..2502
	mod_base = OTHER
	note = thymine
modified_base	2505..2508
	mod_base = OTHER
	note = thymine
modified_base	2510
	mod_base = OTHER
	note = thymine
modified_base	2524..2525
	mod_base = OTHER
	note = thymine
modified_base	2527
	mod_base = OTHER
	note = thymine
modified_base	2529
	mod_base = OTHER
	note = thymine
modified_base	2534
	mod_base = OTHER
	note = thymine
modified_base	2538
	mod_base = OTHER
	note = thymine
modified_base	2540
	mod_base = OTHER
	note = thymine
modified_base	2543..2544
	mod_base = OTHER
	note = thymine
modified_base	2546
	mod_base = OTHER
	note = thymine
modified_base	2548
	mod_base = OTHER
	note = thymine
modified_base	2555
	mod_base = OTHER
	note = thymine
modified_base	2557..2559
	mod_base = OTHER

-continued

modified_base	note = thymine 2566 mod_base = OTHER note = thymine
modified_base	2573 mod_base = OTHER note = thymine
modified_base	2577 mod_base = OTHER note = thymine
modified_base	2587..2589 mod_base = OTHER note = thymine
modified_base	2592 mod_base = OTHER note = thymine
modified_base	2598 mod_base = OTHER note = thymine
modified_base	2606 mod_base = OTHER note = thymine
modified_base	2611..2612 mod_base = OTHER note = thymine
modified_base	2614 mod_base = OTHER note = thymine
modified_base	2624 mod_base = OTHER note = thymine
modified_base	2626..2627 mod_base = OTHER note = thymine
modified_base	2630 mod_base = OTHER note = thymine
modified_base	2636..2638 mod_base = OTHER note = thymine
modified_base	2641 mod_base = OTHER note = thymine
modified_base	2644..2645 mod_base = OTHER note = thymine
modified_base	2648 mod_base = OTHER note = thymine
modified_base	2654 mod_base = OTHER note = thymine
modified_base	2656 mod_base = OTHER note = thymine
modified_base	2665 mod_base = OTHER note = thymine
modified_base	2668 mod_base = OTHER note = thymine
modified_base	2670..2671 mod_base = OTHER note = thymine
modified_base	2674 mod_base = OTHER note = thymine
modified_base	2678 mod_base = OTHER note = thymine
modified_base	2682 mod_base = OTHER note = thymine
modified_base	2684 mod_base = OTHER note = thymine
modified_base	2686 mod_base = OTHER note = thymine

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modified_base	2692
	mod_base = OTHER
	note = thymine
modified_base	2696
	mod_base = OTHER
	note = thymine
modified_base	2701..2703
	mod_base = OTHER
	note = thymine
modified_base	2707
	mod_base = OTHER
	note = thymine
modified_base	2710..2711
	mod_base = OTHER
	note = thymine
modified_base	2717
	mod_base = OTHER
	note = thymine
modified_base	2719
	mod_base = OTHER
	note = thymine
modified_base	2721
	mod_base = OTHER
	note = thymine
modified_base	2738..2739
	mod_base = OTHER
	note = thymine
modified_base	2746
	mod_base = OTHER
	note = thymine
modified_base	2748
	mod_base = OTHER
	note = thymine
modified_base	2750
	mod_base = OTHER
	note = thymine
modified_base	2756
	mod_base = OTHER
	note = thymine
modified_base	2758..2759
	mod_base = OTHER
	note = thymine
modified_base	2762..2763
	mod_base = OTHER
	note = thymine
modified_base	2765
	mod_base = OTHER
	note = thymine
modified_base	2767
	mod_base = OTHER
	note = thymine
modified_base	2772
	mod_base = OTHER
	note = thymine
modified_base	2775..2776
	mod_base = OTHER
	note = thymine
modified_base	2783
	mod_base = OTHER
	note = thymine
modified_base	2786
	mod_base = OTHER
	note = thymine
modified_base	2790
	mod_base = OTHER
	note = thymine
modified_base	2795..2796
	mod_base = OTHER
	note = thymine
modified_base	2798
	mod_base = OTHER
	note = thymine
modified_base	2802
	mod_base = OTHER
	note = thymine
modified_base	2821
	mod_base = OTHER
	note = thymine
modified_base	2824..2825

-continued

	mod_base = OTHER
	note = thymine
modified_base	2827..2828
	mod_base = OTHER
	note = thymine
modified_base	2832..2835
	mod_base = OTHER
	note = thymine
modified_base	2842
	mod_base = OTHER
	note = thymine
modified_base	2844..2846
	mod_base = OTHER
	note = thymine
modified_base	2851
	mod_base = OTHER
	note = thymine
modified_base	2856
	mod_base = OTHER
	note = thymine
modified_base	2861
	mod_base = OTHER
	note = thymine
modified_base	2872..2874
	mod_base = OTHER
	note = thymine
modified_base	2882..2883
	mod_base = OTHER
	note = thymine
modified_base	2886..2887
	mod_base = OTHER
	note = thymine
modified_base	2890
	mod_base = OTHER
	note = thymine
modified_base	2892..2893
	mod_base = OTHER
	note = thymine
modified_base	2898
	mod_base = OTHER
	note = thymine
modified_base	2904
	mod_base = OTHER
	note = thymine
modified_base	2906..2907
	mod_base = OTHER
	note = thymine
modified_base	2909
	mod_base = OTHER
	note = thymine
modified_base	2913
	mod_base = OTHER
	note = thymine
modified_base	2915
	mod_base = OTHER
	note = thymine
modified_base	2917
	mod_base = OTHER
	note = thymine
modified_base	2919
	mod_base = OTHER
	note = thymine
modified_base	2922
	mod_base = OTHER
	note = thymine
modified_base	2924
	mod_base = OTHER
	note = thymine
modified_base	2926..2927
	mod_base = OTHER
	note = thymine
modified_base	2934..2935
	mod_base = OTHER
	note = thymine
modified_base	2937
	mod_base = OTHER
	note = thymine
modified_base	2940..2941
	mod_base = OTHER

-continued

modified_base	note = thymine 2950..2951 mod_base = OTHER note = thymine
modified_base	2957 mod_base = OTHER note = thymine
modified_base	2959 mod_base = OTHER note = thymine
modified_base	2967..2968 mod_base = OTHER note = thymine
modified_base	2976 mod_base = OTHER note = thymine
modified_base	2978 mod_base = OTHER note = thymine
modified_base	2980 mod_base = OTHER note = thymine
modified_base	2982 mod_base = OTHER note = thymine
modified_base	2985 mod_base = OTHER note = thymine
modified_base	2987..2988 mod_base = OTHER note = thymine
modified_base	2990 mod_base = OTHER note = thymine
modified_base	2993 mod_base = OTHER note = thymine
modified_base	2995..2996 mod_base = OTHER note = thymine
modified_base	2998..3003 mod_base = OTHER note = thymine
modified_base	3006..3008 mod_base = OTHER note = thymine
modified_base	3010 mod_base = OTHER note = thymine
modified_base	3016 mod_base = OTHER note = thymine
modified_base	3028 mod_base = OTHER note = thymine
modified_base	3032 mod_base = OTHER note = thymine
modified_base	3036..3037 mod_base = OTHER note = thymine
modified_base	3040 mod_base = OTHER note = thymine
modified_base	3044 mod_base = OTHER note = thymine
modified_base	3052 mod_base = OTHER note = thymine
modified_base	3054 mod_base = OTHER note = thymine
modified_base	3061..3062 mod_base = OTHER note = thymine
modified_base	3069 mod_base = OTHER note = thymine

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modified_base	3071
	mod_base = OTHER
	note = thymine
modified_base	3073..3075
	mod_base = OTHER
	note = thymine
modified_base	3078..3079
	mod_base = OTHER
	note = thymine
modified_base	3086
	mod_base = OTHER
	note = thymine
modified_base	3089
	mod_base = OTHER
	note = thymine
modified_base	3092
	mod_base = OTHER
	note = thymine
modified_base	3098
	mod_base = OTHER
	note = thymine
modified_base	3102
	mod_base = OTHER
	note = thymine
modified_base	3104
	mod_base = OTHER
	note = thymine
modified_base	3106
	mod_base = OTHER
	note = thymine
modified_base	3112
	mod_base = OTHER
	note = thymine
modified_base	3114..3116
	mod_base = OTHER
	note = thymine
modified_base	3118..3119
	mod_base = OTHER
	note = thymine
modified_base	3124
	mod_base = OTHER
	note = thymine
modified_base	3128
	mod_base = OTHER
	note = thymine
modified_base	3133..3134
	mod_base = OTHER
	note = thymine
modified_base	3136..3137
	mod_base = OTHER
	note = thymine
modified_base	3142
	mod_base = OTHER
	note = thymine
modified_base	3144..3145
	mod_base = OTHER
	note = thymine
modified_base	3147..3148
	mod_base = OTHER
	note = thymine
modified_base	3155..3158
	mod_base = OTHER
	note = thymine
modified_base	3161
	mod_base = OTHER
	note = thymine
modified_base	3163..3165
	mod_base = OTHER
	note = thymine
modified_base	3167
	mod_base = OTHER
	note = thymine
modified_base	3171
	mod_base = OTHER
	note = thymine
modified_base	3178
	mod_base = OTHER
	note = thymine
modified_base	3184

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	mod_base = OTHER
	note = thymine
modified_base	3186
	mod_base = OTHER
	note = thymine
modified_base	3190
	mod_base = OTHER
	note = thymine
modified_base	3192..3193
	mod_base = OTHER
	note = thymine
modified_base	3196..3197
	mod_base = OTHER
	note = thymine
modified_base	3202
	mod_base = OTHER
	note = thymine
modified_base	3206
	mod_base = OTHER
	note = thymine
modified_base	3209
	mod_base = OTHER
	note = thymine
modified_base	3211
	mod_base = OTHER
	note = thymine
modified_base	3218..3219
	mod_base = OTHER
	note = thymine
modified_base	3223..3224
	mod_base = OTHER
	note = thymine
modified_base	3229
	mod_base = OTHER
	note = thymine
modified_base	3232..3233
	mod_base = OTHER
	note = thymine
modified_base	3241
	mod_base = OTHER
	note = thymine
modified_base	3247..3248
	mod_base = OTHER
	note = thymine
modified_base	3254
	mod_base = OTHER
	note = thymine
modified_base	3256
	mod_base = OTHER
	note = thymine
modified_base	3260
	mod_base = OTHER
	note = thymine
modified_base	3264..3265
	mod_base = OTHER
	note = thymine
modified_base	3269
	mod_base = OTHER
	note = thymine
modified_base	3276
	mod_base = OTHER
	note = thymine
modified_base	3278
	mod_base = OTHER
	note = thymine
modified_base	3281..3282
	mod_base = OTHER
	note = thymine
modified_base	3289
	mod_base = OTHER
	note = thymine
modified_base	3293
	mod_base = OTHER
	note = thymine
modified_base	3297
	mod_base = OTHER
	note = thymine
modified_base	3299
	mod_base = OTHER

-continued

modified_base	note = thymine 3305..3306 mod_base = OTHER note = thymine
modified_base	3313 mod_base = OTHER note = thymine
modified_base	3315..3317 mod_base = OTHER note = thymine
modified_base	3322 mod_base = OTHER note = thymine
modified_base	3324 mod_base = OTHER note = thymine
modified_base	3329 mod_base = OTHER note = thymine
modified_base	3338..3339 mod_base = OTHER note = thymine
modified_base	3341 mod_base = OTHER note = thymine
modified_base	3344 mod_base = OTHER note = thymine
modified_base	3347..3350 mod_base = OTHER note = thymine
modified_base	3353 mod_base = OTHER note = thymine
modified_base	3355 mod_base = OTHER note = thymine
modified_base	3360 mod_base = OTHER note = thymine
modified_base	3365..3366 mod_base = OTHER note = thymine
modified_base	3369 mod_base = OTHER note = thymine
modified_base	3373 mod_base = OTHER note = thymine
modified_base	3379 mod_base = OTHER note = thymine
modified_base	3381..3382 mod_base = OTHER note = thymine
modified_base	3389 mod_base = OTHER note = thymine
modified_base	3391 mod_base = OTHER note = thymine
modified_base	3395 mod_base = OTHER note = thymine
modified_base	3400 mod_base = OTHER note = thymine
modified_base	3405 mod_base = OTHER note = thymine
modified_base	3407 mod_base = OTHER note = thymine
misc_feature	1..3410 note = Junin virus strain Candid #1 segment S, completesequences-ACCESSION FJ969442-VERSION FJ969442.1-GI:264665554 - Viruses; ssRNA viruses; ssRNA negative-strand viruses; Arenaviridae; Mammarenavirus.
source	1..3410

-continued

mol_type = other RNA							
organism = unidentified							
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aacctttttg	caggaggctc	tgaacattgc	ttctgttgca	ctgagttcca	ttgccatcat	180	
taagggtgta	gtgaacctgt	acaaaagtgg	tttgttccaa	ttcttttgat	tcctagcact	240	
cgtaggaaga	tctgtccagc	aaagaagcttt	taaaatcgga	ctgcacacag	agttccgacag	300	
ttgtgtcttc	tcaatggtgt	gtgtcttttc	caacaatcca	ctgacactgc	ctctgtttgtg	360	
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gggacatgat	tggcatctag	accaccattt	tctgtgtagg	aaccgtgcaa	agacagaagg	600	
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gtgcttaaat	aaggcacaca	ctaccagttg	gcctctccaa	tgctccactg	accacgttaa	780	
cacattacac	ttccttacc	gaggtaaaaa	cttcaacttt	ccaaggagct	ccttgaagac	840	
attcttctcc	tggtctttga	cagactcacc	cggcaaggat	accctctggg	gctatttgtct	900	
agaagtagtg	atgtctgttc	cagccaaaaa	gaagtgcttt	ggcaattctt	ctgtagcaaaa	960	
atgcaatctg	aatacatgct	ctgaaattct	tgacatgctg	aggcttttgc	attacacaaa	1020	
aaatgctatc	aaaaccttaa	atgatgaaac	taagaaacaa	gtaaatctga	tgggacagac	1080	
aatgaactgc	tgtatatctg	acaaatttat	gatgaaaaac	aaaattaggg	aactgtatgag	1140	
tgctccttat	ctgcaattaca	caaaattttg	gtatgtcaac	cacacacttt	caggacaaca	1200	
ctcattacca	agggtctggt	taataaaaaa	caacagctat	tgaacatct	ctgacttcgc	1260	
taatgatcgg	atatttagaaa	gtgactcttt	aattttctgaa	atgtctaagca	gcagagtattc	1320	
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cgaagcatcg	ctcttggccc	acaggtttgaa	cagctctgggt	gggttgcagct	tgggttaagta	1500	
ccccactcta	aagtaaaccaa	cagtttggcg	tagaggacac	tgaagactcc	tgaaggtctcc	1560	
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aattgagggt	tagggttcac	ctcatagagc	ttccactcta	gcactgactg	aaacattatgt	1740	
cagtcattga	gagcacagtg	tggttctactg	gagggccaact	tgaagggaat	atctctttcc	1800	
ctcttttttc	tattgacaac	accctcatgt	tgtatattgc	ataagtgacc	atattttctc	1860	
cagacctgtt	gatcaaaact	cctggcttgt	tcagatgtga	gcttaaacat	aaccagttta	1920	
agatctcttc	tctcatagag	gtcacaacaac	ttctcatgct	catcgagatc	ttgagtagtc	1980	
acaaccatgt	cttgaggcag	caagccgatc	tcgttaactaa	gaactctcgg	catgtgactt	2040	
tctatgtctt	tcattaagat	gccgtgagag	tgtctgctac	cattttttaa	ccctttctca	2100	
tcagtgtggt	ttctgaagca	gtcgaatatc	tgctttacctg	agcgttggaa	caacgccatc	2160	
tcaacagggt	cagtagactg	ttcttcaatg	tcagaccaaa	gggtatgtgt	ggggtcgagt	2220	
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ctctgtgtgt	cccatgacct	gcctatgatt	tgtatcttgg	aaccaataata	agggccaaca	2400	
tcgccagaaa	ggcaaaagt	gtacagaagg	ttttcataag	ggtttctatt	gctctggttt	2460	
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aagttatact	gggtgactct	gattgtctcc	acaatatttc	accacataag	aatggaatga	2580	
ccagctttta	ctgcaagctg	aagactaaa	ttgtagccag	aaattgtgat	ggaagcttca	2640	
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tttaattctc	caaggtcaga	gcaccaactc	atcagttcat	ctctcccacg	atctcccacc	3060	
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acctgtgcct	cttgggttga	gtcaatgctg	tcagcaataa	ctctggcacc	cttcagaaca	3240	
tccgacttga	cagtcctgagt	aaattgactc	aaacctctcc	ttaaggactg	agtcacatcta	3300	
aagcttgaaa	actgttttga	gtgtgccagt	ccagaagttc	tgggtgatatt			

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SEQ ID NO: 4                               multype = RNA   length = 7114
FEATURE                                     Location/Qualifiers
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                                           note = thymine
modified_base                               12
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                                           note = thymine
modified_base                               15
                                           mod_base = OTHER
                                           note = thymine
modified_base                               21
                                           mod_base = OTHER
                                           note = thymine
modified_base                               25..26
                                           mod_base = OTHER
                                           note = thymine
modified base                               29

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-continued

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	note = thymine
modified_base	32..33
	mod_base = OTHER
	note = thymine
modified_base	38
	mod_base = OTHER
	note = thymine
modified_base	40
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	note = thymine
modified_base	45..46
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modified_base	48
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modified_base	51
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modified_base	53
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modified_base	67
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modified_base	85
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modified_base	93
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modified_base	105
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modified_base	111
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modified_base	113
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	note = thymine
modified_base	126
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	note = thymine
modified_base	129
	mod_base = OTHER
	note = thymine
modified_base	156..158
	mod_base = OTHER
	note = thymine
modified_base	166
	mod_base = OTHER
	note = thymine
modified_base	170
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modified_base	185
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	note = thymine
modified_base	188
	mod_base = OTHER

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modified_base	194 mod_base = OTHER note = thymine
modified_base	198 mod_base = OTHER note = thymine
modified_base	200 mod_base = OTHER note = thymine
modified_base	204 mod_base = OTHER note = thymine
modified_base	207 mod_base = OTHER note = thymine
modified_base	210 mod_base = OTHER note = thymine
modified_base	213..215 mod_base = OTHER note = thymine
modified_base	218 mod_base = OTHER note = thymine
modified_base	221 mod_base = OTHER note = thymine
modified_base	227..229 mod_base = OTHER note = thymine
modified_base	232 mod_base = OTHER note = thymine
modified_base	237 mod_base = OTHER note = thymine
modified_base	239 mod_base = OTHER note = thymine
modified_base	242 mod_base = OTHER note = thymine
modified_base	245 mod_base = OTHER note = thymine
modified_base	249 mod_base = OTHER note = thymine
modified_base	253..255 mod_base = OTHER note = thymine
modified_base	257..259 mod_base = OTHER note = thymine
modified_base	264 mod_base = OTHER note = thymine
modified_base	269 mod_base = OTHER note = thymine
modified_base	275 mod_base = OTHER note = thymine
modified_base	277 mod_base = OTHER note = thymine
modified_base	279..280 mod_base = OTHER note = thymine
modified_base	287..288 mod_base = OTHER note = thymine
modified_base	293 mod_base = OTHER note = thymine

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modified_base	295
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modified_base	297
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	note = thymine
modified_base	302
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modified_base	304
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modified_base	471
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	note = thymine
modified_base	476..478
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modified_base	480
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	note = thymine
modified_base	483..484

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modified_base	488
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modified_base	493
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modified_base	497
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modified_base	499
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	note = thymine
modified_base	506..508
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modified_base	512
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	note = thymine
modified_base	514
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modified_base	519
	mod_base = OTHER
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modified_base	536
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modified_base	538
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modified_base	546
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	note = thymine
modified_base	549..551
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modified_base	557
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modified_base	559
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	note = thymine
modified_base	562
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	note = thymine
modified_base	566
	mod_base = OTHER
	note = thymine
modified_base	573..574
	mod_base = OTHER
	note = thymine
modified_base	579..581
	mod_base = OTHER
	note = thymine
modified_base	587..588
	mod_base = OTHER
	note = thymine
modified_base	597
	mod_base = OTHER
	note = thymine
modified_base	605..607
	mod_base = OTHER
	note = thymine
modified_base	609
	mod_base = OTHER
	note = thymine
modified_base	614
	mod_base = OTHER

-continued

modified_base	note = thymine 622 mod_base = OTHER note = thymine
modified_base	627 mod_base = OTHER note = thymine
modified_base	630 mod_base = OTHER note = thymine
modified_base	632..634 mod_base = OTHER note = thymine
modified_base	636 mod_base = OTHER note = thymine
modified_base	638 mod_base = OTHER note = thymine
modified_base	642 mod_base = OTHER note = thymine
modified_base	645 mod_base = OTHER note = thymine
modified_base	660 mod_base = OTHER note = thymine
modified_base	663 mod_base = OTHER note = thymine
modified_base	665..666 mod_base = OTHER note = thymine
modified_base	669 mod_base = OTHER note = thymine
modified_base	674 mod_base = OTHER note = thymine
modified_base	677 mod_base = OTHER note = thymine
modified_base	679..680 mod_base = OTHER note = thymine
modified_base	682..684 mod_base = OTHER note = thymine
modified_base	686 mod_base = OTHER note = thymine
modified_base	689 mod_base = OTHER note = thymine
modified_base	691..692 mod_base = OTHER note = thymine
modified_base	694..698 mod_base = OTHER note = thymine
modified_base	702 mod_base = OTHER note = thymine
modified_base	710 mod_base = OTHER note = thymine
modified_base	713..714 mod_base = OTHER note = thymine
modified_base	718..719 mod_base = OTHER note = thymine
modified_base	722 mod_base = OTHER note = thymine
modified_base	727..729 mod_base = OTHER note = thymine

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modified_base	738
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	note = thymine
modified_base	742
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	note = thymine
modified_base	744
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modified_base	755
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	note = thymine
modified_base	760..761
	mod_base = OTHER
	note = thymine
modified_base	765..766
	mod_base = OTHER
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modified_base	770
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modified_base	772
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	note = thymine
modified_base	777
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	note = thymine
modified_base	794
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modified_base	796
	mod_base = OTHER
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modified_base	798
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modified_base	800
	mod_base = OTHER
	note = thymine
modified_base	802..803
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	note = thymine
modified_base	811..813
	mod_base = OTHER
	note = thymine
modified_base	816..817
	mod_base = OTHER
	note = thymine
modified_base	821
	mod_base = OTHER
	note = thymine
modified_base	823
	mod_base = OTHER
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modified_base	825
	mod_base = OTHER
	note = thymine
modified_base	832
	mod_base = OTHER
	note = thymine
modified_base	837
	mod_base = OTHER
	note = thymine
modified_base	839..841

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modified_base	854..855
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	note = thymine
modified_base	857
	mod_base = OTHER
	note = thymine
modified_base	860
	mod_base = OTHER
	note = thymine
modified_base	864..865
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modified_base	867
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modified_base	869
	mod_base = OTHER
	note = thymine
modified_base	877
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modified_base	891
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	note = thymine
modified_base	894..895
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modified_base	897..898
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modified_base	901..902
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modified_base	905..906
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modified_base	912
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	note = thymine
modified_base	933..934
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modified_base	939
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modified_base	943
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	note = thymine
modified_base	949
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modified_base	951..952
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	note = thymine
modified_base	954
	mod_base = OTHER
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modified_base	957
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modified_base	960..962
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modified_base	note = thymine 1014 mod_base = OTHER
modified_base	note = thymine 1019 mod_base = OTHER
modified_base	note = thymine 1021 mod_base = OTHER
modified_base	note = thymine 1028 mod_base = OTHER
modified_base	note = thymine 1031 mod_base = OTHER
modified_base	note = thymine 1036 mod_base = OTHER
modified_base	note = thymine 1042 mod_base = OTHER
modified_base	note = thymine 1049 mod_base = OTHER
modified_base	note = thymine 1056 mod_base = OTHER
modified_base	note = thymine 1064 mod_base = OTHER
modified_base	note = thymine 1068 mod_base = OTHER
modified_base	note = thymine 1072..1073 mod_base = OTHER
modified_base	note = thymine 1077 mod_base = OTHER
modified_base	note = thymine 1094..1095 mod_base = OTHER
modified_base	note = thymine 1099 mod_base = OTHER
modified_base	note = thymine 1102..1103 mod_base = OTHER
modified_base	note = thymine 1107 mod_base = OTHER
	note = thymine

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modified_base	1119
	mod_base = OTHER
	note = thymine
modified_base	1121
	mod_base = OTHER
	note = thymine
modified_base	1124..1125
	mod_base = OTHER
	note = thymine
modified_base	1132
	mod_base = OTHER
	note = thymine
modified_base	1140
	mod_base = OTHER
	note = thymine
modified_base	1143
	mod_base = OTHER
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modified_base	1165
	mod_base = OTHER
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modified_base	1170
	mod_base = OTHER
	note = thymine
modified_base	1174
	mod_base = OTHER
	note = thymine
modified_base	1176
	mod_base = OTHER
	note = thymine
modified_base	1183
	mod_base = OTHER
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modified_base	1185..1187
	mod_base = OTHER
	note = thymine
modified_base	1189..1190
	mod_base = OTHER
	note = thymine
modified_base	1192
	mod_base = OTHER
	note = thymine
modified_base	1196..1198
	mod_base = OTHER
	note = thymine
modified_base	1202..1205
	mod_base = OTHER
	note = thymine
modified_base	1207
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	note = thymine
modified_base	1214
	mod_base = OTHER
	note = thymine
modified_base	1217
	mod_base = OTHER
	note = thymine
modified_base	1219
	mod_base = OTHER
	note = thymine
modified_base	1221

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modified_base	1223..1224
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	note = thymine
modified_base	1226
	mod_base = OTHER
	note = thymine
modified_base	1229..1230
	mod_base = OTHER
	note = thymine
modified_base	1234
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	note = thymine
modified_base	1237
	mod_base = OTHER
	note = thymine
modified_base	1241..1244
	mod_base = OTHER
	note = thymine
modified_base	1247..1249
	mod_base = OTHER
	note = thymine
modified_base	1256..1257
	mod_base = OTHER
	note = thymine
modified_base	1259..1260
	mod_base = OTHER
	note = thymine
modified_base	1272..1273
	mod_base = OTHER
	note = thymine
modified_base	1285
	mod_base = OTHER
	note = thymine
modified_base	1287
	mod_base = OTHER
	note = thymine
modified_base	1293
	mod_base = OTHER
	note = thymine
modified_base	1297
	mod_base = OTHER
	note = thymine
modified_base	1300
	mod_base = OTHER
	note = thymine
modified_base	1302
	mod_base = OTHER
	note = thymine
modified_base	1304..1305
	mod_base = OTHER
	note = thymine
modified_base	1308
	mod_base = OTHER
	note = thymine
modified_base	1324
	mod_base = OTHER
	note = thymine
modified_base	1326..1327
	mod_base = OTHER
	note = thymine
modified_base	1329
	mod_base = OTHER
	note = thymine
modified_base	1344
	mod_base = OTHER
	note = thymine
modified_base	1356..1357
	mod_base = OTHER
	note = thymine
modified_base	1360
	mod_base = OTHER
	note = thymine
modified_base	1362..1363
	mod_base = OTHER
	note = thymine
modified_base	1365
	mod_base = OTHER

-continued

modified_base	note = thymine 1372 mod_base = OTHER note = thymine
modified_base	1379 mod_base = OTHER note = thymine
modified_base	1383 mod_base = OTHER note = thymine
modified_base	1390..1391 mod_base = OTHER note = thymine
modified_base	1396 mod_base = OTHER note = thymine
modified_base	1398 mod_base = OTHER note = thymine
modified_base	1400..1401 mod_base = OTHER note = thymine
modified_base	1411..1412 mod_base = OTHER note = thymine
modified_base	1415 mod_base = OTHER note = thymine
modified_base	1417..1418 mod_base = OTHER note = thymine
modified_base	1421 mod_base = OTHER note = thymine
modified_base	1424..1426 mod_base = OTHER note = thymine
modified_base	1428..1429 mod_base = OTHER note = thymine
modified_base	1431..1432 mod_base = OTHER note = thymine
modified_base	1435 mod_base = OTHER note = thymine
modified_base	1439 mod_base = OTHER note = thymine
modified_base	1447 mod_base = OTHER note = thymine
modified_base	1452 mod_base = OTHER note = thymine
modified_base	1454 mod_base = OTHER note = thymine
modified_base	1456 mod_base = OTHER note = thymine
modified_base	1460..1461 mod_base = OTHER note = thymine
modified_base	1466 mod_base = OTHER note = thymine
modified_base	1470 mod_base = OTHER note = thymine
modified_base	1472 mod_base = OTHER note = thymine
modified_base	1475..1476 mod_base = OTHER note = thymine
modified_base	1480 mod_base = OTHER note = thymine

-continued

modified_base	1482
	mod_base = OTHER
	note = thymine
modified_base	1485
	mod_base = OTHER
	note = thymine
modified_base	1487..1488
	mod_base = OTHER
	note = thymine
modified_base	1490
	mod_base = OTHER
	note = thymine
modified_base	1494
	mod_base = OTHER
	note = thymine
modified_base	1496
	mod_base = OTHER
	note = thymine
modified_base	1502..1503
	mod_base = OTHER
	note = thymine
modified_base	1506..1507
	mod_base = OTHER
	note = thymine
modified_base	1523..1525
	mod_base = OTHER
	note = thymine
modified_base	1535
	mod_base = OTHER
	note = thymine
modified_base	1537..1538
	mod_base = OTHER
	note = thymine
modified_base	1543
	mod_base = OTHER
	note = thymine
modified_base	1547
	mod_base = OTHER
	note = thymine
modified_base	1550
	mod_base = OTHER
	note = thymine
modified_base	1554..1556
	mod_base = OTHER
	note = thymine
modified_base	1558
	mod_base = OTHER
	note = thymine
modified_base	1561
	mod_base = OTHER
	note = thymine
modified_base	1568
	mod_base = OTHER
	note = thymine
modified_base	1577..1578
	mod_base = OTHER
	note = thymine
modified_base	1585
	mod_base = OTHER
	note = thymine
modified_base	1590..1591
	mod_base = OTHER
	note = thymine
modified_base	1594
	mod_base = OTHER
	note = thymine
modified_base	1601..1602
	mod_base = OTHER
	note = thymine
modified_base	1605
	mod_base = OTHER
	note = thymine
modified_base	1607
	mod_base = OTHER
	note = thymine
modified_base	1612
	mod_base = OTHER
	note = thymine
modified_base	1615

-continued

	mod_base = OTHER
	note = thymine
modified_base	1622..1623
	mod_base = OTHER
	note = thymine
modified_base	1629..1631
	mod_base = OTHER
	note = thymine
modified_base	1633
	mod_base = OTHER
	note = thymine
modified_base	1635
	mod_base = OTHER
	note = thymine
modified_base	1638
	mod_base = OTHER
	note = thymine
modified_base	1640
	mod_base = OTHER
	note = thymine
modified_base	1644
	mod_base = OTHER
	note = thymine
modified_base	1646
	mod_base = OTHER
	note = thymine
modified_base	1648
	mod_base = OTHER
	note = thymine
modified_base	1653
	mod_base = OTHER
	note = thymine
modified_base	1658
	mod_base = OTHER
	note = thymine
modified_base	1668
	mod_base = OTHER
	note = thymine
modified_base	1672
	mod_base = OTHER
	note = thymine
modified_base	1674
	mod_base = OTHER
	note = thymine
modified_base	1676..1678
	mod_base = OTHER
	note = thymine
modified_base	1684
	mod_base = OTHER
	note = thymine
modified_base	1686
	mod_base = OTHER
	note = thymine
modified_base	1692..1693
	mod_base = OTHER
	note = thymine
modified_base	1695..1696
	mod_base = OTHER
	note = thymine
modified_base	1698
	mod_base = OTHER
	note = thymine
modified_base	1700
	mod_base = OTHER
	note = thymine
modified_base	1702..1703
	mod_base = OTHER
	note = thymine
modified_base	1706..1708
	mod_base = OTHER
	note = thymine
modified_base	1712
	mod_base = OTHER
	note = thymine
modified_base	1716
	mod_base = OTHER
	note = thymine
modified_base	1720..1721
	mod_base = OTHER

-continued

modified_base	note = thymine 1723 mod_base = OTHER note = thymine
modified_base	1726 mod_base = OTHER note = thymine
modified_base	1730 mod_base = OTHER note = thymine
modified_base	1734..1737 mod_base = OTHER note = thymine
modified_base	1740..1741 mod_base = OTHER note = thymine
modified_base	1744 mod_base = OTHER note = thymine
modified_base	1747 mod_base = OTHER note = thymine
modified_base	1749 mod_base = OTHER note = thymine
modified_base	1751 mod_base = OTHER note = thymine
modified_base	1767 mod_base = OTHER note = thymine
modified_base	1769 mod_base = OTHER note = thymine
modified_base	1776..1777 mod_base = OTHER note = thymine
modified_base	1781 mod_base = OTHER note = thymine
modified_base	1783..1784 mod_base = OTHER note = thymine
modified_base	1789 mod_base = OTHER note = thymine
modified_base	1791 mod_base = OTHER note = thymine
modified_base	1799 mod_base = OTHER note = thymine
modified_base	1801 mod_base = OTHER note = thymine
modified_base	1803..1804 mod_base = OTHER note = thymine
modified_base	1808..1809 mod_base = OTHER note = thymine
modified_base	1815 mod_base = OTHER note = thymine
modified_base	1818..1819 mod_base = OTHER note = thymine
modified_base	1822 mod_base = OTHER note = thymine
modified_base	1829 mod_base = OTHER note = thymine
modified_base	1836 mod_base = OTHER note = thymine
modified_base	1838..1839 mod_base = OTHER note = thymine

-continued

modified_base	1841..1842 mod_base = OTHER note = thymine
modified_base	1847 mod_base = OTHER note = thymine
modified_base	1850..1852 mod_base = OTHER note = thymine
modified_base	1857 mod_base = OTHER note = thymine
modified_base	1860..1862 mod_base = OTHER note = thymine
modified_base	1864..1865 mod_base = OTHER note = thymine
modified_base	1868 mod_base = OTHER note = thymine
modified_base	1870 mod_base = OTHER note = thymine
modified_base	1875 mod_base = OTHER note = thymine
modified_base	1880 mod_base = OTHER note = thymine
modified_base	1884 mod_base = OTHER note = thymine
modified_base	1887..1888 mod_base = OTHER note = thymine
modified_base	1891 mod_base = OTHER note = thymine
modified_base	1895 mod_base = OTHER note = thymine
modified_base	1898..1899 mod_base = OTHER note = thymine
modified_base	1902 mod_base = OTHER note = thymine
modified_base	1904..1905 mod_base = OTHER note = thymine
modified_base	1907 mod_base = OTHER note = thymine
modified_base	1909 mod_base = OTHER note = thymine
modified_base	1911 mod_base = OTHER note = thymine
modified_base	1913 mod_base = OTHER note = thymine
modified_base	1916 mod_base = OTHER note = thymine
modified_base	1919 mod_base = OTHER note = thymine
modified_base	1922..1924 mod_base = OTHER note = thymine
modified_base	1928..1930 mod_base = OTHER note = thymine
modified_base	1947..1948 mod_base = OTHER note = thymine
modified_base	1950..1951

-continued

	mod_base = OTHER
	note = thymine
modified_base	1953..1954
	mod_base = OTHER
	note = thymine
modified_base	1958
	mod_base = OTHER
	note = thymine
modified_base	1974
	mod_base = OTHER
	note = thymine
modified_base	1976
	mod_base = OTHER
	note = thymine
modified_base	1981
	mod_base = OTHER
	note = thymine
modified_base	1983..1984
	mod_base = OTHER
	note = thymine
modified_base	1987
	mod_base = OTHER
	note = thymine
modified_base	2004
	mod_base = OTHER
	note = thymine
modified_base	2009
	mod_base = OTHER
	note = thymine
modified_base	2013
	mod_base = OTHER
	note = thymine
modified_base	2022..2024
	mod_base = OTHER
	note = thymine
modified_base	2026
	mod_base = OTHER
	note = thymine
modified_base	2043
	mod_base = OTHER
	note = thymine
modified_base	2046
	mod_base = OTHER
	note = thymine
modified_base	2056..2057
	mod_base = OTHER
	note = thymine
modified_base	2060..2062
	mod_base = OTHER
	note = thymine
modified_base	2069..2070
	mod_base = OTHER
	note = thymine
modified_base	2073..2074
	mod_base = OTHER
	note = thymine
modified_base	2078..2079
	mod_base = OTHER
	note = thymine
modified_base	2081..2083
	mod_base = OTHER
	note = thymine
modified_base	2087
	mod_base = OTHER
	note = thymine
modified_base	2090..2094
	mod_base = OTHER
	note = thymine
modified_base	2100
	mod_base = OTHER
	note = thymine
modified_base	2104
	mod_base = OTHER
	note = thymine
modified_base	2108..2110
	mod_base = OTHER
	note = thymine
modified_base	2115..2117
	mod_base = OTHER

-continued

modified_base	note = thymine 2119 mod_base = OTHER note = thymine
modified_base	2121 mod_base = OTHER note = thymine
modified_base	2128..2129 mod_base = OTHER note = thymine
modified_base	2141 mod_base = OTHER note = thymine
modified_base	2146..2147 mod_base = OTHER note = thymine
modified_base	2149..2150 mod_base = OTHER note = thymine
modified_base	2155 mod_base = OTHER note = thymine
modified_base	2159 mod_base = OTHER note = thymine
modified_base	2161 mod_base = OTHER note = thymine
modified_base	2163 mod_base = OTHER note = thymine
modified_base	2166..2168 mod_base = OTHER note = thymine
modified_base	2170 mod_base = OTHER note = thymine
modified_base	2172 mod_base = OTHER note = thymine
modified_base	2180 mod_base = OTHER note = thymine
modified_base	2186..2188 mod_base = OTHER note = thymine
modified_base	2193 mod_base = OTHER note = thymine
modified_base	2197 mod_base = OTHER note = thymine
modified_base	2205..2206 mod_base = OTHER note = thymine
modified_base	2209 mod_base = OTHER note = thymine
modified_base	2215 mod_base = OTHER note = thymine
modified_base	2218 mod_base = OTHER note = thymine
modified_base	2220..2221 mod_base = OTHER note = thymine
modified_base	2233 mod_base = OTHER note = thymine
modified_base	2240 mod_base = OTHER note = thymine
modified_base	2256 mod_base = OTHER note = thymine
modified_base	2261 mod_base = OTHER note = thymine

-continued

modified_base	2267..2269
	mod_base = OTHER
	note = thymine
modified_base	2271..2272
	mod_base = OTHER
	note = thymine
modified_base	2275..2276
	mod_base = OTHER
	note = thymine
modified_base	2280
	mod_base = OTHER
	note = thymine
modified_base	2282
	mod_base = OTHER
	note = thymine
modified_base	2290
	mod_base = OTHER
	note = thymine
modified_base	2292..2293
	mod_base = OTHER
	note = thymine
modified_base	2295..2296
	mod_base = OTHER
	note = thymine
modified_base	2303..2306
	mod_base = OTHER
	note = thymine
modified_base	2308
	mod_base = OTHER
	note = thymine
modified_base	2313..2315
	mod_base = OTHER
	note = thymine
modified_base	2317..2318
	mod_base = OTHER
	note = thymine
modified_base	2326
	mod_base = OTHER
	note = thymine
modified_base	2331
	mod_base = OTHER
	note = thymine
modified_base	2333
	mod_base = OTHER
	note = thymine
modified_base	2335..2336
	mod_base = OTHER
	note = thymine
modified_base	2338
	mod_base = OTHER
	note = thymine
modified_base	2341..2342
	mod_base = OTHER
	note = thymine
modified_base	2344
	mod_base = OTHER
	note = thymine
modified_base	2348
	mod_base = OTHER
	note = thymine
modified_base	2350..2351
	mod_base = OTHER
	note = thymine
modified_base	2356
	mod_base = OTHER
	note = thymine
modified_base	2358
	mod_base = OTHER
	note = thymine
modified_base	2360..2361
	mod_base = OTHER
	note = thymine
modified_base	2363..2365
	mod_base = OTHER
	note = thymine
modified_base	2367
	mod_base = OTHER
	note = thymine
modified_base	2369

-continued

	mod_base = OTHER
	note = thymine
modified_base	2375
	mod_base = OTHER
	note = thymine
modified_base	2377..2378
	mod_base = OTHER
	note = thymine
modified_base	2383..2384
	mod_base = OTHER
	note = thymine
modified_base	2386
	mod_base = OTHER
	note = thymine
modified_base	2389
	mod_base = OTHER
	note = thymine
modified_base	2393..2395
	mod_base = OTHER
	note = thymine
modified_base	2398
	mod_base = OTHER
	note = thymine
modified_base	2401
	mod_base = OTHER
	note = thymine
modified_base	2413
	mod_base = OTHER
	note = thymine
modified_base	2418
	mod_base = OTHER
	note = thymine
modified_base	2428
	mod_base = OTHER
	note = thymine
modified_base	2432
	mod_base = OTHER
	note = thymine
modified_base	2436..2437
	mod_base = OTHER
	note = thymine
modified_base	2444
	mod_base = OTHER
	note = thymine
modified_base	2446
	mod_base = OTHER
	note = thymine
modified_base	2448
	mod_base = OTHER
	note = thymine
modified_base	2451
	mod_base = OTHER
	note = thymine
modified_base	2453
	mod_base = OTHER
	note = thymine
modified_base	2458..2460
	mod_base = OTHER
	note = thymine
modified_base	2462
	mod_base = OTHER
	note = thymine
modified_base	2464
	mod_base = OTHER
	note = thymine
modified_base	2467..2469
	mod_base = OTHER
	note = thymine
modified_base	2472
	mod_base = OTHER
	note = thymine
modified_base	2474..2475
	mod_base = OTHER
	note = thymine
modified_base	2479
	mod_base = OTHER
	note = thymine
modified_base	2490
	mod_base = OTHER

-continued

modified_base	note = thymine 2492..2493 mod_base = OTHER note = thymine
modified_base	2501 mod_base = OTHER note = thymine
modified_base	2507 mod_base = OTHER note = thymine
modified_base	2514 mod_base = OTHER note = thymine
modified_base	2517 mod_base = OTHER note = thymine
modified_base	2522 mod_base = OTHER note = thymine
modified_base	2527 mod_base = OTHER note = thymine
modified_base	2530..2531 mod_base = OTHER note = thymine
modified_base	2533 mod_base = OTHER note = thymine
modified_base	2538..2539 mod_base = OTHER note = thymine
modified_base	2542 mod_base = OTHER note = thymine
modified_base	2547..2550 mod_base = OTHER note = thymine
modified_base	2553 mod_base = OTHER note = thymine
modified_base	2560..2563 mod_base = OTHER note = thymine
modified_base	2567 mod_base = OTHER note = thymine
modified_base	2569 mod_base = OTHER note = thymine
modified_base	2571..2572 mod_base = OTHER note = thymine
modified_base	2575 mod_base = OTHER note = thymine
modified_base	2577 mod_base = OTHER note = thymine
modified_base	2580..2581 mod_base = OTHER note = thymine
modified_base	2588..2590 mod_base = OTHER note = thymine
modified_base	2593 mod_base = OTHER note = thymine
modified_base	2601..2602 mod_base = OTHER note = thymine
modified_base	2605..2606 mod_base = OTHER note = thymine
modified_base	2608 mod_base = OTHER note = thymine
modified_base	2614..2615 mod_base = OTHER note = thymine

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modified_base	2617
	mod_base = OTHER
	note = thymine
modified_base	2619
	mod_base = OTHER
	note = thymine
modified_base	2622..2623
	mod_base = OTHER
	note = thymine
modified_base	2625
	mod_base = OTHER
	note = thymine
modified_base	2628
	mod_base = OTHER
	note = thymine
modified_base	2638
	mod_base = OTHER
	note = thymine
modified_base	2642..2643
	mod_base = OTHER
	note = thymine
modified_base	2647..2650
	mod_base = OTHER
	note = thymine
modified_base	2653
	mod_base = OTHER
	note = thymine
modified_base	2655
	mod_base = OTHER
	note = thymine
modified_base	2657
	mod_base = OTHER
	note = thymine
modified_base	2664
	mod_base = OTHER
	note = thymine
modified_base	2669
	mod_base = OTHER
	note = thymine
modified_base	2671
	mod_base = OTHER
	note = thymine
modified_base	2679
	mod_base = OTHER
	note = thymine
modified_base	2688
	mod_base = OTHER
	note = thymine
modified_base	2690..2692
	mod_base = OTHER
	note = thymine
modified_base	2697
	mod_base = OTHER
	note = thymine
modified_base	2699..2700
	mod_base = OTHER
	note = thymine
modified_base	2702..2703
	mod_base = OTHER
	note = thymine
modified_base	2705..2706
	mod_base = OTHER
	note = thymine
modified_base	2710
	mod_base = OTHER
	note = thymine
modified_base	2714
	mod_base = OTHER
	note = thymine
modified_base	2725
	mod_base = OTHER
	note = thymine
modified_base	2727
	mod_base = OTHER
	note = thymine
modified_base	2729
	mod_base = OTHER
	note = thymine
modified_base	2731

-continued

	mod_base = OTHER
	note = thymine
modified_base	2733
	mod_base = OTHER
	note = thymine
modified_base	2735
	mod_base = OTHER
	note = thymine
modified_base	2742
	mod_base = OTHER
	note = thymine
modified_base	2744
	mod_base = OTHER
	note = thymine
modified_base	2746
	mod_base = OTHER
	note = thymine
modified_base	2749
	mod_base = OTHER
	note = thymine
modified_base	2753..2754
	mod_base = OTHER
	note = thymine
modified_base	2757..2758
	mod_base = OTHER
	note = thymine
modified_base	2762
	mod_base = OTHER
	note = thymine
modified_base	2764
	mod_base = OTHER
	note = thymine
modified_base	2767
	mod_base = OTHER
	note = thymine
modified_base	2773
	mod_base = OTHER
	note = thymine
modified_base	2775..2777
	mod_base = OTHER
	note = thymine
modified_base	2783
	mod_base = OTHER
	note = thymine
modified_base	2785
	mod_base = OTHER
	note = thymine
modified_base	2787
	mod_base = OTHER
	note = thymine
modified_base	2791
	mod_base = OTHER
	note = thymine
modified_base	2794
	mod_base = OTHER
	note = thymine
modified_base	2802..2803
	mod_base = OTHER
	note = thymine
modified_base	2809
	mod_base = OTHER
	note = thymine
modified_base	2813..2814
	mod_base = OTHER
	note = thymine
modified_base	2817
	mod_base = OTHER
	note = thymine
modified_base	2824..2825
	mod_base = OTHER
	note = thymine
modified_base	2827
	mod_base = OTHER
	note = thymine
modified_base	2829
	mod_base = OTHER
	note = thymine
modified_base	2831
	mod_base = OTHER

-continued

modified_base	note = thymine 2833 mod_base = OTHER note = thymine
modified_base	2836 mod_base = OTHER note = thymine
modified_base	2838 mod_base = OTHER note = thymine
modified_base	2840 mod_base = OTHER note = thymine
modified_base	2846..2847 mod_base = OTHER note = thymine
modified_base	2849 mod_base = OTHER note = thymine
modified_base	2851..2852 mod_base = OTHER note = thymine
modified_base	2857..2860 mod_base = OTHER note = thymine
modified_base	2865..2866 mod_base = OTHER note = thymine
modified_base	2871..2872 mod_base = OTHER note = thymine
modified_base	2874 mod_base = OTHER note = thymine
modified_base	2876 mod_base = OTHER note = thymine
modified_base	2895..2896 mod_base = OTHER note = thymine
modified_base	2899 mod_base = OTHER note = thymine
modified_base	2903 mod_base = OTHER note = thymine
modified_base	2911 mod_base = OTHER note = thymine
modified_base	2913 mod_base = OTHER note = thymine
modified_base	2915..2916 mod_base = OTHER note = thymine
modified_base	2923 mod_base = OTHER note = thymine
modified_base	2933 mod_base = OTHER note = thymine
modified_base	2935..2936 mod_base = OTHER note = thymine
modified_base	2940..2941 mod_base = OTHER note = thymine
modified_base	2946 mod_base = OTHER note = thymine
modified_base	2959..2960 mod_base = OTHER note = thymine
modified_base	2963 mod_base = OTHER note = thymine
modified_base	2965 mod_base = OTHER note = thymine

-continued

modified_base	2971
	mod_base = OTHER
	note = thymine
modified_base	2973..2975
	mod_base = OTHER
	note = thymine
modified_base	2980
	mod_base = OTHER
	note = thymine
modified_base	2983
	mod_base = OTHER
	note = thymine
modified_base	2988..2989
	mod_base = OTHER
	note = thymine
modified_base	2991..2993
	mod_base = OTHER
	note = thymine
modified_base	2996..2998
	mod_base = OTHER
	note = thymine
modified_base	3005
	mod_base = OTHER
	note = thymine
modified_base	3011..3012
	mod_base = OTHER
	note = thymine
modified_base	3015
	mod_base = OTHER
	note = thymine
modified_base	3028..3030
	mod_base = OTHER
	note = thymine
modified_base	3038..3039
	mod_base = OTHER
	note = thymine
modified_base	3041
	mod_base = OTHER
	note = thymine
modified_base	3043
	mod_base = OTHER
	note = thymine
modified_base	3046
	mod_base = OTHER
	note = thymine
modified_base	3048
	mod_base = OTHER
	note = thymine
modified_base	3050..3051
	mod_base = OTHER
	note = thymine
modified_base	3057..3060
	mod_base = OTHER
	note = thymine
modified_base	3063
	mod_base = OTHER
	note = thymine
modified_base	3067
	mod_base = OTHER
	note = thymine
modified_base	3079..3083
	mod_base = OTHER
	note = thymine
modified_base	3085
	mod_base = OTHER
	note = thymine
modified_base	3091
	mod_base = OTHER
	note = thymine
modified_base	3096
	mod_base = OTHER
	note = thymine
modified_base	3099
	mod_base = OTHER
	note = thymine
modified_base	3102
	mod_base = OTHER
	note = thymine
modified_base	3106

-continued

	mod_base = OTHER
	note = thymine
modified_base	3112
	mod_base = OTHER
	note = thymine
modified_base	3114
	mod_base = OTHER
	note = thymine
modified_base	3118
	mod_base = OTHER
	note = thymine
modified_base	3125
	mod_base = OTHER
	note = thymine
modified_base	3127
	mod_base = OTHER
	note = thymine
modified_base	3132
	mod_base = OTHER
	note = thymine
modified_base	3135
	mod_base = OTHER
	note = thymine
modified_base	3140
	mod_base = OTHER
	note = thymine
modified_base	3144
	mod_base = OTHER
	note = thymine
modified_base	3149
	mod_base = OTHER
	note = thymine
modified_base	3153
	mod_base = OTHER
	note = thymine
modified_base	3164..3165
	mod_base = OTHER
	note = thymine
modified_base	3167..3168
	mod_base = OTHER
	note = thymine
modified_base	3172
	mod_base = OTHER
	note = thymine
modified_base	3175..3176
	mod_base = OTHER
	note = thymine
modified_base	3183
	mod_base = OTHER
	note = thymine
modified_base	3185
	mod_base = OTHER
	note = thymine
modified_base	3189
	mod_base = OTHER
	note = thymine
modified_base	3191
	mod_base = OTHER
	note = thymine
modified_base	3194
	mod_base = OTHER
	note = thymine
modified_base	3196
	mod_base = OTHER
	note = thymine
modified_base	3198..3199
	mod_base = OTHER
	note = thymine
modified_base	3201
	mod_base = OTHER
	note = thymine
modified_base	3208..3209
	mod_base = OTHER
	note = thymine
modified_base	3213
	mod_base = OTHER
	note = thymine
modified_base	3220
	mod_base = OTHER

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modified_base	note = thymine 3222 mod_base = OTHER
modified_base	note = thymine 3235..3236 mod_base = OTHER
modified_base	note = thymine 3238 mod_base = OTHER
modified_base	note = thymine 3240 mod_base = OTHER
modified_base	note = thymine 3244 mod_base = OTHER
modified_base	note = thymine 3252 mod_base = OTHER
modified_base	note = thymine 3255 mod_base = OTHER
modified_base	note = thymine 3259..3260 mod_base = OTHER
modified_base	note = thymine 3264 mod_base = OTHER
modified_base	note = thymine 3270 mod_base = OTHER
modified_base	note = thymine 3273 mod_base = OTHER
modified_base	note = thymine 3276 mod_base = OTHER
modified_base	note = thymine 3285 mod_base = OTHER
modified_base	note = thymine 3287..3288 mod_base = OTHER
modified_base	note = thymine 3298 mod_base = OTHER
modified_base	note = thymine 3301 mod_base = OTHER
modified_base	note = thymine 3306 mod_base = OTHER
modified_base	note = thymine 3316..3317 mod_base = OTHER
modified_base	note = thymine 3323 mod_base = OTHER
modified_base	note = thymine 3329 mod_base = OTHER
modified_base	note = thymine 3331 mod_base = OTHER
modified_base	note = thymine 3333..3334 mod_base = OTHER
modified_base	note = thymine 3338..3340 mod_base = OTHER
modified_base	note = thymine 3344 mod_base = OTHER
modified_base	note = thymine 3351 mod_base = OTHER
modified_base	note = thymine 3356..3357 mod_base = OTHER
	note = thymine

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modified_base	3359
	mod_base = OTHER
	note = thymine
modified_base	3366..3367
	mod_base = OTHER
	note = thymine
modified_base	3370..3371
	mod_base = OTHER
	note = thymine
modified_base	3377..3378
	mod_base = OTHER
	note = thymine
modified_base	3387..3388
	mod_base = OTHER
	note = thymine
modified_base	3390
	mod_base = OTHER
	note = thymine
modified_base	3396
	mod_base = OTHER
	note = thymine
modified_base	3401..3403
	mod_base = OTHER
	note = thymine
modified_base	3409
	mod_base = OTHER
	note = thymine
modified_base	3411..3412
	mod_base = OTHER
	note = thymine
modified_base	3415..3416
	mod_base = OTHER
	note = thymine
modified_base	3419
	mod_base = OTHER
	note = thymine
modified_base	3422..3423
	mod_base = OTHER
	note = thymine
modified_base	3429..3430
	mod_base = OTHER
	note = thymine
modified_base	3435..3437
	mod_base = OTHER
	note = thymine
modified_base	3439..3440
	mod_base = OTHER
	note = thymine
modified_base	3442
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	note = thymine
modified_base	3444
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modified_base	3446
	mod_base = OTHER
	note = thymine
modified_base	3448
	mod_base = OTHER
	note = thymine
modified_base	3453..3454
	mod_base = OTHER
	note = thymine
modified_base	3456..3457
	mod_base = OTHER
	note = thymine
modified_base	3461..3462
	mod_base = OTHER
	note = thymine
modified_base	3464
	mod_base = OTHER
	note = thymine
modified_base	3467
	mod_base = OTHER
	note = thymine
modified_base	3469
	mod_base = OTHER
	note = thymine
modified_base	3471

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	mod_base = OTHER
	note = thymine
modified_base	3491
	mod_base = OTHER
	note = thymine
modified_base	3494
	mod_base = OTHER
	note = thymine
modified_base	3499
	mod_base = OTHER
	note = thymine
modified_base	3502
	mod_base = OTHER
	note = thymine
modified_base	3505..3506
	mod_base = OTHER
	note = thymine
modified_base	3509
	mod_base = OTHER
	note = thymine
modified_base	3515..3517
	mod_base = OTHER
	note = thymine
modified_base	3522
	mod_base = OTHER
	note = thymine
modified_base	3525
	mod_base = OTHER
	note = thymine
modified_base	3528
	mod_base = OTHER
	note = thymine
modified_base	3530
	mod_base = OTHER
	note = thymine
modified_base	3543
	mod_base = OTHER
	note = thymine
modified_base	3549..3550
	mod_base = OTHER
	note = thymine
modified_base	3552..3553
	mod_base = OTHER
	note = thymine
modified_base	3562..3565
	mod_base = OTHER
	note = thymine
modified_base	3568
	mod_base = OTHER
	note = thymine
modified_base	3570
	mod_base = OTHER
	note = thymine
modified_base	3577..3578
	mod_base = OTHER
	note = thymine
modified_base	3584..3585
	mod_base = OTHER
	note = thymine
modified_base	3590..3591
	mod_base = OTHER
	note = thymine
modified_base	3594..3597
	mod_base = OTHER
	note = thymine
modified_base	3599
	mod_base = OTHER
	note = thymine
modified_base	3603..3604
	mod_base = OTHER
	note = thymine
modified_base	3611
	mod_base = OTHER
	note = thymine
modified_base	3613
	mod_base = OTHER
	note = thymine
modified_base	3615
	mod_base = OTHER

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modified_base	note = thymine 3617..3619 mod_base = OTHER
modified_base	note = thymine 3622..3623 mod_base = OTHER
modified_base	note = thymine 3628 mod_base = OTHER
modified_base	note = thymine 3638..3639 mod_base = OTHER
modified_base	note = thymine 3641 mod_base = OTHER
modified_base	note = thymine 3648 mod_base = OTHER
modified_base	note = thymine 3650..3651 mod_base = OTHER
modified_base	note = thymine 3656 mod_base = OTHER
modified_base	note = thymine 3661 mod_base = OTHER
modified_base	note = thymine 3664 mod_base = OTHER
modified_base	note = thymine 3667 mod_base = OTHER
modified_base	note = thymine 3672..3673 mod_base = OTHER
modified_base	note = thymine 3676 mod_base = OTHER
modified_base	note = thymine 3678..3679 mod_base = OTHER
modified_base	note = thymine 3684 mod_base = OTHER
modified_base	note = thymine 3686 mod_base = OTHER
modified_base	note = thymine 3693 mod_base = OTHER
modified_base	note = thymine 3698..3699 mod_base = OTHER
modified_base	note = thymine 3701 mod_base = OTHER
modified_base	note = thymine 3703 mod_base = OTHER
modified_base	note = thymine 3705..3707 mod_base = OTHER
modified_base	note = thymine 3717 mod_base = OTHER
modified_base	note = thymine 3720 mod_base = OTHER
modified_base	note = thymine 3723..3724 mod_base = OTHER
modified_base	note = thymine 3726 mod_base = OTHER
modified_base	note = thymine 3730..3731 mod_base = OTHER
	note = thymine

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modified_base	3734
	mod_base = OTHER
	note = thymine
modified_base	3740..3742
	mod_base = OTHER
	note = thymine
modified_base	3747..3748
	mod_base = OTHER
	note = thymine
modified_base	3752
	mod_base = OTHER
	note = thymine
modified_base	3754
	mod_base = OTHER
	note = thymine
modified_base	3756..3758
	mod_base = OTHER
	note = thymine
modified_base	3763
	mod_base = OTHER
	note = thymine
modified_base	3766
	mod_base = OTHER
	note = thymine
modified_base	3769..3770
	mod_base = OTHER
	note = thymine
modified_base	3772
	mod_base = OTHER
	note = thymine
modified_base	3777
	mod_base = OTHER
	note = thymine
modified_base	3779
	mod_base = OTHER
	note = thymine
modified_base	3781
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	note = thymine
modified_base	3789
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	note = thymine
modified_base	3792
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	note = thymine
modified_base	3794
	mod_base = OTHER
	note = thymine
modified_base	3798
	mod_base = OTHER
	note = thymine
modified_base	3806
	mod_base = OTHER
	note = thymine
modified_base	3810..3811
	mod_base = OTHER
	note = thymine
modified_base	3815
	mod_base = OTHER
	note = thymine
modified_base	3821
	mod_base = OTHER
	note = thymine
modified_base	3825..3827
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	note = thymine
modified_base	3829
	mod_base = OTHER
	note = thymine
modified_base	3834
	mod_base = OTHER
	note = thymine
modified_base	3838..3841
	mod_base = OTHER
	note = thymine
modified_base	3843
	mod_base = OTHER
	note = thymine
modified_base	3845

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modified_base	3849..3850
	mod_base = OTHER
	note = thymine
modified_base	3852
	mod_base = OTHER
	note = thymine
modified_base	3859
	mod_base = OTHER
	note = thymine
modified_base	3863..3864
	mod_base = OTHER
	note = thymine
modified_base	3870..3871
	mod_base = OTHER
	note = thymine
modified_base	3874..3876
	mod_base = OTHER
	note = thymine
modified_base	3880
	mod_base = OTHER
	note = thymine
modified_base	3883
	mod_base = OTHER
	note = thymine
modified_base	3886
	mod_base = OTHER
	note = thymine
modified_base	3897
	mod_base = OTHER
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modified_base	3900
	mod_base = OTHER
	note = thymine
modified_base	3903..3905
	mod_base = OTHER
	note = thymine
modified_base	3908
	mod_base = OTHER
	note = thymine
modified_base	3910
	mod_base = OTHER
	note = thymine
modified_base	3917..3918
	mod_base = OTHER
	note = thymine
modified_base	3922
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	note = thymine
modified_base	3935
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modified_base	3937
	mod_base = OTHER
	note = thymine
modified_base	3941
	mod_base = OTHER
	note = thymine
modified_base	3943
	mod_base = OTHER
	note = thymine
modified_base	3946..3947
	mod_base = OTHER
	note = thymine
modified_base	3949
	mod_base = OTHER
	note = thymine
modified_base	3951
	mod_base = OTHER
	note = thymine
modified_base	3954
	mod_base = OTHER
	note = thymine
modified_base	3957
	mod_base = OTHER
	note = thymine
modified_base	3961
	mod_base = OTHER

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modified_base	note = thymine 3968..3970 mod_base = OTHER note = thymine
modified_base	3972 mod_base = OTHER note = thymine
modified_base	3975 mod_base = OTHER note = thymine
modified_base	3978 mod_base = OTHER note = thymine
modified_base	3981 mod_base = OTHER note = thymine
modified_base	3983 mod_base = OTHER note = thymine
modified_base	3986..3987 mod_base = OTHER note = thymine
modified_base	3991..3992 mod_base = OTHER note = thymine
modified_base	3999 mod_base = OTHER note = thymine
modified_base	4002..4003 mod_base = OTHER note = thymine
modified_base	4010..4011 mod_base = OTHER note = thymine
modified_base	4019..4020 mod_base = OTHER note = thymine
modified_base	4023 mod_base = OTHER note = thymine
modified_base	4025..4026 mod_base = OTHER note = thymine
modified_base	4028..4029 mod_base = OTHER note = thymine
modified_base	4032 mod_base = OTHER note = thymine
modified_base	4037 mod_base = OTHER note = thymine
modified_base	4039..4040 mod_base = OTHER note = thymine
modified_base	4042..4043 mod_base = OTHER note = thymine
modified_base	4045 mod_base = OTHER note = thymine
modified_base	4050..4055 mod_base = OTHER note = thymine
modified_base	4058 mod_base = OTHER note = thymine
modified_base	4078 mod_base = OTHER note = thymine
modified_base	4080 mod_base = OTHER note = thymine
modified_base	4083 mod_base = OTHER note = thymine
modified_base	4088 mod_base = OTHER note = thymine

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modified_base	4091..4092
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modified_base	4101
	mod_base = OTHER
	note = thymine
modified_base	4105
	mod_base = OTHER
	note = thymine
modified_base	4109..4111
	mod_base = OTHER
	note = thymine
modified_base	4115
	mod_base = OTHER
	note = thymine
modified_base	4121
	mod_base = OTHER
	note = thymine
modified_base	4125
	mod_base = OTHER
	note = thymine
modified_base	4134
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	note = thymine
modified_base	4139
	mod_base = OTHER
	note = thymine
modified_base	4143
	mod_base = OTHER
	note = thymine
modified_base	4145..4147
	mod_base = OTHER
	note = thymine
modified_base	4149
	mod_base = OTHER
	note = thymine
modified_base	4155
	mod_base = OTHER
	note = thymine
modified_base	4162
	mod_base = OTHER
	note = thymine
modified_base	4170
	mod_base = OTHER
	note = thymine
modified_base	4172
	mod_base = OTHER
	note = thymine
modified_base	4176
	mod_base = OTHER
	note = thymine
modified_base	4182
	mod_base = OTHER
	note = thymine
modified_base	4185..4186
	mod_base = OTHER
	note = thymine
modified_base	4191
	mod_base = OTHER
	note = thymine
modified_base	4194
	mod_base = OTHER
	note = thymine
modified_base	4198..4199
	mod_base = OTHER
	note = thymine
modified_base	4206
	mod_base = OTHER
	note = thymine
modified_base	4208
	mod_base = OTHER
	note = thymine
modified_base	4211..4213
	mod_base = OTHER
	note = thymine
modified_base	4216
	mod_base = OTHER
	note = thymine
modified_base	4218

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	mod_base = OTHER
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modified_base	4220
	mod_base = OTHER
	note = thymine
modified_base	4222
	mod_base = OTHER
	note = thymine
modified_base	4229
	mod_base = OTHER
	note = thymine
modified_base	4247
	mod_base = OTHER
	note = thymine
modified_base	4251
	mod_base = OTHER
	note = thymine
modified_base	4260
	mod_base = OTHER
	note = thymine
modified_base	4262
	mod_base = OTHER
	note = thymine
modified_base	4266
	mod_base = OTHER
	note = thymine
modified_base	4269
	mod_base = OTHER
	note = thymine
modified_base	4277
	mod_base = OTHER
	note = thymine
modified_base	4281..4286
	mod_base = OTHER
	note = thymine
modified_base	4289..4291
	mod_base = OTHER
	note = thymine
modified_base	4296..4297
	mod_base = OTHER
	note = thymine
modified_base	4300
	mod_base = OTHER
	note = thymine
modified_base	4303
	mod_base = OTHER
	note = thymine
modified_base	4305
	mod_base = OTHER
	note = thymine
modified_base	4310..4313
	mod_base = OTHER
	note = thymine
modified_base	4315
	mod_base = OTHER
	note = thymine
modified_base	4320
	mod_base = OTHER
	note = thymine
modified_base	4324
	mod_base = OTHER
	note = thymine
modified_base	4326..4327
	mod_base = OTHER
	note = thymine
modified_base	4333
	mod_base = OTHER
	note = thymine
modified_base	4335..4336
	mod_base = OTHER
	note = thymine
modified_base	4340
	mod_base = OTHER
	note = thymine
modified_base	4343
	mod_base = OTHER
	note = thymine
modified_base	4345
	mod_base = OTHER

-continued

modified_base	note = thymine 4348..4350 mod_base = OTHER note = thymine
modified_base	4352 mod_base = OTHER note = thymine
modified_base	4356 mod_base = OTHER note = thymine
modified_base	4359..4361 mod_base = OTHER note = thymine
modified_base	4363 mod_base = OTHER note = thymine
modified_base	4366 mod_base = OTHER note = thymine
modified_base	4371 mod_base = OTHER note = thymine
modified_base	4381..4382 mod_base = OTHER note = thymine
modified_base	4385..4386 mod_base = OTHER note = thymine
modified_base	4389 mod_base = OTHER note = thymine
modified_base	4392 mod_base = OTHER note = thymine
modified_base	4394 mod_base = OTHER note = thymine
modified_base	4396..4397 mod_base = OTHER note = thymine
modified_base	4402 mod_base = OTHER note = thymine
modified_base	4404 mod_base = OTHER note = thymine
modified_base	4407 mod_base = OTHER note = thymine
modified_base	4411 mod_base = OTHER note = thymine
modified_base	4415 mod_base = OTHER note = thymine
modified_base	4418 mod_base = OTHER note = thymine
modified_base	4421..4422 mod_base = OTHER note = thymine
modified_base	4424..4425 mod_base = OTHER note = thymine
modified_base	4427..4428 mod_base = OTHER note = thymine
modified_base	4431..4432 mod_base = OTHER note = thymine
modified_base	4435..4436 mod_base = OTHER note = thymine
modified_base	4438 mod_base = OTHER note = thymine
modified_base	4441..4442 mod_base = OTHER note = thymine

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modified_base	4444..4445 mod_base = OTHER note = thymine
modified_base	4453 mod_base = OTHER note = thymine
modified_base	4456 mod_base = OTHER note = thymine
modified_base	4466 mod_base = OTHER note = thymine
modified_base	4468..4469 mod_base = OTHER note = thymine
modified_base	4473..4478 mod_base = OTHER note = thymine
modified_base	4481 mod_base = OTHER note = thymine
modified_base	4483..4484 mod_base = OTHER note = thymine
modified_base	4488 mod_base = OTHER note = thymine
modified_base	4491 mod_base = OTHER note = thymine
modified_base	4495 mod_base = OTHER note = thymine
modified_base	4500 mod_base = OTHER note = thymine
modified_base	4506 mod_base = OTHER note = thymine
modified_base	4512..4513 mod_base = OTHER note = thymine
modified_base	4515 mod_base = OTHER note = thymine
modified_base	4519 mod_base = OTHER note = thymine
modified_base	4523..4526 mod_base = OTHER note = thymine
modified_base	4528..4531 mod_base = OTHER note = thymine
modified_base	4537 mod_base = OTHER note = thymine
modified_base	4539 mod_base = OTHER note = thymine
modified_base	4546 mod_base = OTHER note = thymine
modified_base	4549 mod_base = OTHER note = thymine
modified_base	4559 mod_base = OTHER note = thymine
modified_base	4561..4562 mod_base = OTHER note = thymine
modified_base	4571..4573 mod_base = OTHER note = thymine
modified_base	4575 mod_base = OTHER note = thymine
modified_base	4578..4579

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	mod_base = OTHER
	note = thymine
modified_base	4581
	mod_base = OTHER
	note = thymine
modified_base	4584
	mod_base = OTHER
	note = thymine
modified_base	4597
	mod_base = OTHER
	note = thymine
modified_base	4599..4601
	mod_base = OTHER
	note = thymine
modified_base	4607..4608
	mod_base = OTHER
	note = thymine
modified_base	4611
	mod_base = OTHER
	note = thymine
modified_base	4617..4618
	mod_base = OTHER
	note = thymine
modified_base	4624..4625
	mod_base = OTHER
	note = thymine
modified_base	4633..4636
	mod_base = OTHER
	note = thymine
modified_base	4638..4639
	mod_base = OTHER
	note = thymine
modified_base	4642..4643
	mod_base = OTHER
	note = thymine
modified_base	4650
	mod_base = OTHER
	note = thymine
modified_base	4658
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	note = thymine
modified_base	4660
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	note = thymine
modified_base	4663
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	note = thymine
modified_base	4666
	mod_base = OTHER
	note = thymine
modified_base	4669
	mod_base = OTHER
	note = thymine
modified_base	4671..4672
	mod_base = OTHER
	note = thymine
modified_base	4678
	mod_base = OTHER
	note = thymine
modified_base	4680
	mod_base = OTHER
	note = thymine
modified_base	4686..4687
	mod_base = OTHER
	note = thymine
modified_base	4690..4693
	mod_base = OTHER
	note = thymine
modified_base	4697..4698
	mod_base = OTHER
	note = thymine
modified_base	4703
	mod_base = OTHER
	note = thymine
modified_base	4705
	mod_base = OTHER
	note = thymine
modified_base	4709..4711
	mod_base = OTHER

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modified_base	note = thymine
	4715
	mod_base = OTHER
modified_base	note = thymine
	4717..4718
	mod_base = OTHER
modified_base	note = thymine
	4721
	mod_base = OTHER
modified_base	note = thymine
	4723
	mod_base = OTHER
modified_base	note = thymine
	4725..4726
	mod_base = OTHER
modified_base	note = thymine
	4738
	mod_base = OTHER
modified_base	note = thymine
	4753
	mod_base = OTHER
modified_base	note = thymine
	4755
	mod_base = OTHER
modified_base	note = thymine
	4757..4759
	mod_base = OTHER
modified_base	note = thymine
	4766
	mod_base = OTHER
modified_base	note = thymine
	4769
	mod_base = OTHER
modified_base	note = thymine
	4774..4776
	mod_base = OTHER
modified_base	note = thymine
	4780
	mod_base = OTHER
modified_base	note = thymine
	4782
	mod_base = OTHER
modified_base	note = thymine
	4784
	mod_base = OTHER
modified_base	note = thymine
	4792
	mod_base = OTHER
modified_base	note = thymine
	4794..4795
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	1..7114
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	organism = unidentified

SEQUENCE: 4

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cagactcctc	aagagccaca	cagccagccg	cagaatttag	gagggtagct	cacagcagtc	180
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atgatcacta	cctttgttta	aggtgccatc	agggtatggt	aaggaattca	gatctctgca	300
atatctgctg	gaagcccctg	cccaccacaa	tcacagtacc	ggtggagcca	acagaccac	360
caccataggc	agactgcaca	gggtcagacc	cgaccccccg	gggggcccc	atggggaccc	420
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gcttcagtg	cctgcatgtc	accoccttca	atctgaactg	cccttgggga	tctgatatca	540
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caaatattat	aagtgaacgg	atcaactgct	ctttgtgtag	atcataaacg	aggacaaaag	660
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ataagggttc	cacactatca	atgtcctcta	gtgctccaat	tgccttgact	atgacatccc	780
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modified_base	note = thymine 140..141 mod_base = OTHER
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modified_base	227
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modified_base	327

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modified_base	830
	mod_base = OTHER
	note = thymine
modified_base	833..835
	mod_base = OTHER
	note = thymine
modified_base	849
	mod_base = OTHER
	note = thymine
modified_base	853..854
	mod_base = OTHER
	note = thymine
modified_base	857
	mod_base = OTHER
	note = thymine
modified_base	861
	mod_base = OTHER
	note = thymine
modified_base	869
	mod_base = OTHER
	note = thymine
modified_base	880..881
	mod_base = OTHER
	note = thymine
modified_base	886
	mod_base = OTHER
	note = thymine
modified_base	891..893
	mod_base = OTHER
	note = thymine
modified_base	895
	mod_base = OTHER
	note = thymine
modified_base	901
	mod_base = OTHER
	note = thymine
modified_base	903..904
	mod_base = OTHER
	note = thymine
modified_base	911
	mod_base = OTHER
	note = thymine
modified_base	918
	mod_base = OTHER
	note = thymine
modified_base	924
	mod_base = OTHER

-continued

modified_base	note = thymine 927..928 mod_base = OTHER note = thymine
modified_base	930..931 mod_base = OTHER note = thymine
modified_base	935 mod_base = OTHER note = thymine
modified_base	943 mod_base = OTHER note = thymine
modified_base	947 mod_base = OTHER note = thymine
modified_base	950..951 mod_base = OTHER note = thymine
modified_base	953..954 mod_base = OTHER note = thymine
modified_base	957 mod_base = OTHER note = thymine
modified_base	965..966 mod_base = OTHER note = thymine
modified_base	970 mod_base = OTHER note = thymine
modified_base	972..974 mod_base = OTHER note = thymine
modified_base	989..990 mod_base = OTHER note = thymine
modified_base	997 mod_base = OTHER note = thymine
modified_base	1002 mod_base = OTHER note = thymine
modified_base	1004 mod_base = OTHER note = thymine
modified_base	1008 mod_base = OTHER note = thymine
modified_base	1011 mod_base = OTHER note = thymine
modified_base	1014 mod_base = OTHER note = thymine
modified_base	1021..1022 mod_base = OTHER note = thymine
modified_base	1024 mod_base = OTHER note = thymine
modified_base	1026 mod_base = OTHER note = thymine
modified_base	1031 mod_base = OTHER note = thymine
modified_base	1034 mod_base = OTHER note = thymine
modified_base	1040 mod_base = OTHER note = thymine
modified_base	1043..1044 mod_base = OTHER note = thymine
modified_base	1048 mod_base = OTHER note = thymine

-continued

modified_base	1059
	mod_base = OTHER
	note = thymine
modified_base	1062..1064
	mod_base = OTHER
	note = thymine
modified_base	1068
	mod_base = OTHER
	note = thymine
modified_base	1072..1073
	mod_base = OTHER
	note = thymine
modified_base	1085
	mod_base = OTHER
	note = thymine
modified_base	1090
	mod_base = OTHER
	note = thymine
modified_base	1092
	mod_base = OTHER
	note = thymine
modified_base	1096..1097
	mod_base = OTHER
	note = thymine
modified_base	1102..1103
	mod_base = OTHER
	note = thymine
modified_base	1105..1106
	mod_base = OTHER
	note = thymine
modified_base	1118
	mod_base = OTHER
	note = thymine
modified_base	1122..1123
	mod_base = OTHER
	note = thymine
modified_base	1125..1127
	mod_base = OTHER
	note = thymine
modified_base	1130..1132
	mod_base = OTHER
	note = thymine
modified_base	1137
	mod_base = OTHER
	note = thymine
modified_base	1142
	mod_base = OTHER
	note = thymine
modified_base	1145
	mod_base = OTHER
	note = thymine
modified_base	1148
	mod_base = OTHER
	note = thymine
modified_base	1159..1160
	mod_base = OTHER
	note = thymine
modified_base	1167
	mod_base = OTHER
	note = thymine
modified_base	1169
	mod_base = OTHER
	note = thymine
modified_base	1172
	mod_base = OTHER
	note = thymine
modified_base	1178
	mod_base = OTHER
	note = thymine
modified_base	1183
	mod_base = OTHER
	note = thymine
modified_base	1185..1186
	mod_base = OTHER
	note = thymine
modified_base	1191..1192
	mod_base = OTHER
	note = thymine
modified_base	1195

-continued

	mod_base = OTHER
	note = thymine
modified_base	1201..1204
	mod_base = OTHER
	note = thymine
modified_base	1207
	mod_base = OTHER
	note = thymine
modified_base	1211
	mod_base = OTHER
	note = thymine
modified_base	1218
	mod_base = OTHER
	note = thymine
modified_base	1236
	mod_base = OTHER
	note = thymine
modified_base	1239
	mod_base = OTHER
	note = thymine
modified_base	1241
	mod_base = OTHER
	note = thymine
modified_base	1249
	mod_base = OTHER
	note = thymine
modified_base	1252
	mod_base = OTHER
	note = thymine
modified_base	1256..1257
	mod_base = OTHER
	note = thymine
modified_base	1259
	mod_base = OTHER
	note = thymine
modified_base	1266
	mod_base = OTHER
	note = thymine
modified_base	1269..1270
	mod_base = OTHER
	note = thymine
modified_base	1272..1273
	mod_base = OTHER
	note = thymine
modified_base	1276..1277
	mod_base = OTHER
	note = thymine
modified_base	1281
	mod_base = OTHER
	note = thymine
modified_base	1291..1292
	mod_base = OTHER
	note = thymine
modified_base	1296
	mod_base = OTHER
	note = thymine
modified_base	1304
	mod_base = OTHER
	note = thymine
modified_base	1320
	mod_base = OTHER
	note = thymine
modified_base	1325
	mod_base = OTHER
	note = thymine
modified_base	1328..1329
	mod_base = OTHER
	note = thymine
modified_base	1337
	mod_base = OTHER
	note = thymine
modified_base	1339..1340
	mod_base = OTHER
	note = thymine
modified_base	1350..1351
	mod_base = OTHER
	note = thymine
modified_base	1355
	mod_base = OTHER

-continued

modified_base	note = thymine 1371 mod_base = OTHER note = thymine
modified_base	1379 mod_base = OTHER note = thymine
modified_base	1384..1385 mod_base = OTHER note = thymine
modified_base	1388 mod_base = OTHER note = thymine
modified_base	1394..1395 mod_base = OTHER note = thymine
modified_base	1397 mod_base = OTHER note = thymine
modified_base	1400 mod_base = OTHER note = thymine
modified_base	1402..1405 mod_base = OTHER note = thymine
modified_base	1411 mod_base = OTHER note = thymine
modified_base	1413 mod_base = OTHER note = thymine
modified_base	1417 mod_base = OTHER note = thymine
modified_base	1419 mod_base = OTHER note = thymine
modified_base	1421 mod_base = OTHER note = thymine
modified_base	1424 mod_base = OTHER note = thymine
modified_base	1430 mod_base = OTHER note = thymine
modified_base	1432..1433 mod_base = OTHER note = thymine
modified_base	1436 mod_base = OTHER note = thymine
modified_base	1442..1443 mod_base = OTHER note = thymine
modified_base	1445 mod_base = OTHER note = thymine
modified_base	1451 mod_base = OTHER note = thymine
modified_base	1469 mod_base = OTHER note = thymine
modified_base	1476 mod_base = OTHER note = thymine
modified_base	1480 mod_base = OTHER note = thymine
modified_base	1483 mod_base = OTHER note = thymine
modified_base	1485 mod_base = OTHER note = thymine
modified_base	1501..1502 mod_base = OTHER note = thymine

-continued

modified_base	1517..1519 mod_base = OTHER note = thymine
modified_base	1521 mod_base = OTHER note = thymine
modified_base	1524..1525 mod_base = OTHER note = thymine
modified_base	1527 mod_base = OTHER note = thymine
modified_base	1530 mod_base = OTHER note = thymine
modified_base	1534..1536 mod_base = OTHER note = thymine
modified_base	1541 mod_base = OTHER note = thymine
modified_base	1545 mod_base = OTHER note = thymine
modified_base	1548 mod_base = OTHER note = thymine
modified_base	1550 mod_base = OTHER note = thymine
modified_base	1559 mod_base = OTHER note = thymine
modified_base	1561 mod_base = OTHER note = thymine
modified_base	1573 mod_base = OTHER note = thymine
modified_base	1587 mod_base = OTHER note = thymine
modified_base	1591 mod_base = OTHER note = thymine
modified_base	1595 mod_base = OTHER note = thymine
modified_base	1597 mod_base = OTHER note = thymine
modified_base	1603 mod_base = OTHER note = thymine
modified_base	1613 mod_base = OTHER note = thymine
modified_base	1620 mod_base = OTHER note = thymine
modified_base	1638 mod_base = OTHER note = thymine
modified_base	1640..1641 mod_base = OTHER note = thymine
modified_base	1646 mod_base = OTHER note = thymine
modified_base	1648 mod_base = OTHER note = thymine
modified_base	1656..1658 mod_base = OTHER note = thymine
modified_base	1664 mod_base = OTHER note = thymine
modified_base	1666

-continued

	mod_base = OTHER
	note = thymine
modified_base	1672..1673
	mod_base = OTHER
	note = thymine
modified_base	1677
	mod_base = OTHER
	note = thymine
modified_base	1680
	mod_base = OTHER
	note = thymine
modified_base	1682
	mod_base = OTHER
	note = thymine
modified_base	1690
	mod_base = OTHER
	note = thymine
modified_base	1692..1693
	mod_base = OTHER
	note = thymine
modified_base	1695
	mod_base = OTHER
	note = thymine
modified_base	1702..1705
	mod_base = OTHER
	note = thymine
modified_base	1710
	mod_base = OTHER
	note = thymine
modified_base	1712
	mod_base = OTHER
	note = thymine
modified_base	1720..1721
	mod_base = OTHER
	note = thymine
modified_base	1724..1726
	mod_base = OTHER
	note = thymine
modified_base	1735..1737
	mod_base = OTHER
	note = thymine
modified_base	1744
	mod_base = OTHER
	note = thymine
modified_base	1747
	mod_base = OTHER
	note = thymine
modified_base	1752
	mod_base = OTHER
	note = thymine
modified_base	1756
	mod_base = OTHER
	note = thymine
modified_base	1760
	mod_base = OTHER
	note = thymine
modified_base	1767
	mod_base = OTHER
	note = thymine
modified_base	1774
	mod_base = OTHER
	note = thymine
modified_base	1777
	mod_base = OTHER
	note = thymine
modified_base	1779
	mod_base = OTHER
	note = thymine
modified_base	1781..1783
	mod_base = OTHER
	note = thymine
modified_base	1785..1786
	mod_base = OTHER
	note = thymine
modified_base	1788..1792
	mod_base = OTHER
	note = thymine
modified_base	1794
	mod_base = OTHER

-continued

modified_base	note = thymine 1798..1799 mod_base = OTHER note = thymine
modified_base	1802 mod_base = OTHER note = thymine
modified_base	1804..1805 mod_base = OTHER note = thymine
modified_base	1810 mod_base = OTHER note = thymine
modified_base	1812 mod_base = OTHER note = thymine
modified_base	1816 mod_base = OTHER note = thymine
modified_base	1818..1819 mod_base = OTHER note = thymine
modified_base	1833 mod_base = OTHER note = thymine
modified_base	1835..1837 mod_base = OTHER note = thymine
modified_base	1839 mod_base = OTHER note = thymine
modified_base	1847..1849 mod_base = OTHER note = thymine
modified_base	1851 mod_base = OTHER note = thymine
modified_base	1853..1854 mod_base = OTHER note = thymine
modified_base	1857 mod_base = OTHER note = thymine
modified_base	1860 mod_base = OTHER note = thymine
modified_base	1864 mod_base = OTHER note = thymine
modified_base	1871..1872 mod_base = OTHER note = thymine
modified_base	1876 mod_base = OTHER note = thymine
modified_base	1879 mod_base = OTHER note = thymine
modified_base	1882..1884 mod_base = OTHER note = thymine
modified_base	1890 mod_base = OTHER note = thymine
modified_base	1894 mod_base = OTHER note = thymine
modified_base	1899..1900 mod_base = OTHER note = thymine
modified_base	1903 mod_base = OTHER note = thymine
modified_base	1905 mod_base = OTHER note = thymine
modified_base	1908..1909 mod_base = OTHER note = thymine

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modified_base	1912
	mod_base = OTHER
	note = thymine
modified_base	1914..1915
	mod_base = OTHER
	note = thymine
modified_base	1917
	mod_base = OTHER
	note = thymine
modified_base	1919
	mod_base = OTHER
	note = thymine
modified_base	1923
	mod_base = OTHER
	note = thymine
modified_base	1932..1934
	mod_base = OTHER
	note = thymine
modified_base	1936
	mod_base = OTHER
	note = thymine
modified_base	1939
	mod_base = OTHER
	note = thymine
modified_base	1941
	mod_base = OTHER
	note = thymine
modified_base	1944
	mod_base = OTHER
	note = thymine
modified_base	1952..1953
	mod_base = OTHER
	note = thymine
modified_base	1960..1961
	mod_base = OTHER
	note = thymine
modified_base	1969
	mod_base = OTHER
	note = thymine
modified_base	1974
	mod_base = OTHER
	note = thymine
modified_base	1988
	mod_base = OTHER
	note = thymine
modified_base	1990
	mod_base = OTHER
	note = thymine
modified_base	1994
	mod_base = OTHER
	note = thymine
modified_base	2002
	mod_base = OTHER
	note = thymine
modified_base	2012..2013
	mod_base = OTHER
	note = thymine
modified_base	2019..2020
	mod_base = OTHER
	note = thymine
modified_base	2028
	mod_base = OTHER
	note = thymine
modified_base	2037
	mod_base = OTHER
	note = thymine
modified_base	2041
	mod_base = OTHER
	note = thymine
modified_base	2046
	mod_base = OTHER
	note = thymine
modified_base	2048
	mod_base = OTHER
	note = thymine
modified_base	2052
	mod_base = OTHER
	note = thymine
modified_base	2055..2056

-continued

	mod_base = OTHER
	note = thymine
modified_base	2061
	mod_base = OTHER
	note = thymine
modified_base	2063..2064
	mod_base = OTHER
	note = thymine
modified_base	2067..2068
	mod_base = OTHER
	note = thymine
modified_base	2072..2073
	mod_base = OTHER
	note = thymine
modified_base	2075..2079
	mod_base = OTHER
	note = thymine
modified_base	2082
	mod_base = OTHER
	note = thymine
modified_base	2090..2091
	mod_base = OTHER
	note = thymine
modified_base	2095
	mod_base = OTHER
	note = thymine
modified_base	2097
	mod_base = OTHER
	note = thymine
modified_base	2103
	mod_base = OTHER
	note = thymine
modified_base	2105
	mod_base = OTHER
	note = thymine
modified_base	2107
	mod_base = OTHER
	note = thymine
modified_base	2112
	mod_base = OTHER
	note = thymine
modified_base	2118..2119
	mod_base = OTHER
	note = thymine
modified_base	2121
	mod_base = OTHER
	note = thymine
modified_base	2123
	mod_base = OTHER
	note = thymine
modified_base	2126..2127
	mod_base = OTHER
	note = thymine
modified_base	2134..2135
	mod_base = OTHER
	note = thymine
modified_base	2138
	mod_base = OTHER
	note = thymine
modified_base	2140..2142
	mod_base = OTHER
	note = thymine
modified_base	2147
	mod_base = OTHER
	note = thymine
modified_base	2151
	mod_base = OTHER
	note = thymine
modified_base	2154..2155
	mod_base = OTHER
	note = thymine
modified_base	2159
	mod_base = OTHER
	note = thymine
modified_base	2161
	mod_base = OTHER
	note = thymine
modified_base	2166
	mod_base = OTHER

-continued

modified_base	note = thymine 2170 mod_base = OTHER note = thymine
modified_base	2172 mod_base = OTHER note = thymine
modified_base	2176 mod_base = OTHER note = thymine
modified_base	2180 mod_base = OTHER note = thymine
modified_base	2182 mod_base = OTHER note = thymine
modified_base	2190..2191 mod_base = OTHER note = thymine
modified_base	2196 mod_base = OTHER note = thymine
modified_base	2200..2201 mod_base = OTHER note = thymine
modified_base	2204 mod_base = OTHER note = thymine
modified_base	2209 mod_base = OTHER note = thymine
modified_base	2214 mod_base = OTHER note = thymine
modified_base	2216..2219 mod_base = OTHER note = thymine
modified_base	2227..2228 mod_base = OTHER note = thymine
modified_base	2230 mod_base = OTHER note = thymine
modified_base	2232 mod_base = OTHER note = thymine
modified_base	2236 mod_base = OTHER note = thymine
modified_base	2238 mod_base = OTHER note = thymine
modified_base	2242..2243 mod_base = OTHER note = thymine
modified_base	2253 mod_base = OTHER note = thymine
modified_base	2258 mod_base = OTHER note = thymine
modified_base	2264 mod_base = OTHER note = thymine
modified_base	2267 mod_base = OTHER note = thymine
modified_base	2270 mod_base = OTHER note = thymine
modified_base	2279 mod_base = OTHER note = thymine
modified_base	2284 mod_base = OTHER note = thymine
modified_base	2288 mod_base = OTHER note = thymine

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modified_base	2292..2293 mod_base = OTHER note = thymine
modified_base	2297 mod_base = OTHER note = thymine
modified_base	2303..2305 mod_base = OTHER note = thymine
modified_base	2307 mod_base = OTHER note = thymine
modified_base	2311..2312 mod_base = OTHER note = thymine
modified_base	2314 mod_base = OTHER note = thymine
modified_base	2319 mod_base = OTHER note = thymine
modified_base	2323..2324 mod_base = OTHER note = thymine
modified_base	2326..2327 mod_base = OTHER note = thymine
modified_base	2329 mod_base = OTHER note = thymine
modified_base	2331..2334 mod_base = OTHER note = thymine
modified_base	2339 mod_base = OTHER note = thymine
modified_base	2342 mod_base = OTHER note = thymine
modified_base	2344 mod_base = OTHER note = thymine
modified_base	2353 mod_base = OTHER note = thymine
modified_base	2357 mod_base = OTHER note = thymine
modified_base	2359..2360 mod_base = OTHER note = thymine
modified_base	2362 mod_base = OTHER note = thymine
modified_base	2369 mod_base = OTHER note = thymine
modified_base	2371 mod_base = OTHER note = thymine
modified_base	2373 mod_base = OTHER note = thymine
modified_base	2375 mod_base = OTHER note = thymine
modified_base	2381 mod_base = OTHER note = thymine
modified_base	2384..2385 mod_base = OTHER note = thymine
modified_base	2390 mod_base = OTHER note = thymine
modified_base	2402..2404 mod_base = OTHER note = thymine
modified_base	2406

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	mod_base = OTHER
	note = thymine
modified_base	2413
	mod_base = OTHER
	note = thymine
modified_base	2415..2418
	mod_base = OTHER
	note = thymine
modified_base	2421
	mod_base = OTHER
	note = thymine
modified_base	2427..2428
	mod_base = OTHER
	note = thymine
modified_base	2431
	mod_base = OTHER
	note = thymine
modified_base	2433
	mod_base = OTHER
	note = thymine
modified_base	2441..2442
	mod_base = OTHER
	note = thymine
modified_base	2445
	mod_base = OTHER
	note = thymine
modified_base	2447
	mod_base = OTHER
	note = thymine
modified_base	2458
	mod_base = OTHER
	note = thymine
modified_base	2460..2461
	mod_base = OTHER
	note = thymine
modified_base	2465..2468
	mod_base = OTHER
	note = thymine
modified_base	2470
	mod_base = OTHER
	note = thymine
modified_base	2472..2473
	mod_base = OTHER
	note = thymine
modified_base	2486
	mod_base = OTHER
	note = thymine
modified_base	2490..2491
	mod_base = OTHER
	note = thymine
modified_base	2499
	mod_base = OTHER
	note = thymine
modified_base	2502
	mod_base = OTHER
	note = thymine
modified_base	2506
	mod_base = OTHER
	note = thymine
modified_base	2508..2509
	mod_base = OTHER
	note = thymine
modified_base	2512..2513
	mod_base = OTHER
	note = thymine
modified_base	2517..2518
	mod_base = OTHER
	note = thymine
modified_base	2521
	mod_base = OTHER
	note = thymine
modified_base	2527..2528
	mod_base = OTHER
	note = thymine
modified_base	2532
	mod_base = OTHER
	note = thymine
modified_base	2534
	mod_base = OTHER

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modified_base	note = thymine 2539 mod_base = OTHER note = thymine
modified_base	2541..2542 mod_base = OTHER note = thymine
modified_base	2550 mod_base = OTHER note = thymine
modified_base	2561 mod_base = OTHER note = thymine
modified_base	2567 mod_base = OTHER note = thymine
modified_base	2571..2572 mod_base = OTHER note = thymine
modified_base	2590 mod_base = OTHER note = thymine
modified_base	2595..2596 mod_base = OTHER note = thymine
modified_base	2598 mod_base = OTHER note = thymine
modified_base	2608 mod_base = OTHER note = thymine
modified_base	2610..2611 mod_base = OTHER note = thymine
modified_base	2614 mod_base = OTHER note = thymine
modified_base	2617 mod_base = OTHER note = thymine
modified_base	2622 mod_base = OTHER note = thymine
modified_base	2625 mod_base = OTHER note = thymine
modified_base	2627..2628 mod_base = OTHER note = thymine
modified_base	2632 mod_base = OTHER note = thymine
modified_base	2635 mod_base = OTHER note = thymine
modified_base	2648 mod_base = OTHER note = thymine
modified_base	2652 mod_base = OTHER note = thymine
modified_base	2655..2656 mod_base = OTHER note = thymine
modified_base	2658 mod_base = OTHER note = thymine
modified_base	2660..2662 mod_base = OTHER note = thymine
modified_base	2668..2670 mod_base = OTHER note = thymine
modified_base	2676 mod_base = OTHER note = thymine
modified_base	2679..2680 mod_base = OTHER note = thymine

-continued

modified_base	2685..2687
	mod_base = OTHER
	note = thymine
modified_base	2691
	mod_base = OTHER
	note = thymine
modified_base	2694..2695
	mod_base = OTHER
	note = thymine
modified_base	2699
	mod_base = OTHER
	note = thymine
modified_base	2701
	mod_base = OTHER
	note = thymine
modified_base	2703
	mod_base = OTHER
	note = thymine
modified_base	2718
	mod_base = OTHER
	note = thymine
modified_base	2720
	mod_base = OTHER
	note = thymine
modified_base	2722
	mod_base = OTHER
	note = thymine
modified_base	2729..2730
	mod_base = OTHER
	note = thymine
modified_base	2732
	mod_base = OTHER
	note = thymine
modified_base	2739
	mod_base = OTHER
	note = thymine
modified_base	2742..2743
	mod_base = OTHER
	note = thymine
modified_base	2752
	mod_base = OTHER
	note = thymine
modified_base	2754
	mod_base = OTHER
	note = thymine
modified_base	2756
	mod_base = OTHER
	note = thymine
modified_base	2760
	mod_base = OTHER
	note = thymine
modified_base	2762..2764
	mod_base = OTHER
	note = thymine
modified_base	2770
	mod_base = OTHER
	note = thymine
modified_base	2779
	mod_base = OTHER
	note = thymine
modified_base	2782..2783
	mod_base = OTHER
	note = thymine
modified_base	2788..2789
	mod_base = OTHER
	note = thymine
modified_base	2794..2795
	mod_base = OTHER
	note = thymine
modified_base	2797
	mod_base = OTHER
	note = thymine
modified_base	2804..2805
	mod_base = OTHER
	note = thymine
modified_base	2808..2809
	mod_base = OTHER
	note = thymine
modified_base	2811..2812

-continued

	mod_base = OTHER
	note = thymine
modified_base	2819
	mod_base = OTHER
	note = thymine
modified_base	2822
	mod_base = OTHER
	note = thymine
modified_base	2826
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modified_base	2828..2830
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	note = thymine
modified_base	2835
	mod_base = OTHER
	note = thymine
modified_base	2843
	mod_base = OTHER
	note = thymine
modified_base	2845..2846
	mod_base = OTHER
	note = thymine
modified_base	2857..2858
	mod_base = OTHER
	note = thymine
modified_base	2865
	mod_base = OTHER
	note = thymine
modified_base	2868
	mod_base = OTHER
	note = thymine
modified_base	2874..2876
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modified_base	2878
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modified_base	2881
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modified_base	2901
	mod_base = OTHER
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modified_base	2906
	mod_base = OTHER
	note = thymine
modified_base	2908..2910
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	note = thymine
modified_base	2913
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	note = thymine
modified_base	2915
	mod_base = OTHER
	note = thymine
modified_base	2918..2919
	mod_base = OTHER
	note = thymine
modified_base	2922
	mod_base = OTHER
	note = thymine
modified_base	2924
	mod_base = OTHER

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modified_base	note = thymine 2929..2930 mod_base = OTHER
modified_base	note = thymine 2934..2935 mod_base = OTHER
modified_base	note = thymine 2941 mod_base = OTHER
modified_base	note = thymine 2943 mod_base = OTHER
modified_base	note = thymine 2945 mod_base = OTHER
modified_base	note = thymine 2951 mod_base = OTHER
modified_base	note = thymine 2958 mod_base = OTHER
modified_base	note = thymine 2965..2967 mod_base = OTHER
modified_base	note = thymine 2974 mod_base = OTHER
modified_base	note = thymine 2977 mod_base = OTHER
modified_base	note = thymine 2980 mod_base = OTHER
modified_base	note = thymine 2982..2983 mod_base = OTHER
modified_base	note = thymine 2988..2989 mod_base = OTHER
modified_base	note = thymine 2994..2995 mod_base = OTHER
modified_base	note = thymine 2997 mod_base = OTHER
modified_base	note = thymine 3003 mod_base = OTHER
modified_base	note = thymine 3019 mod_base = OTHER
modified_base	note = thymine 3023..3024 mod_base = OTHER
modified_base	note = thymine 3026..3027 mod_base = OTHER
modified_base	note = thymine 3029 mod_base = OTHER
modified_base	note = thymine 3034 mod_base = OTHER
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modified_base	note = thymine 3047..3049 mod_base = OTHER
modified_base	note = thymine 3057..3058 mod_base = OTHER
modified_base	note = thymine 3060..3061 mod_base = OTHER
	note = thymine

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modified_base	3068
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modified_base	3070..3071
	mod_base = OTHER
	note = thymine
modified_base	3075..3077
	mod_base = OTHER
	note = thymine
modified_base	3081
	mod_base = OTHER
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	note = thymine
modified_base	3095
	mod_base = OTHER
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modified_base	3100
	mod_base = OTHER
	note = thymine
modified_base	3102
	mod_base = OTHER
	note = thymine
modified_base	3105..3106
	mod_base = OTHER
	note = thymine
modified_base	3112..3113
	mod_base = OTHER
	note = thymine
modified_base	3115
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modified_base	3131..3133
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modified_base	3135
	mod_base = OTHER
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	mod_base = OTHER
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modified_base	3140
	mod_base = OTHER
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modified_base	3142
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	note = thymine
modified_base	3144..3147
	mod_base = OTHER
	note = thymine
modified_base	3150..3151
	mod_base = OTHER
	note = thymine
modified_base	3154
	mod_base = OTHER
	note = thymine
modified_base	3157
	mod_base = OTHER
	note = thymine
modified_base	3160

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modified_base	3163
	mod_base = OTHER
	note = thymine
modified_base	3165
	mod_base = OTHER
	note = thymine
modified_base	3171..3172
	mod_base = OTHER
	note = thymine
modified_base	3175
	mod_base = OTHER
	note = thymine
modified_base	3180
	mod_base = OTHER
	note = thymine
modified_base	3189
	mod_base = OTHER
	note = thymine
modified_base	3198..3199
	mod_base = OTHER
	note = thymine
modified_base	3207..3208
	mod_base = OTHER
	note = thymine
modified_base	3211..3212
	mod_base = OTHER
	note = thymine
modified_base	3216
	mod_base = OTHER
	note = thymine
modified_base	3219..3220
	mod_base = OTHER
	note = thymine
modified_base	3223
	mod_base = OTHER
	note = thymine
modified_base	3234..3235
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	mod_base = OTHER
	note = thymine
modified_base	3242
	mod_base = OTHER
	note = thymine
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	mod_base = OTHER
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	note = thymine
modified_base	3253
	mod_base = OTHER
	note = thymine
modified_base	3255
	mod_base = OTHER
	note = thymine
modified_base	3260..3261
	mod_base = OTHER
	note = thymine
modified_base	3263
	mod_base = OTHER
	note = thymine
modified_base	3265..3266
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modified_base	3268
	mod_base = OTHER
	note = thymine
modified_base	3272
	mod_base = OTHER
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modified_base	3275..3276
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modified_base	3280
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modified_base      note = thymine
                   3292
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modified_base      note = thymine
                   3294..3295
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modified_base      note = thymine
                   3299..3300
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modified_base      note = thymine
                   3303..3304
                   mod_base = OTHER
modified_base      note = thymine
                   3316
                   mod_base = OTHER
modified_base      note = thymine
                   3318..3319
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modified_base      note = thymine
                   3321..3322
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modified_base      note = thymine
                   3325
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modified_base      note = thymine
                   3329
                   mod_base = OTHER
modified_base      note = thymine
                   3332..3334
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modified_base      note = thymine
                   3336
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modified_base      note = thymine
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modified_base      note = thymine
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modified_base      note = thymine
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misc_feature       1..3377
                   note = Lymphocytic choriomeningitis mammarenavirus Viruses
                   (LCMV); Lymphocytic choriomeningitis virus clone 13
                   segment S, complete sequence.ACCESION DQ361065-VERSION
                   DQ361065.2-GI:116563461.-ssRNA viruses; ssRNA
                   negative-strand viruses; Arenaviridae; Mammarenavirus.
source             1..3377
                   mol_type = other RNA
                   organism = unidentified

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gtctacaatt ttgccacctg tgggatattc gcattgatca gtttctact tctggctggc 240
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aagtcagtgg agtttgatat gtcacatctg aacctgacca tgcccaacgc atgttcagcc 360
aacaactccc accattacat cagtatgggg acttctggac tagaattgac cttcaccaat 420
gattccatca tcagtcacaa cttttgcaat ctgacctctg ccttcaacaa aaagacctt 480
gaccacacac tcatgagtat agtttcgagc ctacacctca gtatcagagg gaactccaac 540
tataaggcag tatcctgcga cttcaacaat ggcataacca tccaatacaa cttgacattc 600
tcagatgcac aaagtgtctc gagccagtgt agaaccttca gaggtagagt ctagatatg 660

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tggaagaaacc actgcacata tgcaggtcct tttaggatgt ccaggtattct cctttcccaa 840
gagaagacta agttctctac taggagacta gcgggcacat tcacctggac ttgtcagac 900
tcttcagggg tggagaatcc aggtgggttat tgcctgacca aatggatgat tcttgctgca 960
gagcttaagt gtttcgggaa cacagcagtt gcgaaatgca atgtaaatca tgatgaagaa 1020
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gacgtagaat ctgccttgca cttattcaaa acaacagtga attctttgat ttcagatcaa 1140
ctactgatga ggaaccactt gagagatctg atgggggtgc catattgcaa ttactcaaag 1200
ttttggtacc tagaacatgc aaagaccggc gaaactagt tccccaagt ctggctgtgc 1260
accaatggtt cttacttaaa tgagacccac ttcagtgaac aaatcgaaac ggaagccgat 1320
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gcattgatgg acctctgat gttttccaca tctgcatac tagtcagcat cttctgcac 1440
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ttaaccaaca aaggaatttg tagttgtggt gcatttaagg tgcttggtgt aaaaaccgtc 1560
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gagcacctat aactgacgag gtcaaacccg gttgcgcat gaagagtgct gcaagatcca 2040
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ctgatgtgaa gctctgcaat tctcttctca atgcttgctg ccattggaag ctcttaactt 3300
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ctaggatcca ctgtgctg 3377

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SEQ ID NO: 6      moltype = RNA length = 7229
FEATURE          Location/Qualifiers
misc_feature      1..7229
                  note = Lymphocytic choriomeningitis mammarenavirus Viruses
                  (LCMV); Lymphocytic choriomeningitis virus clone 13
                  segment L, complete sequence.-ACCESSION DQ361066-VERSION
                  DQ361066.1-GI:86440167.-ssRNA viruses; ssRNA
                  negative-strand viruses; Arenaviridae; Mammarenavirus
source            1..7229
                  mol_type = other RNA
                  organism = unidentified
modified_base     13
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modified_base     16
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                  note = thymine
modified_base     22..24
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                  note = thymine
modified_base     27..28
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                  note = thymine
modified_base     33
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modified_base     35..37
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                  note = thymine
modified_base     40..41
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                  note = thymine

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modified_base	49..51 mod_base = OTHER note = thymine
modified_base	53..54 mod_base = OTHER note = thymine
modified_base	57 mod_base = OTHER note = thymine
modified_base	63 mod_base = OTHER note = thymine
modified_base	65 mod_base = OTHER note = thymine
modified_base	72 mod_base = OTHER note = thymine
modified_base	78 mod_base = OTHER note = thymine
modified_base	82 mod_base = OTHER note = thymine
modified_base	85 mod_base = OTHER note = thymine
modified_base	91 mod_base = OTHER note = thymine
modified_base	95 mod_base = OTHER note = thymine
modified_base	105 mod_base = OTHER note = thymine
modified_base	128 mod_base = OTHER note = thymine
modified_base	131 mod_base = OTHER note = thymine
modified_base	148 mod_base = OTHER note = thymine
modified_base	151 mod_base = OTHER note = thymine
modified_base	158 mod_base = OTHER note = thymine
modified_base	165 mod_base = OTHER note = thymine
modified_base	167 mod_base = OTHER note = thymine
modified_base	169..170 mod_base = OTHER note = thymine
modified_base	176..178 mod_base = OTHER note = thymine
modified_base	183 mod_base = OTHER note = thymine
modified_base	189 mod_base = OTHER note = thymine
modified_base	191..192 mod_base = OTHER note = thymine
modified_base	195 mod_base = OTHER note = thymine
modified_base	204..206 mod_base = OTHER note = thymine
modified_base	213..214

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	note = thymine
modified_base	222
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	note = thymine
modified_base	227
	mod_base = OTHER
	note = thymine
modified_base	234
	mod_base = OTHER
	note = thymine
modified_base	238..240
	mod_base = OTHER
	note = thymine
modified_base	249
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	note = thymine
modified_base	251..253
	mod_base = OTHER
	note = thymine
modified_base	259..260
	mod_base = OTHER
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modified_base	284
	mod_base = OTHER
	note = thymine
modified_base	287
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	note = thymine
modified_base	289..291
	mod_base = OTHER
	note = thymine
modified_base	293
	mod_base = OTHER
	note = thymine
modified_base	297
	mod_base = OTHER
	note = thymine
modified_base	299
	mod_base = OTHER
	note = thymine
modified_base	303..304
	mod_base = OTHER
	note = thymine
modified_base	315..316
	mod_base = OTHER
	note = thymine
modified_base	322
	mod_base = OTHER
	note = thymine
modified_base	324
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	note = thymine
modified_base	339
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	note = thymine
modified_base	341
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-continued

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modified_base	note = thymine 620..621 mod_base = OTHER
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modified_base	note = thymine 650 mod_base = OTHER
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modified_base	note = thymine 660 mod_base = OTHER
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modified_base	note = thymine 666 mod_base = OTHER
modified_base	note = thymine 670..672 mod_base = OTHER
	note = thymine

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modified_base	685
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modified_base	687
	mod_base = OTHER
	note = thymine
modified_base	692
	mod_base = OTHER
	note = thymine
modified_base	695..696
	mod_base = OTHER
	note = thymine
modified_base	710
	mod_base = OTHER
	note = thymine
modified_base	712
	mod_base = OTHER
	note = thymine
modified_base	727
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modified_base	732
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modified_base	741
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	mod_base = OTHER
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modified_base	757
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modified_base	760..761
	mod_base = OTHER
	note = thymine
modified_base	764
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modified_base	771..772
	mod_base = OTHER
	note = thymine
modified_base	777..778
	mod_base = OTHER
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modified_base	780
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modified_base	782
	mod_base = OTHER
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modified_base	797
	mod_base = OTHER
	note = thymine
modified_base	799..800
	mod_base = OTHER
	note = thymine
modified_base	807..808
	mod_base = OTHER
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modified_base	810

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	note = thymine
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modified_base	846
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modified_base	849
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modified_base	851..854
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modified_base	863
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modified_base	874
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modified_base	888
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modified_base	893..894
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modified_base	905..906
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modified_base	917..920
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modified_base	929..931
	mod_base = OTHER
	note = thymine
modified_base	933..934
	mod_base = OTHER
	note = thymine
modified_base	937..938
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modified_base	954
	mod_base = OTHER

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modified_base	963 mod_base = OTHER note = thymine
modified_base	967 mod_base = OTHER note = thymine
modified_base	977 mod_base = OTHER note = thymine
modified_base	983 mod_base = OTHER note = thymine
modified_base	988 mod_base = OTHER note = thymine
modified_base	994 mod_base = OTHER note = thymine
modified_base	996 mod_base = OTHER note = thymine
modified_base	998 mod_base = OTHER note = thymine
modified_base	1004 mod_base = OTHER note = thymine
modified_base	1008 mod_base = OTHER note = thymine
modified_base	1012 mod_base = OTHER note = thymine
modified_base	1014 mod_base = OTHER note = thymine
modified_base	1017 mod_base = OTHER note = thymine
modified_base	1022 mod_base = OTHER note = thymine
modified_base	1027 mod_base = OTHER note = thymine
modified_base	1029..1030 mod_base = OTHER note = thymine
modified_base	1032..1035 mod_base = OTHER note = thymine
modified_base	1049 mod_base = OTHER note = thymine
modified_base	1055 mod_base = OTHER note = thymine
modified_base	1064 mod_base = OTHER note = thymine
modified_base	1072 mod_base = OTHER note = thymine
modified_base	1074 mod_base = OTHER note = thymine
modified_base	1082 mod_base = OTHER note = thymine
modified_base	1084..1085 mod_base = OTHER note = thymine
modified_base	1089 mod_base = OTHER note = thymine

-continued

modified_base	1091
	mod_base = OTHER
	note = thymine
modified_base	1094
	mod_base = OTHER
	note = thymine
modified_base	1097
	mod_base = OTHER
	note = thymine
modified_base	1103
	mod_base = OTHER
	note = thymine
modified_base	1105..1106
	mod_base = OTHER
	note = thymine
modified_base	1111
	mod_base = OTHER
	note = thymine
modified_base	1113
	mod_base = OTHER
	note = thymine
modified_base	1116
	mod_base = OTHER
	note = thymine
modified_base	1118
	mod_base = OTHER
	note = thymine
modified_base	1130
	mod_base = OTHER
	note = thymine
modified_base	1133..1134
	mod_base = OTHER
	note = thymine
modified_base	1139..1140
	mod_base = OTHER
	note = thymine
modified_base	1151
	mod_base = OTHER
	note = thymine
modified_base	1153
	mod_base = OTHER
	note = thymine
modified_base	1155
	mod_base = OTHER
	note = thymine
modified_base	1158
	mod_base = OTHER
	note = thymine
modified_base	1162
	mod_base = OTHER
	note = thymine
modified_base	1166
	mod_base = OTHER
	note = thymine
modified_base	1174..1175
	mod_base = OTHER
	note = thymine
modified_base	1178
	mod_base = OTHER
	note = thymine
modified_base	1185..1186
	mod_base = OTHER
	note = thymine
modified_base	1188..1189
	mod_base = OTHER
	note = thymine
modified_base	1191
	mod_base = OTHER
	note = thymine
modified_base	1193
	mod_base = OTHER
	note = thymine
modified_base	1195
	mod_base = OTHER
	note = thymine
modified_base	1197
	mod_base = OTHER
	note = thymine
modified_base	1199

-continued

	mod_base = OTHER
	note = thymine
modified_base	1205..1206
	mod_base = OTHER
	note = thymine
modified_base	1214
	mod_base = OTHER
	note = thymine
modified_base	1222
	mod_base = OTHER
	note = thymine
modified_base	1225
	mod_base = OTHER
	note = thymine
modified_base	1231..1232
	mod_base = OTHER
	note = thymine
modified_base	1243
	mod_base = OTHER
	note = thymine
modified_base	1247..1248
	mod_base = OTHER
	note = thymine
modified_base	1250
	mod_base = OTHER
	note = thymine
modified_base	1252
	mod_base = OTHER
	note = thymine
modified_base	1255..1256
	mod_base = OTHER
	note = thymine
modified_base	1263
	mod_base = OTHER
	note = thymine
modified_base	1273
	mod_base = OTHER
	note = thymine
modified_base	1275
	mod_base = OTHER
	note = thymine
modified_base	1279
	mod_base = OTHER
	note = thymine
modified_base	1285..1287
	mod_base = OTHER
	note = thymine
modified_base	1290..1291
	mod_base = OTHER
	note = thymine
modified_base	1298
	mod_base = OTHER
	note = thymine
modified_base	1302
	mod_base = OTHER
	note = thymine
modified_base	1304..1305
	mod_base = OTHER
	note = thymine
modified_base	1307..1308
	mod_base = OTHER
	note = thymine
modified_base	1310
	mod_base = OTHER
	note = thymine
modified_base	1313..1314
	mod_base = OTHER
	note = thymine
modified_base	1316
	mod_base = OTHER
	note = thymine
modified_base	1320
	mod_base = OTHER
	note = thymine
modified_base	1325
	mod_base = OTHER
	note = thymine
modified_base	1330
	mod_base = OTHER

-continued

modified_base	note = thymine 1335..1336 mod_base = OTHER note = thymine
modified_base	1343 mod_base = OTHER note = thymine
modified_base	1348 mod_base = OTHER note = thymine
modified_base	1354..1357 mod_base = OTHER note = thymine
modified_base	1359 mod_base = OTHER note = thymine
modified_base	1361 mod_base = OTHER note = thymine
modified_base	1364..1365 mod_base = OTHER note = thymine
modified_base	1371 mod_base = OTHER note = thymine
modified_base	1373 mod_base = OTHER note = thymine
modified_base	1376 mod_base = OTHER note = thymine
modified_base	1385 mod_base = OTHER note = thymine
modified_base	1388..1390 mod_base = OTHER note = thymine
modified_base	1396..1398 mod_base = OTHER note = thymine
modified_base	1402 mod_base = OTHER note = thymine
modified_base	1406..1408 mod_base = OTHER note = thymine
modified_base	1412..1414 mod_base = OTHER note = thymine
modified_base	1416 mod_base = OTHER note = thymine
modified_base	1424 mod_base = OTHER note = thymine
modified_base	1428 mod_base = OTHER note = thymine
modified_base	1431 mod_base = OTHER note = thymine
modified_base	1437 mod_base = OTHER note = thymine
modified_base	1445..1446 mod_base = OTHER note = thymine
modified_base	1457 mod_base = OTHER note = thymine
modified_base	1459 mod_base = OTHER note = thymine
modified_base	1464..1465 mod_base = OTHER note = thymine
modified_base	1469 mod_base = OTHER note = thymine

-continued

modified_base	1493 mod_base = OTHER note = thymine
modified_base	1499 mod_base = OTHER note = thymine
modified_base	1502..1503 mod_base = OTHER note = thymine
modified_base	1506 mod_base = OTHER note = thymine
modified_base	1512 mod_base = OTHER note = thymine
modified_base	1518 mod_base = OTHER note = thymine
modified_base	1520 mod_base = OTHER note = thymine
modified_base	1526..1530 mod_base = OTHER note = thymine
modified_base	1532..1533 mod_base = OTHER note = thymine
modified_base	1541..1543 mod_base = OTHER note = thymine
modified_base	1552 mod_base = OTHER note = thymine
modified_base	1555..1556 mod_base = OTHER note = thymine
modified_base	1570 mod_base = OTHER note = thymine
modified_base	1574 mod_base = OTHER note = thymine
modified_base	1583 mod_base = OTHER note = thymine
modified_base	1585..1586 mod_base = OTHER note = thymine
modified_base	1591 mod_base = OTHER note = thymine
modified_base	1596 mod_base = OTHER note = thymine
modified_base	1599..1601 mod_base = OTHER note = thymine
modified_base	1603..1605 mod_base = OTHER note = thymine
modified_base	1607 mod_base = OTHER note = thymine
modified_base	1618 mod_base = OTHER note = thymine
modified_base	1625..1626 mod_base = OTHER note = thymine
modified_base	1630 mod_base = OTHER note = thymine
modified_base	1633 mod_base = OTHER note = thymine
modified_base	1635 mod_base = OTHER note = thymine
modified_base	1638 mod_base = OTHER note = thymine

-continued

	mod_base = OTHER
	note = thymine
modified_base	1641..1642
	mod_base = OTHER
	note = thymine
modified_base	1644..1645
	mod_base = OTHER
	note = thymine
modified_base	1654..1655
	mod_base = OTHER
	note = thymine
modified_base	1657
	mod_base = OTHER
	note = thymine
modified_base	1666
	mod_base = OTHER
	note = thymine
modified_base	1669..1670
	mod_base = OTHER
	note = thymine
modified_base	1676
	mod_base = OTHER
	note = thymine
modified_base	1682
	mod_base = OTHER
	note = thymine
modified_base	1684..1687
	mod_base = OTHER
	note = thymine
modified_base	1694..1695
	mod_base = OTHER
	note = thymine
modified_base	1697..1698
	mod_base = OTHER
	note = thymine
modified_base	1706
	mod_base = OTHER
	note = thymine
modified_base	1710
	mod_base = OTHER
	note = thymine
modified_base	1713
	mod_base = OTHER
	note = thymine
modified_base	1716
	mod_base = OTHER
	note = thymine
modified_base	1726..1727
	mod_base = OTHER
	note = thymine
modified_base	1730
	mod_base = OTHER
	note = thymine
modified_base	1740
	mod_base = OTHER
	note = thymine
modified_base	1745
	mod_base = OTHER
	note = thymine
modified_base	1750
	mod_base = OTHER
	note = thymine
modified_base	1754
	mod_base = OTHER
	note = thymine
modified_base	1757
	mod_base = OTHER
	note = thymine
modified_base	1763..1764
	mod_base = OTHER
	note = thymine
modified_base	1772
	mod_base = OTHER
	note = thymine
modified_base	1788
	mod_base = OTHER
	note = thymine
modified_base	1790
	mod_base = OTHER

-continued

modified_base	note = thymine 1793 mod_base = OTHER note = thymine
modified_base	1795..1796 mod_base = OTHER note = thymine
modified_base	1799 mod_base = OTHER note = thymine
modified_base	1803..1806 mod_base = OTHER note = thymine
modified_base	1811..1812 mod_base = OTHER note = thymine
modified_base	1820 mod_base = OTHER note = thymine
modified_base	1827 mod_base = OTHER note = thymine
modified_base	1831 mod_base = OTHER note = thymine
modified_base	1840..1841 mod_base = OTHER note = thymine
modified_base	1843 mod_base = OTHER note = thymine
modified_base	1847..1848 mod_base = OTHER note = thymine
modified_base	1851 mod_base = OTHER note = thymine
modified_base	1853 mod_base = OTHER note = thymine
modified_base	1857..1858 mod_base = OTHER note = thymine
modified_base	1871 mod_base = OTHER note = thymine
modified_base	1879..1881 mod_base = OTHER note = thymine
modified_base	1884..1885 mod_base = OTHER note = thymine
modified_base	1887 mod_base = OTHER note = thymine
modified_base	1893 mod_base = OTHER note = thymine
modified_base	1899 mod_base = OTHER note = thymine
modified_base	1902 mod_base = OTHER note = thymine
modified_base	1904..1909 mod_base = OTHER note = thymine
modified_base	1914 mod_base = OTHER note = thymine
modified_base	1917 mod_base = OTHER note = thymine
modified_base	1919 mod_base = OTHER note = thymine
modified_base	1926 mod_base = OTHER note = thymine

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modified_base	1930
	mod_base = OTHER
	note = thymine
modified_base	1940
	mod_base = OTHER
	note = thymine
modified_base	1943
	mod_base = OTHER
	note = thymine
modified_base	1946
	mod_base = OTHER
	note = thymine
modified_base	1953
	mod_base = OTHER
	note = thymine
modified_base	1955..1956
	mod_base = OTHER
	note = thymine
modified_base	1961
	mod_base = OTHER
	note = thymine
modified_base	1963
	mod_base = OTHER
	note = thymine
modified_base	1971
	mod_base = OTHER
	note = thymine
modified_base	1978..1979
	mod_base = OTHER
	note = thymine
modified_base	1983
	mod_base = OTHER
	note = thymine
modified_base	1987..1988
	mod_base = OTHER
	note = thymine
modified_base	1991..1992
	mod_base = OTHER
	note = thymine
modified_base	1997..1998
	mod_base = OTHER
	note = thymine
modified_base	2002..2004
	mod_base = OTHER
	note = thymine
modified_base	2006
	mod_base = OTHER
	note = thymine
modified_base	2009
	mod_base = OTHER
	note = thymine
modified_base	2014
	mod_base = OTHER
	note = thymine
modified_base	2019..2020
	mod_base = OTHER
	note = thymine
modified_base	2025
	mod_base = OTHER
	note = thymine
modified_base	2027
	mod_base = OTHER
	note = thymine
modified_base	2033..2035
	mod_base = OTHER
	note = thymine
modified_base	2037..2038
	mod_base = OTHER
	note = thymine
modified_base	2041
	mod_base = OTHER
	note = thymine
modified_base	2045
	mod_base = OTHER
	note = thymine
modified_base	2048
	mod_base = OTHER
	note = thymine
modified_base	2053

-continued

	mod_base = OTHER
	note = thymine
modified_base	2057..2059
	mod_base = OTHER
	note = thymine
modified_base	2063
	mod_base = OTHER
	note = thymine
modified_base	2066..2067
	mod_base = OTHER
	note = thymine
modified_base	2073..2074
	mod_base = OTHER
	note = thymine
modified_base	2078..2079
	mod_base = OTHER
	note = thymine
modified_base	2081
	mod_base = OTHER
	note = thymine
modified_base	2085
	mod_base = OTHER
	note = thymine
modified_base	2087..2088
	mod_base = OTHER
	note = thymine
modified_base	2090..2091
	mod_base = OTHER
	note = thymine
modified_base	2093..2094
	mod_base = OTHER
	note = thymine
modified_base	2098..2100
	mod_base = OTHER
	note = thymine
modified_base	2103
	mod_base = OTHER
	note = thymine
modified_base	2107..2108
	mod_base = OTHER
	note = thymine
modified_base	2110
	mod_base = OTHER
	note = thymine
modified_base	2113
	mod_base = OTHER
	note = thymine
modified_base	2132
	mod_base = OTHER
	note = thymine
modified_base	2141..2142
	mod_base = OTHER
	note = thymine
modified_base	2145
	mod_base = OTHER
	note = thymine
modified_base	2149
	mod_base = OTHER
	note = thymine
modified_base	2154
	mod_base = OTHER
	note = thymine
modified_base	2160
	mod_base = OTHER
	note = thymine
modified_base	2162
	mod_base = OTHER
	note = thymine
modified_base	2165
	mod_base = OTHER
	note = thymine
modified_base	2181..2182
	mod_base = OTHER
	note = thymine
modified_base	2191
	mod_base = OTHER
	note = thymine
modified_base	2193
	mod_base = OTHER

-continued

modified_base	note = thymine 2195 mod_base = OTHER note = thymine
modified_base	2198 mod_base = OTHER note = thymine
modified_base	2204 mod_base = OTHER note = thymine
modified_base	2209..2210 mod_base = OTHER note = thymine
modified_base	2212 mod_base = OTHER note = thymine
modified_base	2216 mod_base = OTHER note = thymine
modified_base	2222 mod_base = OTHER note = thymine
modified_base	2224..2226 mod_base = OTHER note = thymine
modified_base	2228 mod_base = OTHER note = thymine
modified_base	2235 mod_base = OTHER note = thymine
modified_base	2237 mod_base = OTHER note = thymine
modified_base	2244..2245 mod_base = OTHER note = thymine
modified_base	2247 mod_base = OTHER note = thymine
modified_base	2252 mod_base = OTHER note = thymine
modified_base	2258..2259 mod_base = OTHER note = thymine
modified_base	2266 mod_base = OTHER note = thymine
modified_base	2268..2269 mod_base = OTHER note = thymine
modified_base	2271 mod_base = OTHER note = thymine
modified_base	2276 mod_base = OTHER note = thymine
modified_base	2279 mod_base = OTHER note = thymine
modified_base	2282 mod_base = OTHER note = thymine
modified_base	2284 mod_base = OTHER note = thymine
modified_base	2286 mod_base = OTHER note = thymine
modified_base	2289..2290 mod_base = OTHER note = thymine
modified_base	2294 mod_base = OTHER note = thymine
modified_base	2297 mod_base = OTHER note = thymine

-continued

modified_base	2306..2310 mod_base = OTHER note = thymine
modified_base	2313 mod_base = OTHER note = thymine
modified_base	2315 mod_base = OTHER note = thymine
modified_base	2321..2324 mod_base = OTHER note = thymine
modified_base	2328 mod_base = OTHER note = thymine
modified_base	2330..2334 mod_base = OTHER note = thymine
modified_base	2336..2338 mod_base = OTHER note = thymine
modified_base	2340 mod_base = OTHER note = thymine
modified_base	2349 mod_base = OTHER note = thymine
modified_base	2352..2353 mod_base = OTHER note = thymine
modified_base	2356 mod_base = OTHER note = thymine
modified_base	2361..2364 mod_base = OTHER note = thymine
modified_base	2374 mod_base = OTHER note = thymine
modified_base	2376 mod_base = OTHER note = thymine
modified_base	2386 mod_base = OTHER note = thymine
modified_base	2391 mod_base = OTHER note = thymine
modified_base	2397 mod_base = OTHER note = thymine
modified_base	2399 mod_base = OTHER note = thymine
modified_base	2407 mod_base = OTHER note = thymine
modified_base	2411..2412 mod_base = OTHER note = thymine
modified_base	2414..2415 mod_base = OTHER note = thymine
modified_base	2418..2419 mod_base = OTHER note = thymine
modified_base	2422..2423 mod_base = OTHER note = thymine
modified_base	2425 mod_base = OTHER note = thymine
modified_base	2435..2436 mod_base = OTHER note = thymine
modified_base	2438 mod_base = OTHER note = thymine
modified_base	2440

-continued

	mod_base = OTHER
	note = thymine
modified_base	2448..2449
	mod_base = OTHER
	note = thymine
modified_base	2451
	mod_base = OTHER
	note = thymine
modified_base	2455..2457
	mod_base = OTHER
	note = thymine
modified_base	2460
	mod_base = OTHER
	note = thymine
modified_base	2467..2469
	mod_base = OTHER
	note = thymine
modified_base	2471..2472
	mod_base = OTHER
	note = thymine
modified_base	2474
	mod_base = OTHER
	note = thymine
modified_base	2479
	mod_base = OTHER
	note = thymine
modified_base	2481
	mod_base = OTHER
	note = thymine
modified_base	2483..2484
	mod_base = OTHER
	note = thymine
modified_base	2487
	mod_base = OTHER
	note = thymine
modified_base	2491
	mod_base = OTHER
	note = thymine
modified_base	2494
	mod_base = OTHER
	note = thymine
modified_base	2499..2500
	mod_base = OTHER
	note = thymine
modified_base	2502..2504
	mod_base = OTHER
	note = thymine
modified_base	2507
	mod_base = OTHER
	note = thymine
modified_base	2509..2510
	mod_base = OTHER
	note = thymine
modified_base	2517
	mod_base = OTHER
	note = thymine
modified_base	2520
	mod_base = OTHER
	note = thymine
modified_base	2524
	mod_base = OTHER
	note = thymine
modified_base	2527
	mod_base = OTHER
	note = thymine
modified_base	2531
	mod_base = OTHER
	note = thymine
modified_base	2540..2543
	mod_base = OTHER
	note = thymine
modified_base	2545
	mod_base = OTHER
	note = thymine
modified_base	2547..2548
	mod_base = OTHER
	note = thymine
modified_base	2556
	mod_base = OTHER

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modified_base	note = thymine 2559..2560 mod_base = OTHER note = thymine
modified_base	2563 mod_base = OTHER note = thymine
modified_base	2571..2573 mod_base = OTHER note = thymine
modified_base	2576..2577 mod_base = OTHER note = thymine
modified_base	2582..2583 mod_base = OTHER note = thymine
modified_base	2591..2592 mod_base = OTHER note = thymine
modified_base	2597 mod_base = OTHER note = thymine
modified_base	2599 mod_base = OTHER note = thymine
modified_base	2602 mod_base = OTHER note = thymine
modified_base	2606 mod_base = OTHER note = thymine
modified_base	2609..2610 mod_base = OTHER note = thymine
modified_base	2615 mod_base = OTHER note = thymine
modified_base	2617..2618 mod_base = OTHER note = thymine
modified_base	2620..2621 mod_base = OTHER note = thymine
modified_base	2636 mod_base = OTHER note = thymine
modified_base	2638 mod_base = OTHER note = thymine
modified_base	2644..2645 mod_base = OTHER note = thymine
modified_base	2652 mod_base = OTHER note = thymine
modified_base	2657 mod_base = OTHER note = thymine
modified_base	2660..2664 mod_base = OTHER note = thymine
modified_base	2666 mod_base = OTHER note = thymine
modified_base	2668 mod_base = OTHER note = thymine
modified_base	2670 mod_base = OTHER note = thymine
modified_base	2672..2673 mod_base = OTHER note = thymine
modified_base	2677 mod_base = OTHER note = thymine
modified_base	2681 mod_base = OTHER note = thymine

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modified_base	2683
	mod_base = OTHER
	note = thymine
modified_base	2692..2693
	mod_base = OTHER
	note = thymine
modified_base	2695..2696
	mod_base = OTHER
	note = thymine
modified_base	2699..2700
	mod_base = OTHER
	note = thymine
modified_base	2704..2705
	mod_base = OTHER
	note = thymine
modified_base	2711..2715
	mod_base = OTHER
	note = thymine
modified_base	2720..2721
	mod_base = OTHER
	note = thymine
modified_base	2723
	mod_base = OTHER
	note = thymine
modified_base	2726
	mod_base = OTHER
	note = thymine
modified_base	2731..2733
	mod_base = OTHER
	note = thymine
modified_base	2736..2737
	mod_base = OTHER
	note = thymine
modified_base	2744..2746
	mod_base = OTHER
	note = thymine
modified_base	2748..2749
	mod_base = OTHER
	note = thymine
modified_base	2756..2758
	mod_base = OTHER
	note = thymine
modified_base	2760..2762
	mod_base = OTHER
	note = thymine
modified_base	2765..2766
	mod_base = OTHER
	note = thymine
modified_base	2776..2777
	mod_base = OTHER
	note = thymine
modified_base	2780
	mod_base = OTHER
	note = thymine
modified_base	2784
	mod_base = OTHER
	note = thymine
modified_base	2788
	mod_base = OTHER
	note = thymine
modified_base	2790..2792
	mod_base = OTHER
	note = thymine
modified_base	2794
	mod_base = OTHER
	note = thymine
modified_base	2796..2797
	mod_base = OTHER
	note = thymine
modified_base	2799
	mod_base = OTHER
	note = thymine
modified_base	2801
	mod_base = OTHER
	note = thymine
modified_base	2806
	mod_base = OTHER
	note = thymine
modified_base	2808

-continued

	mod_base = OTHER
	note = thymine
modified_base	2816
	mod_base = OTHER
	note = thymine
modified_base	2825
	mod_base = OTHER
	note = thymine
modified_base	2827..2829
	mod_base = OTHER
	note = thymine
modified_base	2834
	mod_base = OTHER
	note = thymine
modified_base	2838
	mod_base = OTHER
	note = thymine
modified_base	2840..2841
	mod_base = OTHER
	note = thymine
modified_base	2844
	mod_base = OTHER
	note = thymine
modified_base	2846..2847
	mod_base = OTHER
	note = thymine
modified_base	2851
	mod_base = OTHER
	note = thymine
modified_base	2857
	mod_base = OTHER
	note = thymine
modified_base	2861
	mod_base = OTHER
	note = thymine
modified_base	2873..2874
	mod_base = OTHER
	note = thymine
modified_base	2879
	mod_base = OTHER
	note = thymine
modified_base	2882..2884
	mod_base = OTHER
	note = thymine
modified_base	2891
	mod_base = OTHER
	note = thymine
modified_base	2896
	mod_base = OTHER
	note = thymine
modified_base	2898..2899
	mod_base = OTHER
	note = thymine
modified_base	2901
	mod_base = OTHER
	note = thymine
modified_base	2904..2906
	mod_base = OTHER
	note = thymine
modified_base	2910
	mod_base = OTHER
	note = thymine
modified_base	2913
	mod_base = OTHER
	note = thymine
modified_base	2915..2916
	mod_base = OTHER
	note = thymine
modified_base	2920
	mod_base = OTHER
	note = thymine
modified_base	2929
	mod_base = OTHER
	note = thymine
modified_base	2939..2941
	mod_base = OTHER
	note = thymine
modified_base	2946
	mod_base = OTHER

-continued

modified_base	note = thymine 2950..2951 mod_base = OTHER note = thymine
modified_base	2954 mod_base = OTHER note = thymine
modified_base	2956 mod_base = OTHER note = thymine
modified_base	2961..2962 mod_base = OTHER note = thymine
modified_base	2964 mod_base = OTHER note = thymine
modified_base	2966 mod_base = OTHER note = thymine
modified_base	2968 mod_base = OTHER note = thymine
modified_base	2970..2971 mod_base = OTHER note = thymine
modified_base	2973..2975 mod_base = OTHER note = thymine
modified_base	2983..2984 mod_base = OTHER note = thymine
modified_base	2987 mod_base = OTHER note = thymine
modified_base	2989 mod_base = OTHER note = thymine
modified_base	2993 mod_base = OTHER note = thymine
modified_base	2995..2997 mod_base = OTHER note = thymine
modified_base	3002..3003 mod_base = OTHER note = thymine
modified_base	3008..3009 mod_base = OTHER note = thymine
modified_base	3011 mod_base = OTHER note = thymine
modified_base	3013 mod_base = OTHER note = thymine
modified_base	3019 mod_base = OTHER note = thymine
modified_base	3025 mod_base = OTHER note = thymine
modified_base	3032..3034 mod_base = OTHER note = thymine
modified_base	3036 mod_base = OTHER note = thymine
modified_base	3049..3050 mod_base = OTHER note = thymine
modified_base	3053 mod_base = OTHER note = thymine
modified_base	3061 mod_base = OTHER note = thymine
modified_base	3065 mod_base = OTHER note = thymine

-continued

modified_base	3070
	mod_base = OTHER
	note = thymine
modified_base	3072
	mod_base = OTHER
	note = thymine
modified_base	3076..3078
	mod_base = OTHER
	note = thymine
modified_base	3082..3083
	mod_base = OTHER
	note = thymine
modified_base	3085
	mod_base = OTHER
	note = thymine
modified_base	3096
	mod_base = OTHER
	note = thymine
modified_base	3108..3112
	mod_base = OTHER
	note = thymine
modified_base	3117
	mod_base = OTHER
	note = thymine
modified_base	3120
	mod_base = OTHER
	note = thymine
modified_base	3124..3126
	mod_base = OTHER
	note = thymine
modified_base	3128..3130
	mod_base = OTHER
	note = thymine
modified_base	3141
	mod_base = OTHER
	note = thymine
modified_base	3146
	mod_base = OTHER
	note = thymine
modified_base	3151..3152
	mod_base = OTHER
	note = thymine
modified_base	3157..3158
	mod_base = OTHER
	note = thymine
modified_base	3175..3176
	mod_base = OTHER
	note = thymine
modified_base	3179
	mod_base = OTHER
	note = thymine
modified_base	3185
	mod_base = OTHER
	note = thymine
modified_base	3189..3190
	mod_base = OTHER
	note = thymine
modified_base	3200
	mod_base = OTHER
	note = thymine
modified_base	3204
	mod_base = OTHER
	note = thymine
modified_base	3210
	mod_base = OTHER
	note = thymine
modified_base	3213
	mod_base = OTHER
	note = thymine
modified_base	3215
	mod_base = OTHER
	note = thymine
modified_base	3220
	mod_base = OTHER
	note = thymine
modified_base	3225
	mod_base = OTHER
	note = thymine
modified_base	3228

-continued

	mod_base = OTHER
	note = thymine
modified_base	3230
	mod_base = OTHER
	note = thymine
modified_base	3233
	mod_base = OTHER
	note = thymine
modified_base	3236
	mod_base = OTHER
	note = thymine
modified_base	3240..3241
	mod_base = OTHER
	note = thymine
modified_base	3244
	mod_base = OTHER
	note = thymine
modified_base	3246
	mod_base = OTHER
	note = thymine
modified_base	3248
	mod_base = OTHER
	note = thymine
modified_base	3254
	mod_base = OTHER
	note = thymine
modified_base	3256
	mod_base = OTHER
	note = thymine
modified_base	3259
	mod_base = OTHER
	note = thymine
modified_base	3261..3263
	mod_base = OTHER
	note = thymine
modified_base	3273
	mod_base = OTHER
	note = thymine
modified_base	3282
	mod_base = OTHER
	note = thymine
modified_base	3287
	mod_base = OTHER
	note = thymine
modified_base	3290..3291
	mod_base = OTHER
	note = thymine
modified_base	3294
	mod_base = OTHER
	note = thymine
modified_base	3300
	mod_base = OTHER
	note = thymine
modified_base	3302
	mod_base = OTHER
	note = thymine
modified_base	3306
	mod_base = OTHER
	note = thymine
modified_base	3317
	mod_base = OTHER
	note = thymine
modified_base	3323
	mod_base = OTHER
	note = thymine
modified_base	3332..3333
	mod_base = OTHER
	note = thymine
modified_base	3335
	mod_base = OTHER
	note = thymine
modified_base	3342..3343
	mod_base = OTHER
	note = thymine
modified_base	3347
	mod_base = OTHER
	note = thymine
modified_base	3354
	mod_base = OTHER

-continued

modified_base	note = thymine 3356 mod_base = OTHER note = thymine
modified_base	3360 mod_base = OTHER note = thymine
modified_base	3366 mod_base = OTHER note = thymine
modified_base	3370 mod_base = OTHER note = thymine
modified_base	3372 mod_base = OTHER note = thymine
modified_base	3374 mod_base = OTHER note = thymine
modified_base	3379 mod_base = OTHER note = thymine
modified_base	3385 mod_base = OTHER note = thymine
modified_base	3387 mod_base = OTHER note = thymine
modified_base	3393..3395 mod_base = OTHER note = thymine
modified_base	3401 mod_base = OTHER note = thymine
modified_base	3403..3404 mod_base = OTHER note = thymine
modified_base	3406 mod_base = OTHER note = thymine
modified_base	3408 mod_base = OTHER note = thymine
modified_base	3414..3415 mod_base = OTHER note = thymine
modified_base	3417 mod_base = OTHER note = thymine
modified_base	3419 mod_base = OTHER note = thymine
modified_base	3423 mod_base = OTHER note = thymine
modified_base	3429..3430 mod_base = OTHER note = thymine
modified_base	3432..3434 mod_base = OTHER note = thymine
modified_base	3436 mod_base = OTHER note = thymine
modified_base	3444 mod_base = OTHER note = thymine
modified_base	3451..3454 mod_base = OTHER note = thymine
modified_base	3458..3459 mod_base = OTHER note = thymine
modified_base	3464..3465 mod_base = OTHER note = thymine
modified_base	3470 mod_base = OTHER note = thymine

-continued

modified_base	3472
	mod_base = OTHER
	note = thymine
modified_base	3475..3477
	mod_base = OTHER
	note = thymine
modified_base	3480
	mod_base = OTHER
	note = thymine
modified_base	3483..3484
	mod_base = OTHER
	note = thymine
modified_base	3488..3489
	mod_base = OTHER
	note = thymine
modified_base	3509
	mod_base = OTHER
	note = thymine
modified_base	3518..3519
	mod_base = OTHER
	note = thymine
modified_base	3521
	mod_base = OTHER
	note = thymine
modified_base	3525
	mod_base = OTHER
	note = thymine
modified_base	3527
	mod_base = OTHER
	note = thymine
modified_base	3534..3535
	mod_base = OTHER
	note = thymine
modified_base	3543
	mod_base = OTHER
	note = thymine
modified_base	3546
	mod_base = OTHER
	note = thymine
modified_base	3551
	mod_base = OTHER
	note = thymine
modified_base	3554
	mod_base = OTHER
	note = thymine
modified_base	3556..3558
	mod_base = OTHER
	note = thymine
modified_base	3572
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	note = thymine
modified_base	3575
	mod_base = OTHER
	note = thymine
modified_base	3578
	mod_base = OTHER
	note = thymine
modified_base	3581
	mod_base = OTHER
	note = thymine
modified_base	3586
	mod_base = OTHER
	note = thymine
modified_base	3589..3591
	mod_base = OTHER
	note = thymine
modified_base	3596..3599
	mod_base = OTHER
	note = thymine
modified_base	3601
	mod_base = OTHER
	note = thymine
modified_base	3609..3610
	mod_base = OTHER
	note = thymine
modified_base	3622
	mod_base = OTHER
	note = thymine
modified_base	3630

-continued

	mod_base = OTHER
	note = thymine
modified_base	3633..3634
	mod_base = OTHER
	note = thymine
modified_base	3637
	mod_base = OTHER
	note = thymine
modified_base	3643..3645
	mod_base = OTHER
	note = thymine
modified_base	3648
	mod_base = OTHER
	note = thymine
modified_base	3650
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	note = thymine
modified_base	3653
	mod_base = OTHER
	note = thymine
modified_base	3657
	mod_base = OTHER
	note = thymine
modified_base	3660
	mod_base = OTHER
	note = thymine
modified_base	3662
	mod_base = OTHER
	note = thymine
modified_base	3665..3667
	mod_base = OTHER
	note = thymine
modified_base	3676..3677
	mod_base = OTHER
	note = thymine
modified_base	3686..3687
	mod_base = OTHER
	note = thymine
modified_base	3690
	mod_base = OTHER
	note = thymine
modified_base	3693
	mod_base = OTHER
	note = thymine
modified_base	3696..3697
	mod_base = OTHER
	note = thymine
modified_base	3705..3706
	mod_base = OTHER
	note = thymine
modified_base	3710..3713
	mod_base = OTHER
	note = thymine
modified_base	3718..3719
	mod_base = OTHER
	note = thymine
modified_base	3722..3723
	mod_base = OTHER
	note = thymine
modified_base	3725
	mod_base = OTHER
	note = thymine
modified_base	3728..3729
	mod_base = OTHER
	note = thymine
modified_base	3731..3733
	mod_base = OTHER
	note = thymine
modified_base	3745
	mod_base = OTHER
	note = thymine
modified_base	3756..3757
	mod_base = OTHER
	note = thymine
modified_base	3767
	mod_base = OTHER
	note = thymine
modified_base	3773
	mod_base = OTHER

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modified_base	note = thymine 3776 mod_base = OTHER
modified_base	note = thymine 3778..3779 mod_base = OTHER
modified_base	note = thymine 3781 mod_base = OTHER
modified_base	note = thymine 3783..3784 mod_base = OTHER
modified_base	note = thymine 3789..3790 mod_base = OTHER
modified_base	note = thymine 3792 mod_base = OTHER
modified_base	note = thymine 3798..3802 mod_base = OTHER
modified_base	note = thymine 3804 mod_base = OTHER
modified_base	note = thymine 3807 mod_base = OTHER
modified_base	note = thymine 3812 mod_base = OTHER
modified_base	note = thymine 3814 mod_base = OTHER
modified_base	note = thymine 3819 mod_base = OTHER
modified_base	note = thymine 3821 mod_base = OTHER
modified_base	note = thymine 3826..3827 mod_base = OTHER
modified_base	note = thymine 3829 mod_base = OTHER
modified_base	note = thymine 3831 mod_base = OTHER
modified_base	note = thymine 3833..3835 mod_base = OTHER
modified_base	note = thymine 3845 mod_base = OTHER
modified_base	note = thymine 3848 mod_base = OTHER
modified_base	note = thymine 3850..3852 mod_base = OTHER
modified_base	note = thymine 3854 mod_base = OTHER
modified_base	note = thymine 3859 mod_base = OTHER
modified_base	note = thymine 3862 mod_base = OTHER
modified_base	note = thymine 3868..3870 mod_base = OTHER
modified_base	note = thymine 3876..3877 mod_base = OTHER
modified_base	note = thymine 3884 mod_base = OTHER
	note = thymine

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modified_base	3892
	mod_base = OTHER
	note = thymine
modified_base	3895
	mod_base = OTHER
	note = thymine
modified_base	3900
	mod_base = OTHER
	note = thymine
modified_base	3902
	mod_base = OTHER
	note = thymine
modified_base	3904
	mod_base = OTHER
	note = thymine
modified_base	3907
	mod_base = OTHER
	note = thymine
modified_base	3909..3910
	mod_base = OTHER
	note = thymine
modified_base	3913
	mod_base = OTHER
	note = thymine
modified_base	3916..3917
	mod_base = OTHER
	note = thymine
modified_base	3919..3920
	mod_base = OTHER
	note = thymine
modified_base	3922
	mod_base = OTHER
	note = thymine
modified_base	3926
	mod_base = OTHER
	note = thymine
modified_base	3938..3942
	mod_base = OTHER
	note = thymine
modified_base	3954
	mod_base = OTHER
	note = thymine
modified_base	3957
	mod_base = OTHER
	note = thymine
modified_base	3961..3962
	mod_base = OTHER
	note = thymine
modified_base	3964
	mod_base = OTHER
	note = thymine
modified_base	3966
	mod_base = OTHER
	note = thymine
modified_base	3979..3981
	mod_base = OTHER
	note = thymine
modified_base	3984
	mod_base = OTHER
	note = thymine
modified_base	3992
	mod_base = OTHER
	note = thymine
modified_base	3998..3999
	mod_base = OTHER
	note = thymine
modified_base	4002..4004
	mod_base = OTHER
	note = thymine
modified_base	4008
	mod_base = OTHER
	note = thymine
modified_base	4010
	mod_base = OTHER
	note = thymine
modified_base	4014
	mod_base = OTHER
	note = thymine
modified_base	4017..4018

-continued

	mod_base = OTHER
	note = thymine
modified_base	4020
	mod_base = OTHER
	note = thymine
modified_base	4025
	mod_base = OTHER
	note = thymine
modified_base	4027..4028
	mod_base = OTHER
	note = thymine
modified_base	4030..4031
	mod_base = OTHER
	note = thymine
modified_base	4036..4038
	mod_base = OTHER
	note = thymine
modified_base	4040
	mod_base = OTHER
	note = thymine
modified_base	4046
	mod_base = OTHER
	note = thymine
modified_base	4051..4053
	mod_base = OTHER
	note = thymine
modified_base	4058..4059
	mod_base = OTHER
	note = thymine
modified_base	4066..4067
	mod_base = OTHER
	note = thymine
modified_base	4071..4073
	mod_base = OTHER
	note = thymine
modified_base	4077
	mod_base = OTHER
	note = thymine
modified_base	4079
	mod_base = OTHER
	note = thymine
modified_base	4083
	mod_base = OTHER
	note = thymine
modified_base	4085
	mod_base = OTHER
	note = thymine
modified_base	4087..4089
	mod_base = OTHER
	note = thymine
modified_base	4093
	mod_base = OTHER
	note = thymine
modified_base	4097
	mod_base = OTHER
	note = thymine
modified_base	4103..4105
	mod_base = OTHER
	note = thymine
modified_base	4109..4110
	mod_base = OTHER
	note = thymine
modified_base	4113..4115
	mod_base = OTHER
	note = thymine
modified_base	4125
	mod_base = OTHER
	note = thymine
modified_base	4128
	mod_base = OTHER
	note = thymine
modified_base	4130..4131
	mod_base = OTHER
	note = thymine
modified_base	4134
	mod_base = OTHER
	note = thymine
modified_base	4148
	mod_base = OTHER

-continued

modified_base	note = thymine 4151..4152 mod_base = OTHER note = thymine
modified_base	4154 mod_base = OTHER note = thymine
modified_base	4166..4168 mod_base = OTHER note = thymine
modified_base	4170 mod_base = OTHER note = thymine
modified_base	4178..4183 mod_base = OTHER note = thymine
modified_base	4188 mod_base = OTHER note = thymine
modified_base	4190 mod_base = OTHER note = thymine
modified_base	4193 mod_base = OTHER note = thymine
modified_base	4205 mod_base = OTHER note = thymine
modified_base	4208 mod_base = OTHER note = thymine
modified_base	4210..4212 mod_base = OTHER note = thymine
modified_base	4215 mod_base = OTHER note = thymine
modified_base	4219 mod_base = OTHER note = thymine
modified_base	4224 mod_base = OTHER note = thymine
modified_base	4229 mod_base = OTHER note = thymine
modified_base	4237 mod_base = OTHER note = thymine
modified_base	4244 mod_base = OTHER note = thymine
modified_base	4246 mod_base = OTHER note = thymine
modified_base	4248 mod_base = OTHER note = thymine
modified_base	4256 mod_base = OTHER note = thymine
modified_base	4265 mod_base = OTHER note = thymine
modified_base	4272 mod_base = OTHER note = thymine
modified_base	4276..4278 mod_base = OTHER note = thymine
modified_base	4281 mod_base = OTHER note = thymine
modified_base	4283 mod_base = OTHER note = thymine
modified_base	4286 mod_base = OTHER note = thymine

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modified_base	4289..4290 mod_base = OTHER note = thymine
modified_base	4293..4294 mod_base = OTHER note = thymine
modified_base	4300 mod_base = OTHER note = thymine
modified_base	4303 mod_base = OTHER note = thymine
modified_base	4307 mod_base = OTHER note = thymine
modified_base	4313 mod_base = OTHER note = thymine
modified_base	4315..4316 mod_base = OTHER note = thymine
modified_base	4318 mod_base = OTHER note = thymine
modified_base	4321 mod_base = OTHER note = thymine
modified_base	4325 mod_base = OTHER note = thymine
modified_base	4333 mod_base = OTHER note = thymine
modified_base	4335..4337 mod_base = OTHER note = thymine
modified_base	4339 mod_base = OTHER note = thymine
modified_base	4347 mod_base = OTHER note = thymine
modified_base	4351..4354 mod_base = OTHER note = thymine
modified_base	4356..4357 mod_base = OTHER note = thymine
modified_base	4361 mod_base = OTHER note = thymine
modified_base	4368 mod_base = OTHER note = thymine
modified_base	4370..4371 mod_base = OTHER note = thymine
modified_base	4375 mod_base = OTHER note = thymine
modified_base	4377..4378 mod_base = OTHER note = thymine
modified_base	4383 mod_base = OTHER note = thymine
modified_base	4388 mod_base = OTHER note = thymine
modified_base	4390 mod_base = OTHER note = thymine
modified_base	4394 mod_base = OTHER note = thymine
modified_base	4397 mod_base = OTHER note = thymine
modified_base	4399

-continued

	mod_base = OTHER
	note = thymine
modified_base	4402
	mod_base = OTHER
	note = thymine
modified_base	4405..4408
	mod_base = OTHER
	note = thymine
modified_base	4410
	mod_base = OTHER
	note = thymine
modified_base	4412..4414
	mod_base = OTHER
	note = thymine
modified_base	4416
	mod_base = OTHER
	note = thymine
modified_base	4420..4424
	mod_base = OTHER
	note = thymine
modified_base	4426
	mod_base = OTHER
	note = thymine
modified_base	4430
	mod_base = OTHER
	note = thymine
modified_base	4432
	mod_base = OTHER
	note = thymine
modified_base	4435..4437
	mod_base = OTHER
	note = thymine
modified_base	4439
	mod_base = OTHER
	note = thymine
modified_base	4442..4443
	mod_base = OTHER
	note = thymine
modified_base	4449
	mod_base = OTHER
	note = thymine
modified_base	4452
	mod_base = OTHER
	note = thymine
modified_base	4456
	mod_base = OTHER
	note = thymine
modified_base	4458..4459
	mod_base = OTHER
	note = thymine
modified_base	4461
	mod_base = OTHER
	note = thymine
modified_base	4463
	mod_base = OTHER
	note = thymine
modified_base	4468
	mod_base = OTHER
	note = thymine
modified_base	4471
	mod_base = OTHER
	note = thymine
modified_base	4473..4474
	mod_base = OTHER
	note = thymine
modified_base	4476
	mod_base = OTHER
	note = thymine
modified_base	4479
	mod_base = OTHER
	note = thymine
modified_base	4481..4482
	mod_base = OTHER
	note = thymine
modified_base	4487..4490
	mod_base = OTHER
	note = thymine
modified_base	4492..4493
	mod_base = OTHER

-continued

modified_base	note = thymine 4500..4503 mod_base = OTHER
modified_base	note = thymine 4512..4513 mod_base = OTHER
modified_base	note = thymine 4515 mod_base = OTHER
modified_base	note = thymine 4517 mod_base = OTHER
modified_base	note = thymine 4519..4520 mod_base = OTHER
modified_base	note = thymine 4522..4523 mod_base = OTHER
modified_base	note = thymine 4525 mod_base = OTHER
modified_base	note = thymine 4529 mod_base = OTHER
modified_base	note = thymine 4537 mod_base = OTHER
modified_base	note = thymine 4539..4541 mod_base = OTHER
modified_base	note = thymine 4543 mod_base = OTHER
modified_base	note = thymine 4550..4551 mod_base = OTHER
modified_base	note = thymine 4553 mod_base = OTHER
modified_base	note = thymine 4555..4556 mod_base = OTHER
modified_base	note = thymine 4561 mod_base = OTHER
modified_base	note = thymine 4563 mod_base = OTHER
modified_base	note = thymine 4567 mod_base = OTHER
modified_base	note = thymine 4569 mod_base = OTHER
modified_base	note = thymine 4571 mod_base = OTHER
modified_base	note = thymine 4573..4574 mod_base = OTHER
modified_base	note = thymine 4576 mod_base = OTHER
modified_base	note = thymine 4580..4581 mod_base = OTHER
modified_base	note = thymine 4590 mod_base = OTHER
modified_base	note = thymine 4597 mod_base = OTHER
modified_base	note = thymine 4599 mod_base = OTHER
modified_base	note = thymine 4606 mod_base = OTHER
	note = thymine

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modified_base	4610..4611 mod_base = OTHER note = thymine
modified_base	4618..4620 mod_base = OTHER note = thymine
modified_base	4622 mod_base = OTHER note = thymine
modified_base	4624..4625 mod_base = OTHER note = thymine
modified_base	4628 mod_base = OTHER note = thymine
modified_base	4631 mod_base = OTHER note = thymine
modified_base	4633 mod_base = OTHER note = thymine
modified_base	4637 mod_base = OTHER note = thymine
modified_base	4641 mod_base = OTHER note = thymine
modified_base	4646..4647 mod_base = OTHER note = thymine
modified_base	4649 mod_base = OTHER note = thymine
modified_base	4652..4655 mod_base = OTHER note = thymine
modified_base	4657..4660 mod_base = OTHER note = thymine
modified_base	4662..4663 mod_base = OTHER note = thymine
modified_base	4671 mod_base = OTHER note = thymine
modified_base	4673 mod_base = OTHER note = thymine
modified_base	4677 mod_base = OTHER note = thymine
modified_base	4680..4682 mod_base = OTHER note = thymine
modified_base	4686 mod_base = OTHER note = thymine
modified_base	4695 mod_base = OTHER note = thymine
modified_base	4705..4707 mod_base = OTHER note = thymine
modified_base	4709..4710 mod_base = OTHER note = thymine
modified_base	4715 mod_base = OTHER note = thymine
modified_base	4718 mod_base = OTHER note = thymine
modified_base	4720..4721 mod_base = OTHER note = thymine
modified_base	4729 mod_base = OTHER note = thymine
modified_base	4732..4733

-continued

	mod_base = OTHER
	note = thymine
modified_base	4735
	mod_base = OTHER
	note = thymine
modified_base	4739..4741
	mod_base = OTHER
	note = thymine
modified_base	4745
	mod_base = OTHER
	note = thymine
modified_base	4748..4749
	mod_base = OTHER
	note = thymine
modified_base	4751..4752
	mod_base = OTHER
	note = thymine
modified_base	4764..4767
	mod_base = OTHER
	note = thymine
modified_base	4769..4770
	mod_base = OTHER
	note = thymine
modified_base	4773
	mod_base = OTHER
	note = thymine
modified_base	4781
	mod_base = OTHER
	note = thymine
modified_base	4783
	mod_base = OTHER
	note = thymine
modified_base	4786
	mod_base = OTHER
	note = thymine
modified_base	4789
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The invention claimed is:

1. A method of treating a tumor comprising administering to a subject in need thereof a lymphocytic choriomeningitis virus (LCMV), wherein the sequences of the L-ribonucleic acids of the LCMV are selected from the group consisting of WE, Clone 13 and Docile L-ribonucleic acid sequences, and the S-ribonucleic acids of the LCMV are selected from the group consisting of WE, Clone 13 and Docile S-ribonucleic acid sequences.

2. The method of claim 1, wherein the method comprises activating innate (congenital) or adaptive immune cells and thereby effecting tumor regression.

3. The method of claim 1, wherein the method comprises effecting increased secretion of anti-tumoral cytokines by immune cells by activating said immune cells.

4. The method of claim 3, wherein the anti-tumoral cytokine is interferon- α , interferon- β or interferon- γ .

5. The method of claim 1, wherein the tumor is selected from the group consisting of carcinoma, melanoma, blastoma, lymphoma, and sarcoma.

6. The method of claim 1, wherein the arenavirus comprises a nucleic acid comprising an S-ribonucleic acid sequence according to SEQ ID No. 5 or ambisense sequence thereof, or a nucleic acid comprising an L-ribonucleic acid sequence according to SEQ ID No. 6 or ambisense sequence thereof.

7. The method of claim 1, wherein the lymphocytic choriomeningitis virus is a lymphocytic choriomeningitis virus strain WE.

8. The method of claim 1, wherein the subject is a human.

9. The method of claim 1, wherein the L- and S-ribonucleic acids of the LCMV are WE L- and S-ribonucleic acids.

10. A method of treating a tumor comprising administering to a subject in need thereof a lymphocytic choriomeningitis virus (LCMV), wherein the sequences of the L-ribonucleic acids of the LCMV are selected from the group consisting of WE, Clone 13 and Docile L-ribonucleic acid sequences, the S-ribonucleic acids of the LCMV are selected from the group consisting of WE, Clone 13 and Docile S-ribonucleic acid sequences, and the tumor is a solid tumor.

11. The method of claim 10, wherein the tumor is carcinoma.

12. The method of claim 10, wherein the tumor is melanoma.

13. The method of claim 10, wherein the subject is a human.

14. A method of treating a tumor comprising administering to a subject in need thereof a lymphocytic choriomeningitis virus (LCMV), wherein the sequences of the L-ribonucleic acids of the LCMV are selected from the group consisting of WE, Clone 13 and Docile L-ribonucleic acid sequences, the S-ribonucleic acids of the LCMV are selected from the group consisting of WE, Clone 13 and Docile S-ribonucleic acid sequences, and the tumor is selected from the group consisting of oropharynx cancer, colon cancer, lung cancer, cervical cancer, liver cancer, melanoma, pharyngeal cancer, lymphoma, thyroid cancer, and bladder cancer.

15. The method of claim 14, wherein the tumor is selected from the group consisting of oropharynx cancer, colon cancer, lung cancer, cervical cancer, liver cancer, melanoma, pharyngeal cancer, and lymphoma.

16. The method of claim 14, wherein the subject is a human.

* * * * *